NEXT GENERATION SCIENCE STANDARDS ALIGNMENT

STEP Energy Activity Set

Exploring Energy and the Environment

A Supplemental Curriculum
for Middle School Learners

A Midwest Geological Sequestration Consortium Program Funded by the U.S. Department of Energy

Exploring Energy and the Envirionment

A STEP Supplemental Curriculum

Activity Set 1: Exploring Energy Resources

- **Activity 1 - Lesson A:** Energy Use in the United States: A Historical Approach
- **Activity 1 - Lesson B:** Energy Resources Throughout Human History
- **Activity 1 - Lesson C:** Electricity: An Energy Carrier
- **Activity 1 - Lesson D:** Electricity Generation
- **Activity 1 - Lesson E:** Plugged into CO_2
- **Activity 1 - Lesson F:** Energy Flows - Trace it to the Source

Activity Set 2: Carbon Dioxide (CO_2) and Climate Change

- **Activity 2 - Lesson G:** The Carbon Cycle
- **Activity 2 - Lesson H:** What is CO_2?
- **Activity 2 - Lesson I:** The Greenhouse Effect
- **Activity 2 - Lesson J-1:** Evaluating the Evidence of Climate Change
- **Activity 2 - Lesson J-2:** Evaluating the Evidence of Climate Change Addressing Misconceptions
- **Activity 2 - Lesson K:** Examining Your Carbon Footprint, Human Impact Through Transportation

Activity Set 3: Exploring the Science of Carbon Capture and Sequestration (CCUS) as One Option in a Balanced Energy Portfolio

- **Activity 3 - Lesson L:** What is Under Your Feet Basic Geology
- **Activity 3 - Lesson M:** Exploring Porosity & Permeability
- **Activity 3 - Lesson N:** What are CCUS & EOR & How Do They work?

Activity Set 4: Taking it to the next STEP

- **Activity 4 - Lesson O:** Local Energy Debate
- **Activity 4 - Lesson P:** Energy Summit- Global Wedge Game
- **Activity 4 - Lesson Q:** Career Opportunities
- **Activity 4 - Lesson R:** Science Fair Opportunities

Activity Set 1

EXPLORING ENERGY RESOURCES

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Activity Set 1

Exploring Energy Resources

Lessons A–F

A Energy Use in the United States: A Historical Approach

B Energy Resources Throughout Human History

C Electricity: An Energy Carrier

D Electricity Generation

E Plugged into CO_2

F Energy Flows—Trace It to the Source

Lesson A

Energy Use in the United States: A Historical Approach

Quick Overview

Lesson Scope: Energy use for transportation through history

Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Photo cards, worksheet, publications/ web resources

Materials Needed per Student

Photo Cards

Worksheet

Publications/ web resources

Each student will need paper and instructions to record trials.

Guiding Question

What is our relationship to energy?

This lesson is designed to initiate student thinking and dialogue regarding the history of **energy** use in the United States. Through authentic historical photo cards, students explore the history of transportation from horse-drawn carriages to personal vehicles. The story behind these images (see Appendix 1.1.1) provides a foundation for engaging students in conversation related to the services or conveniences provided by society's increased ability to harness energy through innovation and technological advancement. This lesson also introduces the concept of energy as a centralizing idea for the basis of this unit.

Key Understandings

- Energy makes change and is present whenever there are moving objects, sound, light, or heat.
- Increases in the global population cause an increase in energy consumption.
- Society's increased ability to harness energy over the last two hundred years has changed the way Americans live.
- Our current lifestyle is largely dependent on the energy stored within **fossil fuels**.
- Over time, people's needs and wants change, as does the need for new and improved technologies.

Objectives

- Students will be able to define energy.
- Students will be able to differentiate between nonrenewable and renewable energy resources.
- Students will create a brief timeline that outlines energy transitions in the United States from the late 1700s to the present.

Vocabulary

- Energy
- Fossil fuels
- Geothermal heat
- Renewable energy
- Solar power
- Wind power

Background Information for the Teacher

Note: *The background information below is meant for the teacher, not to pass along as reading or vocabulary work for the students. Through the activities, students will encounter the vocabulary after they have had a chance to work "hands on" with the materials, which will allow them to tie the vocabulary to their experiences.*

Energy is the main topic of the unit; therefore, it is necessary for students and teachers to understand what **energy** actually is. For starters, energy makes change (such as the change from a liquid to a gas, or the motion of a car producing sound energy) and can take different forms, such as energy in a field, thermal energy, and energy in motion. Energy is present whenever there are moving objects, sound, light, or heat. When objects collide, energy can be transferred from one object to another, thereby changing their motion. In such collisions, some energy is typically also transferred to the surrounding air; as a result, the air is heated and sound is produced.

The sources of all energy on Earth are the Sun and the Earth's core, both of which emit energy in the form of heat. Energy is spontaneously transferred out of hotter regions or objects and into cooler ones. Energy from these two primary sources is transferred in several ways to other entities on Earth. One example is plants extracting energy from the sun through photosynthesis. Plants harness light energy and convert it into fuel by storing it as carbohydrates in plant tissue, which then is consumed by primary consumers. In another example, heat omitted from Earth's iron-rich core is moved via convection through the mantle, which drives the rock cycle and emits energy in the form of geothermal energy, earthquakes, and volcanoes. The topic of energy transfer will be highlighted in later lessons. Teachers are encouraged to read about energy and energy transfers to aid in facilitating these lessons. Several resources are provided in the supplemental materials and references sections at the end of this lesson.

Helping students develop an awareness of the capabilities that energy provides society is imperative as we tackle energy-related discussions. Too often, energy and related climate change studies are brought into classrooms with an overall bias that the use of **fossil fuels** is bad. However, it is important to dig deeper into the energy source to help students understand the significance of energy use, including our society's dependence on and increasing use of energy resources. Understanding energy use throughout history, including the concept of energy transitions, is important as we consider the current challenges and the need for more sustainable systems.

Additionally, it is important to note the significance of population and human consumption and/or behavior. The significant increase in population in the past century coupled with the demands of postindustrialized societies intensifies the exploitation of resources to meet growing energy demands. It is therefore important to identify that population growth and the resulting overconsumption of resources by industrialized nations may be the primary source of environmental problems in the world. These issues can be used to start conversations about the source of the environmental challenges we are facing. Changes in policy and human behavior along with the development of new technologies will play a large role in solving these issues.

In this lesson, we target a brief and focused aspect of the United States' energy history—transportation. Just two hundred years ago, horses provided the primary means for transportation. Today, personal vehicles are the norm. This lesson also provides a brief overview of energy resource use in the United States. By examining energy consumption from 1775 to the present time, students will gain understandings related to energy transitions and our current dependence on fossil fuels.

In brief, the two primary energy transitions in United States history are from (a) wood to coal (1760–1885) and (b) coal to petroleum (1946). Two hundred years ago, wood was the primary fuel used for heating, and horses provided the means for transportation. With the onset of the Industrial Revolution in the eighteenth and nineteenth century, a significant transition occurred, which changed society and the course of life in Western Europe and the United States. Prior to the Industrial Revolution, manufacturing was often done in people's homes, using hand tools or basic machines. Industrialization marked a shift to powered, special-purpose machinery, factories, and the mass production of products. The iron and textile industries, along with the development of the steam engine, played central roles in the Industrial Revolution, which also saw improved systems of transportation, communication, and banking. By 1885, coal was the primary energy source. By 1946, petroleum and natural gas overtook coal. With each energy transition, the daily lives of people shifted. This lesson explores these transitions and the changes that occurred.

It is important to distinguish between energy sources for transportation (petroleum) and energy sources for electrical generation (coal and natural gas).

Common Student Misconceptions/ Student Challenges

Students may not realize that there are different fossil fuels and that they can be used for different purposes. Petroleum is primarily used in the transportation sector, whereas coal and natural gas are used for electrical generation.

Students may not realize the nature of our dependence on fossil fuels. Many believe that renewables, such as wind and solar, already constitute a significant amount of our overall primary sources, but this is often not the case.

Students may think that **renewable energy** will never run out. However, renewable energy does not necessarily mean that energy is permanent or sustainable; it means that the energy can be quickly replenished. For example, although biomass is a renewable energy source, it is possible to deplete it faster than we can replenish it, meaning it could possibly run out. Additionally, **wind power**, **solar power**, and geothermal power, while renewable, can be inconsistent sources of energy, depending on the time of year and location.

Students may also think that renewable energy does not pollute the environment. Transportation, extraction, and materials used in the process of obtaining renewable energy sources inevitably cause some sort of environmental impact that is often not factored into the final usage. The environmental impacts from renewable energy sources depend on how the energy is harnessed and used. For example, using geothermal heat from a hot spring is usually a pollution-free process, but drilling for **geothermal heat** can release hazardous gases, such as hydrogen sulfide, into the environment.

Next Generation Science Standards

Performance Standard.HS-ESS3-1:

Construct an explanation based on evidence for how the availability of natural resources, the occurrence of natural hazards, and changes in climate have influenced human activity.

Disciplinary Core Idea

Resource availability has guided the development of human society.

Science and Engineering Practices

Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, or peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.

Crosscutting Concepts

Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Modern civilization depends on major technological systems.

Materials

Intermediate Energy Infobook: (http://www.need.org/files/curriculum/guides/Intermediate%20Energy%20Infobook.pdf)

- Historical Photo Cards (Appendix 1.1.1), divided into two sets, Set A, Set B
- Worksheet 1. Energy Use in the United States: A Historical Approach (Appendix 1.1.2)
- U.S. Primary Consumption Estimates by Source, 1775–2011 (Appendix 1.1.2)
- U.S. Energy Consumption 2011 by Source (Appendix 1.1.2)

Setup

It will take approximately 20 minutes to makes copies and cut out the historical images using the materials from Appendix A. Students should have paper and pencils.

Length of Lesson

This lesson will take two 45-minute class periods.

Suggested Procedure

First Class Period

Introduction

1. Introduction to Energy: Silent Conversation and Reading. Begin the lesson by writing the word *energy* on the board. Explain the point of this activity, which is a silent conversation. During the activity, the class will develop a definition of energy based on the responses that are written on the board. Instruct students to first spend some time thinking and writing down ideas at their seats. When they are ready and have a single idea to contribute, they may walk up to the board and write a word or short sentence that explains what they think it means, all of which is done silently. After several words or phrases are posted, students will be encouraged to contribute to each other's ideas in a constructive manner. The end result should resemble a mind map.
2. After this, the teacher can facilitate a discussion by asking students about certain parts of the mind map. In this way, a definition can be created from the contributions on the board. This can be accomplished by underlining or circling agreed upon words and phrases. If the discussion is not leading to an accurate definition, the teacher should intervene and guide students to the correct answer

when appropriate. Refer to the Background Information for the Teacher section for an example definition (i.e. "Energy produces change and is the capacity to do work. Energy comes in the form of light, heat, sound, and motion".)

3. Next, students will need paper and pencils. Have students create a T-chart. Nonrenewable resources will be written in one column and renewable resources will be written in the other. Ask students to brainstorm the different types of resources and place them in one of the two columns. Have students partner with a classmate. Tell them that they will have a few minutes to compare and discuss their lists. Ask them to discuss what the difference is between a renewable and a nonrenewable resource. Finally, provide students with the proper definitions of renewable and nonrenewable resources.

4. Read pages 6 and 7 of the *Intermediate Energy Infobook* (National Energy Education Development Project, 2013) as a class or in small reading groups. "Close reading" allows students to pull evidence from the *Energy Infobook* to support their definitions and ideas.

5. Reiterate the definition of energy. Ask students to provide examples of potential energy and kinetic energy and nonrenewable and renewable resources as shown on page 7 of the *Infobook*. This is a good time to have students talk with partners, discuss as a class, and take notes. These are important concepts that build in the unit. Students may use page 6 of the *Infobook* to guide their answers.

6. As a class, revisit the mind map created at the beginning of the activity. Have students add words and phrases to the map that further deepen the concept and definition of energy.

Teaching Strategy: Think-Pair-Share

Second Class Period

Ask students to consider the following discussion questions individually and then share with a designated neighbor. Remind students that they will be asked to report a brief summary of their findings with the class.

Questions to Pose to Students

Note: *Encourage students to reflect back on their activities of daily living as they consider the energy services available to those living in the 1800s and 1900s. Allow students time to brainstorm the energy services that are provided by 24-hour access to electricity and fuels. These services are conveniences such as lighting, home heating, and transportation.*

The following questions are designed to encourage students to explore the importance of energy resources in modern society through an examination of the services and conveniences that are provided. These discussion questions should be revisited following the timeline activity.

1. What do you think life was like for Americans in the 1800s and 1900s?
2. What was the source of light for reading at night, and how did they get heat for cooking food?
3. What was the primary means of transportation?
4. What would your life be like without electricity and fuels for transportation?

Activity

Place students in small working groups (three to four students) and provide each team with a set of historical photo cards (see Appendix 1.1.1). The historical images are divided into two similar sets and primarily represent the transportation sector from the late 1800s to early 1900s.

1. Ask students to examine the historical photos and then place them in chronological order from earliest to latest. Emphasize that the goal is to create a relative timeline.
2. Once teams have their relative timelines completed, hand out Worksheet 1 (Appendix 1.1.2). The worksheet includes data on the energy consumed in the United States from 1775 to 2011 (in quadrillion British thermal units [BTUs]), Figure 1.
3. The U.S. Energy Information Administration (EIA) graph (Figure 2) documents the primary energy sources utilized in the United States. First ask students to determine the two primary energy transitions in American history. This is important for students to understand as the unit develops because (1) students need to understand we are still dependent on fossil fuels and (2) energy transitions cannot occur overnight.
4. Finally, have a discussion about the world population and its effects on energy consumption. You can visit a number of sites that show population trends and energy use to help initiate this conversation: *http://www.census.gov/popclock/* and *http://investeddevelopment.com/wp-content/uploads/2012/08/Per-Capita-Electric-Consumption-Growth-vs-Population-Growth.png* and *http://oilprice.com/Energy/Energy-General/A-Look-at-World-Energy-Consumption-Over-the-Last-200-Years.html*

   Discussions might include the following:

   a. How does population affect energy consumption?
   b. Which countries consume the most energy and why?

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

Following the class discussion related to the energy transitions, ask students to answer these questions independently:

1. How do you think the energy sources that we depend on most today (for our electricity, home heating, and transportation) are similar to or different from those represented in your historical timeline? Give evidence from today's lesson to support your answer.

   We are largely still dependent on fossil fuels.

2. Do you think there will be another energy transition in the future? Why or why not? Give evidence to support your answer.

   It is important to ask students to consider the next energy transition. Fuel resources and carbon constraints have affected and will continue to affect the future generations. Transitioning to diverse energy sources is considered a grand challenge presented to humankind. Transitions may also require major behavior changes and technological advances. The following questions will lead students to engage in these conversations.

   Are there societies in the world today that exist without the modern conveniences that the majority of Americans have access to? What does this indicate regarding the world's energy consumption? The United States accounts for most of the world's use of resources per capita. Because population and energy demands are increasing, there must be new changes and transitions to solve these current problems.

3. How has the population increase played a role in the world's energy consumption?

   As the population increases, so does energy consumption. The current population is estimated to be over seven billion people

   Beyond technological advances, what are potential solutions to the world's energy consumption demands?

   Answers can vary, but may include solutions around sustainability, policy and governmental regulation of the population and energy use, varying degrees of behavioral change that reduce consumption, or potential government and structural changes of economies that do not focus on heavy production and consumption of goods and services.

Supplemental Materials and References

The National Academies. (2014). *What you need to know about energy.* Retrieved from http://needtoknow.nas.edu/energy/

- This website provides a current outlook on how the United States. uses energy, our energy sources, and the cost of energy, and it outlines energy efficiency techniques.

National Energy Education Development Project (NEED). (2018). Intermediate Energy Infobook. Retrieved from http://www.need.org/files/curriculum/guides/Intermediate%20Energy%20Infobook.pdf

- This resource is an excellent source for adding content to lessons for students as well as background information for the teacher. The *Infobook* outlines the various types of energy sources, electricity use and generation, and a brief history of energy.

O'Connor, P. (2010). *Energy transitions, the Pardee papers*. Retrieved from http://www.bu.edu/pardee/files/2010/11/12-PP-Nov2010.pdf

- This analysis examines historical energy transitions, focusing on how energy resources and technologies emerged to meet the changing demands for energy services in the United States.

Rhodes, R. (2007). *Energy transitions: A curious history.* Retrieved from http://iisdb.stanford.edu/pubs/21972/Rhodes-Energy_Transitions.pdf

- A talk given at the Security Implications of Increased Global Reliance on Nuclear Power Conference outlining the problems historic energy transitions faced in the United States.

U.S. Energy Information Administration. (2012). *Monthly energy review.* Retrieved from http://www.eia.gov/totalenergy/data/monthly/pdf/mer.pdf

Set A.1

Frank Meyers Papers, 1899 UW American Heritage Center

Set B.1

Frank Meyers Papers, 1900 UW American Heritage Center

Set A.2

Frank Meyers Papers, 1869 UW American Heritage Center

Set B.2

Frank J. Meyers Papers, 1920 UW American Heritage Center

Set A.3

Francis E. Warren Papers, 1916 UW American Heritage Center

Set B.3

Edith S. and Frederic E. Clements Papers, 1918 UW American Heritage Center

Set A.4

Frank J. Meyers Papers, 1927 UW American Heritage Center

Set B.4

Frank J. Meyers Papers, 1868 UW American Heritage Center

Set A.5

Frank J. Meyers Papers, 1927 UW American Heritage Center

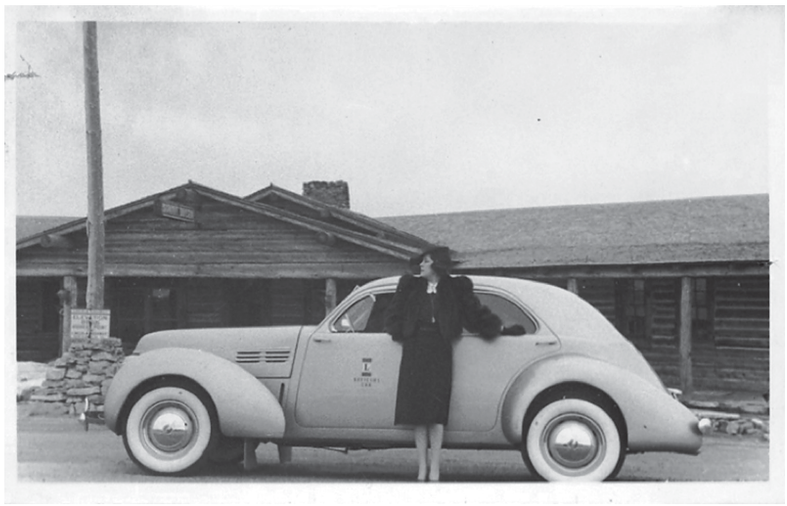
Set B.5

Frank J. Meyers Papers, 1920 UW American Heritage Center

Set A.6

Frank J. Meyers Papers, 1962 UW American Heritage Center

Set B.6



Frank J. Meyers Papers, 1947 -UW American Heritage

1. What energy sources were represented in historical images of your relative timeline?

2. Examine the data presented by the U.S. Energy Information Administration figures on the following pages (Figures 1 and 2).

   What primary energy sources are listed?

   1.

   2.

   3.

   4.

   5.

   6.

   7.

3. Of these, which energy sources were Americans most dependent on in the early 1800s?

4. What can account for the increase in energy use in the mid-1800s?

   a. What energy transitions were taking place at that time?

   b. How did the economic activity and human lifestyles or behaviors change?

5. The second energy transition occurred in approximately 1946. Describe this transition.

6. Now, think about the cars and buses you rode in to get to school today? What primary energy source was used?

7. Think ahead to the year 2040. What energy source will likely fuel our vehicles? Why do you think this?

8. Do we already have the technologies necessary to allow for this transition? If so, what are they? If not, what discoveries or innovations must be made? Can you think of any other changes that must occur for us to adapt?

U.S. Primary Energy Consumption Estimates by Source, 1775–2011

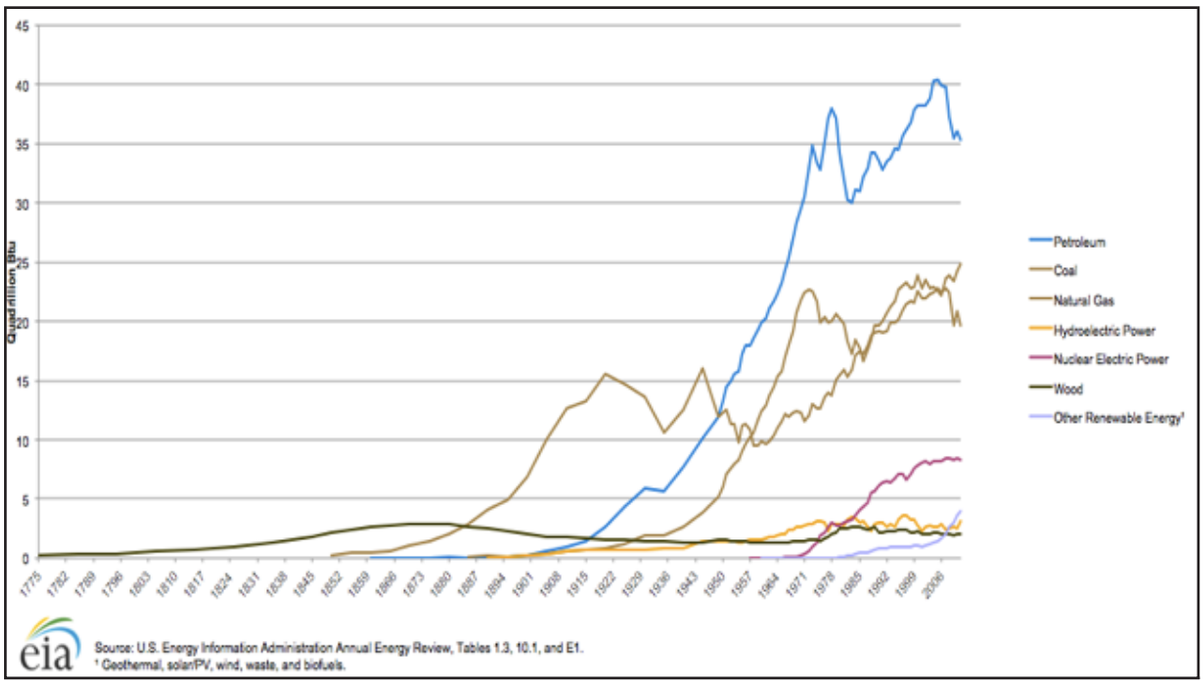


Figure 1. U.S. Primary Energy Consumption Estimates by Source, 1775–2011. From the U.S. Energy Information Administration (EIA).

United States Energy Consumption

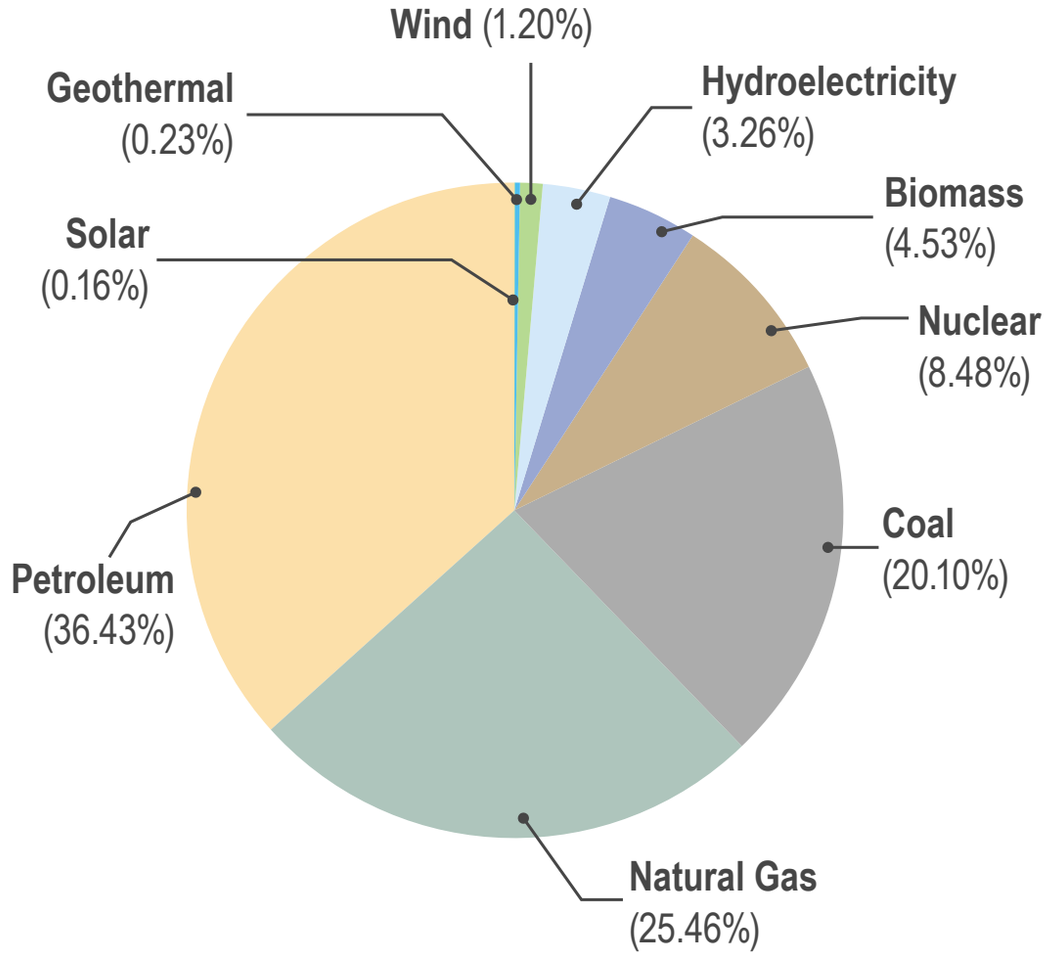


Figure 2. U.S. energy consumption 2011 by source. From the U.S. Energy Information Administration (EIA).

Lesson B

Energy Resources Throughout Human History

Quick Overview

Lesson Scope: Determining how energy sources are formed. Mapping energy sources through history.

Length: Three ~45-minute class periods

Grade Level: 5–8

Materials: Butcher or poster paper, adding machine tape, colored pencils

Materials Needed Per Student

Butcher/ poster paper

Adding machine tape

Colored pencils

Each student will need paper and instructions to record trials.

This lesson includes information from the University of Waikato Web site.

Guiding Question

What is our relationship to energy?

This lesson introduces students to a set of diverse energy sources (both **renewable** and **nonrenewable**) and asks students to explore how these energy sources were formed and how they have been utilized throughout human history.

Key Understandings

- Energy and fuels are derived from **natural resources** and their uses affect the environment.
- Technology has changed the way humans have used primary energy sources throughout history.

Objectives

- Students will be able to draw and describe how a particular energy source is created and formed.
- Students will construct a detailed timeline outlining a resource's historical use and how that use has changed because of technological advances.
- Students will write a summary about how a resource affects the **environment** and how the resource has influenced human activity.

- **Vocabulary**
- Climate change
- Coal
- Environment
- Erosion
- Evolution
- Fossil fuels
- Hydrocarbons
- Natural resources
- Nonrenewable
- Plate tectonics
- Renewable

Background Information for the Teacher

This lesson allows students the opportunity to dive into the specifics of where a particular resource comes from, the history of this resource, and how it has changed over time. It may be a good idea to have books and computers available (in addition to the *Intermediate Energy Infobook*) to allow students to access more information on each resource to aid in the creation of their posters. Further lessons will expand on the steps the energy source takes to get to homes and businesses, but it is important to first understand where the energy source comes from and how it is originally created.

Common Student Misconceptions/ Student Challenges

Students may not realize that humans have been using various forms of **fossil fuels** as energy for a long time, although the way humans have used them has changed dramatically over time. These changes can be tied to advances in technology. In addition, the fuels we use today may change in the future, depending on technology and demand.

Furthermore, students may have misconceptions about where fossil fuels are located and how they are formed. For example, textbook diagrams sometimes show oil below the surface of the Earth in a large cavern. Figures such as these can lead students to the assumption that oil forms big black pools in caves. It is important that students are aware that although oil occasionally does fill in large cracks and reservoirs, the majority of oil is found in porous rocks (such as sandstone or limestone) and in narrow cracks between small particles such as sand.

Students may also think that fossil fuels cannot be found under the ocean, desert, or forest. This idea originates from two sources. The first is a limited understanding regarding the origin of oil and coal (that it is from fossilized marine or swamp matter). The second is the belief that the Earth's surface is static—that it looks the same now as it did millions of years ago. Without an understanding of ideas such as

plate tectonics, climate change, evolution, and erosion, it is difficult to imagine that what is now a desert could have once been an ocean.

Additionally, although many students understand that **hydrocarbons** or fossil fuels are the result of decaying organisms, they often lack an understanding of the actual material of origin. Fossil fuels did not originate from dinosaurs. Oil is predominantly the result of decaying marine organisms that were buried beneath ocean sediment. **Coal** is the result of decaying swamp vegetation. The misconception that oil originates from dinosaurs may be linked to the many cartoons in which it appears the formation of fossil fuels or a time period when swamps (the material that eventually forms coal) were present coincides with when dinosaurs roamed the Earth. Another misconception among some students is that fossil fuels have always been present in the Earth. This idea may stem from the idea that oil and coal take a long time to form. Alternatively, students may think oil and coal are made in much shorter time periods (hundreds, instead of millions of years). In this lesson, consider utilizing the readings from *Oils and Natural Gas* as a resource (located in the Materials section as a book and pdf) for students to address these misconceptions.

The University of Waikato website (see the Materials and Supplemental Materials and References sections) also contains a useful survey that identifies student misconceptions surrounding fossil fuels.

Next Generation Science Standards

Performance Standard.4-ESS3-1:

Obtain and combine information to describe that energy and fuels are derived from natural resources and their uses affect the **environment**.

Disciplinary Core Idea

Energy and fuels that humans use are derived from natural sources, and their use affects the environment in multiple ways. Some resources are **renewable** over time, and others are **nonrenewable**.

Science and Engineering Practices

Obtain and combine information from books and other reliable media to explain these phenomena.

Crosscutting Concepts

- Cause and effect relationships are routinely identified and used to explain change.
- Over time, people's needs and wants change, as do their demands for new and improved technologies.

Performance Standard.HS-ESS3-1:

Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

Disciplinary Core Idea

Resource availability has guided the development of human society.

Science and Engineering Practices

Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including students' own investigations, models, theories, simulations, and peer review) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.

Crosscutting Concepts

- Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.
- Modern civilization depends on major technological systems.

Materials

- Computers and book resources
- *Historical Timeline: History of Alternative Energy and Fossil Fuels*. (*http://alternativeenergy.procon.org/view.timeline.php?timelineID=000015*)
- *Intermediate Energy Infobook* (*http://www.need.org/files/curriculum/guides/Intermediate%20Energy%20Infobook.pdf*)
- *Introduction to Energy Sources Worksheet* (Appendix 1.2.2)
- Poster instructions (Appendix 1.2.3)
- Poster rubric (Appendix 1.2.4)
- Poster presentation questions (Appendix 1.2.5)
- Large pieces of paper, such as newsprint or butcher paper
- Adding machine tape
- Markers or colored pencils

Setup

It will take approximately 1 to 2 hours to gather and print the materials. These materials can be reused in future years. The materials can also be accessed electronically instead of printed.

Print out or use electronically a classroom set of each of the following:

- *Intermediate Energy Infobook* (this was also used in Activity Set 1-A)
- *Historical Timeline: History of Alternative Energy and Fossil Fuels* (color coded)
- *Introduction to Energy Sources Trivia*

Gather art supplies and paper for students' posters.

Group members can be given separate eras to investigate if desirable (e.g., 2000 BCE–1499, 1500–1799, 1800–1899, 1900–1950, 1951–1999, and 2000–present). **Note:** *BCE refers to before the Common Era.*

Length of Lesson

This lesson will take three 45-minute class periods.

Suggested Procedure

1. Form nine student groups, with each group representing one of the primary fuel sources on the timeline from the list below. (Page numbers refer to the location of the primary fuel source in the *Intermediate Energy Infobook*.)
    a. **Coal** (page 10)
    b. **Natural gas** (page 16)
    c. **Hydropower** (page 14)
    d. **Petroleum** (page 18)
    e. **Wind** (page 26)
    f. **Biomass** (page 8)
    g. **Solar** (page 22)
    h. **Uranium (Nuclear)** (page 24)
    i. **Geothermal** (page 12)
2. Explain that each group will be making informational posters, which will be used to teach the rest of the class about energy sources. In addition to the posters, student groups will be giving brief (5-minute) presentations to the rest of the class about their energy sources and the historical uses of those sources by using a gallery walk format.
3. Instruct the students that the newsprint posters should consist of drawings showing how the energy source is extracted and used to produce electricity. Posters also need to have a written overview and a timeline showing the historical uses of the energy source. Instructions for each are given below (and in Appendix 1.2.3) and could be printed out for students. (See Appendix 1.2.4 for the poster rubric.)
    a. **Drawings**: Draw the origin of the resource and describe how it is formed. For example, coal is formed from past swamps that decayed millions of years ago. After sediments accumulated on top of the swamps, heat and pressure caused a chemical change. Wind is created from various weather systems. When different air masses with varying pressures collide, wind is created.
    b. **Overview paragraph**: Write a paragraph that describes the environmental impacts or natural hazards of the energy source and how the energy source has been used by humans.

c. **Historical timeline**: Create a timeline on another piece of butcher paper or adding machine paper highlighting the significant events (in time) for their fuel type based on the *Historical Timeline: History of Alternative Energy and Fossil Fuels*. Have at least four points on the timeline.

4. In a gallery walk format, set up posters around the classroom. As a group, walk from poster to poster. Each student group should give a 5-minute presentation when the class arrives at their poster.

5. Have each student walk from poster to poster to gather the information necessary to answer the following questions on the worksheet provided in Appendix 1.2.5. *These are optional questions and can be changed to fit the needs of the teacher and class.*

   a. What is the energy source?

   b. How is the energy source formed?

   c. When was it first discovered as an energy source?

   d. How has technology altered the way the resource is used?

   e. What are the environmental impacts of this energy source?

   f. How has this resource influenced human activity?

6. The *Energy Resources Throughout Human History Trivia Sheet* (Appendix 1.2.2) is optional and can be used as homework or as an additional learning activity.

7. The teacher should facilitate a class discussion in which she or he asks the following questions:

   a. Look back over the *Historical Timeline: History of Alternative Energy and Fossil Fuels*. When were fossil fuels first used, and by whom?

   The Chinese first used coal in 2000 BCE, natural gas in 200 BCE, and refined petroleum in the 1st century.

   b. What were fossil fuels first used for? How does this compare to the way fossil fuels are used today?

   Coal was first used for heating and cooking, although it was considered inferior to wood. Both natural gas and petroleum were used for evaporating water from salt brines. Petroleum was also used as a lubricant, lamp fuel, and medicine. Today, coal is used mainly for producing electricity to power lights, heating and cooling, cooking, and powering industries. Natural gas is mainly used for heating, cooking, and producing electricity. Petroleum is still used as a lubricant and has many modern applications. It is mostly used for transportation.

   c. Why are modern uses of fossil fuels different from uses in the past?

   Technology has allowed us to use fossil fuels in different ways over time. We no longer must burn fossil

fuels in our houses to create light, but can burn them in a power plant and use the electricity they generate to light our homes. This is also true with transportation. Only in modern times have humans learned to take the energy from fossil fuels and make it available for use in transportation.

d. What technology has gone into modern wind turbines and hydroelectric dams?

Modern wind turbines use advanced materials and engineering to create electricity. Modern hydroelectric dams are a huge engineering feat involving damming large reservoirs and building huge concrete structures to hold the water back.

e. Compared with fossil fuels, wind, and water, why would nuclear energy be considered a young energy?

Nuclear technology did not come about until the mid-1900s, compared with the other energies, which have been around for thousands of years.

f. It is predicted that in the future, humans will find new technologies to keep up with their demand for energy. What evidence from the timeline shows this?

Students may provide various answers using evidence from the timelines to back up their claims.

Wrap-Up/Student Feedback

Pick two different energy sources and compare and contrast how they have been used throughout human history and how their use affects the environment.

Supplemental Materials and References

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Energy Resources Throughout Human History

Scores

Over 18	Advanced
16 to 17	Proficient
14 to 15	Basic
Below 14	Needs Improvement

Questions

1. This fossil fuel was first used by the Chinese as a heat source to evaporate water from a salt-water brine.

2. The Romans used this source of energy for early industrial processes.

3. In the 1st century, the Chinese had already discovered petroleum. What did they first use it for?

4. What are three original uses of wind power?

5. The Industrial Revolution in the 1700s and 1800s was transformed by what fuel source?

6. What year did coal become the primary fuel for train locomotives?

7. In 1859, with the first commercial oil well drilled, what product began to replace other lamp fuels?

8. The first solar power was developed in France to do what?

9. In 1876, William Gryllis Adams demonstrated that sunlight could be converted directly into this power source.

10. Abraham Lincoln severely hampered this fuel industry by imposing a $2.08 per gallon tax on it to fund the American Civil War.

11. How long have windmills that generate electricity been around?

12. In 1892, the town of Boise, Idaho, installed a heating system that provided heated water to 200 homes and 40 downtown businesses that was powered by what earthly product?

13. Diesel engines are extremely versatile. In 1900, it was demonstrated they could run off what fuel with no modifications?

14. Those flex fuel vehicles you see on the road are nothing new. In fact, Henry Ford introduced the first flex fuel vehicle that can run on gasoline and ethanol quite a while ago. What year was it?

15. Seepage from tankers contaminating water and fisheries caused the federal government to establish the first laws controlling pollution from what fossil fuel source?

16. In 1950, with the popularity of the automobile, this fuel source became the most popular in the United States and has been ever since.

17. This is one of the newest sources of electricity generation. The first power plant using this fuel source was not built until 1951.

18. The first U.S. satellite in orbit utilized this power source.

19. In 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded this organization whose objective is to co-ordinate and unify petroleum policies among member countries.

20. In 1979, an accident at this nuclear power plant created widespread public opposition.

Energy Resources Throughout Human History

1. This fossil fuel was first used by the Chinese as a heat source to evaporate water from a salt-water brine. ***Natural gas***
2. The Romans used this source of energy for early industrial processes. ***Hydropower***
3. In the 1st century, the Chinese had already discovered petroleum. What did they first use it for? ***Lamp oil***
4. What are three original uses of wind power? ***Grinding grains, pumping water, crushing sugar cane***
5. The Industrial Revolution in the 1700s and 1800s was transformed by what fuel source? ***Coal***
6. What year did coal become the primary fuel for train locomotives? ***1830***
7. In 1859, with the first commercial oil well drilled what product began to replace other lamp fuels? ***Kerosene***
8. The first solar power was developed in France to do what? ***Create steam to power machines***
9. In 1876, William Gryllis Adams demonstrated that sunlight could be converted directly into this power source. ***Electricity***
10. Abraham Lincoln severely hampered this fuel industry by imposing a $2.08 per gallon tax on it to fund the American Civil War. ***Ethanol***
11. How long have windmills that generate electricity been around? ***Since 1888***
12. In 1892, the town of Boise, Idaho, installed a heating system that provided heated water to 200 homes and 40 downtown businesses that were powered by what earthly product? ***Geothermal energy***
13. Diesel engines are extremely versatile. In 1900, it was demonstrated they could run off of what fuel with no modifications? ***Vegetable oil***
14. Those flex fuel vehicles you see on the road are nothing new. In fact, Henry Ford introduced the first flex fuel vehicle that can run on gasoline and ethanol quite a while ago. What year was it? ***1908***

15. Seepage from tankers contaminating water and fisheries caused the federal government to establish the first laws controlling pollution from what fossil fuel source? ***Petroleum***

16. In 1950, with the popularity of the automobile, this fuel source became the most popular in the United States and has been ever since. ***Petroleum***

17. This is one of the newest sources of electricity generation. The first power plant using this fuel source was not built until 1951. ***Nuclear***

18. The first U.S. satellite in orbit utilized this power source. ***Solar***

19. In 1960, Iran, Iraq, Kuwait, Saudi Arabia, and Venezuela founded this organization whose objective is to co-ordinate and unify petroleum policies among member countries. ***OPEC***

20. In 1979, an accident at this nuclear power Plant created widespread public opposition. ***Three Mile Island***

Poster Instructions

Posters should include drawings, diagrams or pictures; an overview paragraph, and a historical timeline.

Drawings:

Create a diagram or picture with words and phrases showing how the energy resource is created. Use resources provided by your teacher to expand your description. For example, if your energy source is uranium, explain and draw how uranium is formed in the Earth. What type of rock is it? Where on Earth is uranium found? What kind of energy does it produce? Is the source renewable or nonrenewable and why? What kind of product does the energy source finally produce (i.e., electricity, heat, light, transportation, manufacturing, etc.)? Drawings should be detailed. This is your chance to teach the class something new about your energy source!

Overview paragraphs:

Write a paragraph that describes the environmental impacts of your energy source and how it has influenced human activity. Have there been any natural disasters or human-caused disasters from extracting, transpiring, or using this energy resource? How has the use of this energy source affected the environment? For example, creating a dam for hydroelectric power causes loss of habitat, whereas the burning of fossil fuels creates air pollution. Furthermore, how has this energy source influenced human life in positive ways? Think about how you use the energy resource and in what way. Expand on your ideas and include them your paragraph.

Historical timeline:

Create a timeline on another piece of butcher paper or adding machine paper highlighting the significant events (in time) for your fuel type based on the Historical Timeline: History of Alternative Energy and Fossile Fuels. Significant events can include when it was first discovered, how it is used, and how it has changed over time; technological discoveries improving its use; natural disasters; or significant laws or policies regulating its usage.

Category	4	3	2	1
Drawings	Drawings or diagrams clearly and accurately show advanced information on and understanding of how the energy source is formed. Drawings are accompanied by detailed words and phrases describing the formation of the energy source.	Drawings or diagrams proficiently show how the energy source is formed. Drawings are accompanied by basic words and phrases describing the formation of the energy source.	Minimal drawings or diagrams show a basic understanding of how the energy source is formed. Drawings are accompanied by basic words and phrases describing the formation of the energy source.	Drawings or diagrams are not fully present or do not show accurate facts relating to the formation of the energy source.
Overview Paragraph	At least 7 accurate facts relating to the energy source, its environmental impacts, and how it has influenced human activity are present. Paragraph is written and well organized with no grammatical errors.	At least 5 to 6 accurate facts relating to the energy source, its environmental impacts, and how it has influenced human activity are present. Paragraph has good organization and minimal grammatical errors.	Three to 4 accurate facts relating to the energy source, its environmental impacts, and how it has influenced human activity are present. Paragraph has some grammatical errors. Organization may be poor.	Fewer than 3 facts are present in the paragraph. Paragraph may have several grammatical and factual errors.
Historical Timeline	Timeline accurately highlights at least 10 significant events for the fuel types in history, and events are clearly labeled on then paper.	Timeline accurately highlights at least 7 to 9 significant events for the fuel types in history, and events are clearly labeled on the paper.	Timeline highlights fewer than 6 significant events for the fuel types in history, and events are clearly labeled on the paper. Some events may be inaccurate.	Timeline shows few events, and events are not clearly labeled or they lack accuracy.
Knowledge Gained	Student can accurately answer all questions related to facts in the poster.	Student can accurately answer most questions related to facts in the poster.	Student can accurately answer about 75% of questions related to facts in the poster.	Student appears to have insufficient knowledge about the fact in the poster.
Attractiveness and Overall Organization	The poster is exceptionally attractive and well organized in terms of design, layout, and neatness.	The poster is attractive and organized in terms of design, layout, and neatness.	The poster is somewhat attractive, although it may be a bit messy or lack organization.	The poster is distractingly messy or very poorly designed. It is not well organized.

Appendix 1.2.4 | Poster Worksheet

Answer the following questions in complete sentences at each poster.

1. What is the energy source?

2. How is the energy source formed?

3. When was it first discovered as an energy source and how is it used?

4. How has technology altered the way the resource is used?

5. What are the environmental impacts of this energy source?

6. How has this resource influenced human activity?

Lesson C

Electricity: An Energy Carrier

Quick Overview

Lesson Scope: Investigating the flow of electricity

Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Light bulbs, batteries, wires, switch, insulator, conductor bags

Materials Needed per Student

Light bulbs

Batteries

Wires

Switch

Bulb holder

Each student will need paper and instructions to record trials.

Guiding Question

What is our relationship to energy?

This lesson helps to develop a basic understanding of electricity and the flow of electrons through materials (current). It is also foundational to understanding the generation and transformation of electrical energy.

Key Understandings

- Direct current (DC) electricity flows in one direction.
- For a circuit to work, the current must be able to flow through a complete (connected) path.
- Some materials conduct the flow of electrons; others inhibit the flow of electrons.

Objectives

- Students will investigate the flow of electrical current to better understand electricity.

Vocabulary

- Alternating current (AC)
- Battery
- Closed circuit
- Conductor
- Direct current (DC)
- Electric circuit
- Insulator
- Resistance

Background Information for the Teacher

Note: *The background information below is meant for the teacher, not to pass along as a reading or vocabulary work for the students. Through the activities, students will encounter the vocabulary after they have had a chance to work "hands on" with the materials, which will allow them to tie the vocabulary to their experiences.*

Circuit Electricity Information

- An **electric circuit** is a complete path through which an electric current can flow. More specifically, a current is the movement of electricity (or electrons) through a material.

Sources of Electricity

- A **battery** is an electrochemical cell that converts chemical energy into electrical energy. In a battery, electricity produced by the chemical reaction flows in only one direction; this is called a **direct current** (**DC**). The major sources of electricity used to power our homes are electricity-producing power plants, which are connected to us via wires and produce the electricity that we use. This is different from the current in a battery because the current switches very rapidly from one direction to the other and is called an **alternating current** (**AC**).
- A functioning circuit has four basic components: a power source (which, as it pertains to this lesson, will be a battery), a job (to light a light bulb), a switch (the place where the wire is connected to the battery to complete the circuit), and a path (the copper wire). In a circuit with a battery, the flow of electricity is in one direction only, and it flows from the negative terminal of the battery to the positive terminal.
- There are standardized ways to draw circuits, which you can teach your students if you feel this is necessary, but it is more important that students understand the parts and the flow of the current more than that they know how to draw the symbols.
- **Resistance** is opposition to the flow of electric current. Wires have a small amount of resistance; "jobs" in a circuit create varying amounts of resistance. In this lesson, the jobs will be the light produced by the light bulbs. If the electric current can go in two directions and one of the directions has much less resistance than the other, the current will use the path of least resistance. A circuit with no job, or bulb to light (only a wire, a switch, or both), will have little resistance and may cause a closed circuit. You need to be able to recognize a **closed circuit** because often students set them up accidentally. A closed circuit is an electrical connection in which the current flows through the wire, and without a job (turning of a motor, lighting of a bulb), the wire gets very hot, hot enough to

burn. A **conductor** is a material through which electricity moves freely. An **insulator** is a material through which an electric current cannot move.

Common Student Misconceptions/ Student Challenges

Students often do not understand the circular path of the flow of electricity through a circuit, even if they have had experience with this in earlier grades.

Next Generation Science Standards

Performance Standard.4-PS3-4:

Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.

Disciplinary Core Idea

- Energy can also be transferred from place to place by electrical currents, which can then be used locally to produce motion, sound, heat, or light. The currents may have been produced to begin with by transforming the energy of motion into electrical energy.
- The expression "produce energy" typically refers to the conversion of stored energy into a desired form for practical use.

Science and Engineering Practices

- Apply scientific ideas to solve design problems.

Crosscutting Concepts

- Energy can be transferred in various ways and between objects.
- Engineers improve existing technologies or develop new ones.
- Most scientists and engineers work in teams.
- Science affects everyday life.

Materials

Part I

- One bulb, one battery, and one wire for every student (or group of students). The wire should have about an inch of the insulation removed from both ends.
- Students also need paper and instructions for recording their trials.

Part II

- Small bulb holder (one for each group of students)
- Insulator and conductor bags. These can be filled with easily found objects, such as erasers, small change, balloons, plastic items, and office supplies. Make sure each bag contains a variety of both metal and nonmetal objects in each one. Fifteen to 20 items per bag should be sufficient. Bags can be identical or different, depending on whether you want students to directly compare experiences.

Part III

- All items from above
- Two batteries in battery holders along with three light bulbs in holders and four to five insulated wires (for each student group)

Part IV

- All items from above plus a switch and some extra materials (bulbs, batteries, wires)

Setup

Print out the assessment probe developed by Page Keeley: "Batteries, Bulbs, and Wires" (Appendix 1.3.1).

Bulbs, batteries, and wires can be placed in plastic bags and handed out fairly quickly. The rest of the materials should be prepped and ready to go. Initially, it will take an hour or more to get everything organized, but after that, very little time will be needed for setup.

The four ways to light the bulb are with the

- Bulb upright on the positive end
- Bulb sideways on the positive end
- Bulb upright on the negative end
- Bulb sideways on the negative end

Students may question whether there are actually four *different* ways to light the bulb. In each case, the current is traveling through the circuit a slightly different way.

For example, if the bulb is on the positive end and held upright, the current travels out the negative end of the battery, into the side of the bulb, through the filament, and back into the positive end of the battery. If the bulb is instead held sideways, the current first enters the bulb through the bottom and then out the side. When the bulb is placed at the bottom of the battery, the current comes out and immediately

enters the bulb before going through the wire to the positive end. Again, although these changes are slight and may seem semantic, they really help students visualize the flow of current and the internal workings of the bulb.

Some students will have a lot of difficulty with this activity. They will want you to show them how to do it, or they may claim that their bulbs are burned out or batteries are dead. This can be dealt with by testing their equipment (without revealing how to light the bulb) and then asking the students to show what they have tried. Most students will realize that they need to complete the circuit (or make a circle), but do not understand that the electricity must also flow through the filament inside the bulb, or that the filament is connected to the side and the bottom of the base of the bulb. Instructing them to look carefully at the bulb can be helpful.

Two rules:

- If it gets HOT, let go!
- Do not do whatever you were doing again*, and NO bragging about burns is allowed!

*__Note:__ *Some students will create closed circuits by trying to connect the wire to both ends of the battery. When you tell them not to do this again, they are confused because, often, a bulb will light because they have actually touched the side of the base. Sometimes they need to be guided to understanding this or they will continue to burn themselves.*

Length of Lesson

This lesson will take approximately two 45-minute class periods or one 90-minute block. A good breaking point is between Parts II and III. Be sure to spend some time reviewing Parts I and II on the second day.

Suggested Procedure

Part I: Completing a Circuit

1. Before handing out the equipment, have students fill in the assessment probe developed by Page Keeley: "Batteries, Bulbs, and Wires" (included as Appendix 1.3.1). This can be downloaded from the National Science Teachers Association (NSTA) website at *http://www.nsta.org/pdfs/UncoveringIdeasCh7.pdf*.
    - Explain that it is important for them to explain their thinking and draw a picture of their ideas. Put these aside for later use. (Estimated time: 5 minutes)
2. Pass out one wire, one bulb, a battery (Figure 1), and a clean sheet of paper to each student.

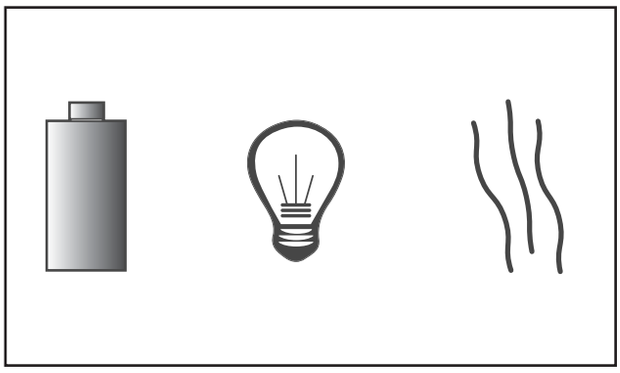


Figure 1. Battery, bulb, and wires.

a. Explain that their job is to light the bulb using these materials.

b. Tell students that there are four ways to light the bulb. This helps students who already have an idea of how to light a bulb think beyond their initial ideas.

3. For the first 10 minutes, ask students to work alone—they are not to share materials or ideas. As they work, encourage them to draw their successes and failures. You can have them divide the sheet in half and put their failures on one side and successes on the other. Often, they need to be reminded to do this.

   a. When students are having difficulty, help them by asking them guiding questions.

   b. Usually, once they have figured out two ways of lighting the bulb, they come up with the third and fourth ways (Figure 2).

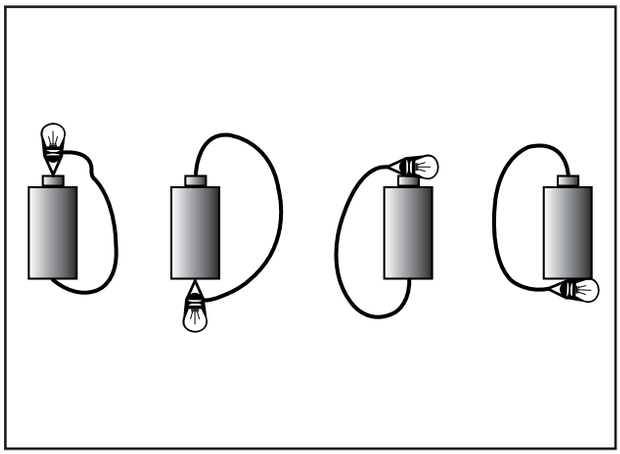


Figure 2. Four ways of lighting the bulb.

4. At this point, it is time to share results. Have four different students draw on the board one of the four ways they found to light the bulb. Ask students to add in arrows showing the flow of current, and begin to label the parts. It is appropriate at this point to use the vocabulary and have them start to put descriptive words into their drawings. Make sure everyone has a clear understanding that to have a complete circuit, electricity needs to be able to make a complete path and flow through the system—which is composed of an electrical source, a job, a path, and a switch (or the place the circuit is closed).

   You can also ask students to draw some of the "failures," and the whole group can examine them and discuss reasons these circuits did not work.

5. Now return to the assessment probe conducted at the start of the lesson. Have students reflect on their thinking and, using the back of the probe, have them write a reflective explanation for how their thinking has changed. (This formative assessment can also wait until after the next section of the lesson if you have time.)

   You are now ready for students to apply their knowledge in Part II, Insulators and Conductors.

Part II: Insulators and Conductors

1. Have students form groups of two to three. They should bring their materials from Part I with them to their groups.

2. Pass out a bulb holder to each group (Figure 3). Demonstrate how to attach the wires to each end and how to insert the bulb so that both the side and bottom are connected. (You could use this opportunity to review or reinforce the idea of the inner workings of the bulb discovered in Part I.) Next, pass out the bags with the materials and have students draw a T-chart on the back of their Completing a Circuit papers. Label the two sides with appropriate descriptors, such as "Bulb lights/Bulb does not light" or "Yes/No". Leave space for the words *conductors* and *insulators*, but do not have them label them as such at this point.

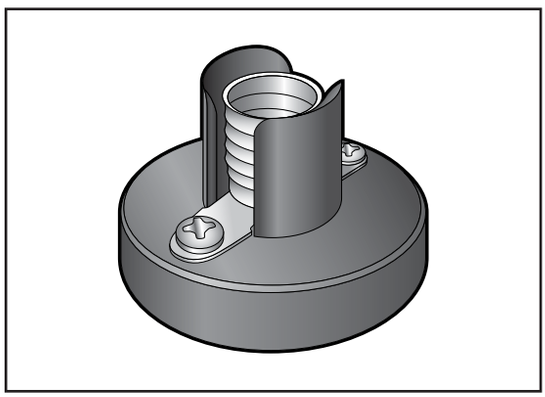


Figure 3. Bulb holder.

3. Instruct the students that they are going to use their knowledge about circuits to identify which materials can be used to complete the circuit and which ones will not work. Ask them to test each of the items and fill in their T-charts. Have them pay careful attention to the similarities among the items that fall into each category.

4. For groups that finish earlier than the rest, have them make predictions about several other objects around the room—give a reason for their prediction, then test them, and write up their results to share.

5. Lead a discussion in which every group has a chance to share what they have learned from this activity. This is the point at which the vocabulary words *insulator* and *conductor* will come up and should be placed in appropriate spots on the T-charts.

6. Other topics of discussion at this point could include those related to safety (impure water as a conductor, insulators around copper wires, differences between AC and DC and amounts of current in household outlets), electricity production, etc.

Part III: Tracing the Flow of Electrons

1. Arrange students in small groups (two to three). Each group will need two batteries in battery holders along with three light bulbs in holders and four to five insulated wires.

2. Instruct students they will need to use their knowledge of electrical circuits to light all the light bulbs in a series circuit. Tell students they can have only one wire connected to the positive terminal of the battery and one wire connected to the negative terminal of the battery. Tell students they can connect only one wire to each terminal on the light bulb and cannot connect wires to each other.

3. Before students begin, they should draw a sketch of how they plan to light their bulbs on their energy journal sheet or a blank sheet of paper.

4. Once students have their supplies, they should try to build their circuit according to their sketch. If their circuits do not work, students should problem solve their design until they get the circuits to work.

5. Have students observe the intensity of the light bulbs. On their drawings, they should note which bulbs are brightest. From their observations, students should hypothesize which direction the current is flowing through the circuits.

6. Create a drawing of a serial circuit for the students to see, like the one in Diagram A. Explain to students that electrons have a negative charge, so current flows from the negative terminal of the battery to the positive terminal. Have students discuss what happens to the electricity as it moves through the light bulb. After the student discussion, ensure they understand that part of the current is converted to light and heat within the light bulb.

7. Explain to students how the electrons flow through the circuit from negative to positive. Ensure students understand that some of the energy is converted to light and heat at each light bulb, so the light bulb closest to the negative terminal will get the most energy, whereas the middle one will get the second most, and the one closest to the positive terminal will get the least.

Part IV: Parallel Circuit

1. Tell students their new goal is to get each light bulb to be the same brightness, using the same materials from Part III.

2. Instruct students they will need to use their knowledge of electric circuits to light all the light bulbs in a series circuit. Tell students they can have only one wire connected to the positive terminal of the battery and one wire connected to the negative terminal of the battery. Tell students they can connect only one wire to each terminal on the light bulb and cannot connect wires to each other.

   Remind the students that once electricity goes through a light bulb, part of the energy is converted to light and heat; therefore, any bulbs in a circuit behind another will not be as bright.

3. Have students create a drawing of their plan to light all the bulbs equally.
4. Students should put their bulbs into a parallel circuit. There are a few different options for a parallel circuit. Each bulb should have a direct path to the negative and positive terminals on the battery.
5. End the session with a whole-group discussion about the flow of electrical current. Have students demonstrate how the current is divided equally into each bulb.

Enrichment Activity

Now that students have created a parallel circuit, introduce them to a switch. Tell them their goal is to wire the switch so that it turns a light bulb on when it is opened.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

Using the strategy "I used to think _____, but now I know _____." Have students write a reflection (including drawings to back up their statements) about a specific part (or one for each part) of the lesson. Use this to gauge their understanding. The enrichment activity could be used as an application.

Supplemental Materials and References

Keeley, P., Eberle, F., & Dorsey, C. (2008). *Uncovering student ideas in science: Another 25 formative assessment probes* (Vol. 3, pp. 57–62). Arlington, VA: NSTA Press.

Assessment Probe Developed by Page Keeley: Batteries, Bulbs, and Wires

Kirsten has a battery and a small bulb. She wonders how many strips of wire she will need to connect the battery and the bulb so that the bulb will light. What is the smallest number of wire strips Kirsten needs to make the bulb light up?

a. One strip of wire

b. Two strips of wire

c. Three strips of wire

d. Four strips of wire

Explain your thinking about how to light the bulb. Draw a picture to support your explanation.

Teacher Notes

Purpose

The purpose of this assessment probe is to elicit students' ideas about complete circuits involving light bulbs. The probe reveals whether students recognize the pathway of electricity in a complete circuit, including its path through a light bulb, to light a bulb using only one strip of wire and a battery.

Explanation

The best answer is A—one strip of wire.

If you closely examine a flashlight bulb, you will see two small wires sticking up in the bulb that are connected by a very fine wire called a *filament*. The two wires on either side of the filament extend downward into the base of the bulb, where you cannot see them through the metal casing surrounding the base of the bulb. One of these wires goes down to the very bottom of the base (the pointed end). The other wire is connected to the side of the metal base (sometimes the side is ridged so that it can screw into a socket). Knowing where these wires end up on the base of the bulb (the tip and the side) is necessary in order to use one wire to make a circuit that lights a bulb. The battery, wire, and bulb need to be connected in such a way as to form a complete circuit. To do this, hold the end of the wire against the negative terminal (the bottom of the battery or smooth end). The other end of the wire should touch or wrap around the side of the metal casing that forms the base of the light bulb. With the wire wrapped around the metal side of the bulb and the other end touching the bottom of the battery, touch the tip of the base of the light bulb to the positive terminal (bumpy end), and the bulb will light. The light bulb lights with just one wire because the electricity flows out of the negative terminal (bottom of the battery), through the wire, into the wire that is attached to the side of the metal casing on the base of the bulb, up through that wire inside the bulb, across the filament, and down the other wire inside the bulb, where it is attached to the point on the base of the bulb that touches the positive terminal (the bump) of the battery, completing a full circuit.

Be aware, however, that students can choose the best answer, "one wire," and still have an incorrect configuration of a circuit. For example, some students may touch one end of the wire to the battery and the other to the bulb, thinking the energy from the battery will flow through the wire to the bulb. Students who choose two wires as their answer may understand that a complete circuit is needed, but may not understand the internal architecture of a light bulb.

Administering the Probe

Make sure students know what you mean by a "strip" of wire. Show them a battery, a bulb, and a coil of wire. Cut a strip from the coil that is long enough to make the circuit so they know what is meant by a strip of wire. Or show them a handful of cut strips, but do not show just one, because it might cue them to the answer. Note that if students choose more than one wire, it is not necessarily incorrect in terms of making a complete circuit. However, the probe is not asking how many wires are needed to make a complete circuit, but rather what the minimum number of wires is. It is important to emphasize drawing as a way of supporting the students' explanations in this probe.

Lesson D

Electricity Generation

Quick Overview

Lesson Scope: Designing an electric generator

Length: 2 to 3 ~45-minute class periods

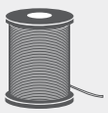
Grade Level: 5–8

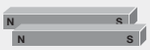
Materials: Multimeter, sample generator, magnetic copper wire, bar magnets, 3-inch steel nails, 1-inch to 3-inch diameter cardboard tube, paper clips, hot glue gun, sandpaper, duct tape

Materials Needed per Student

Multimeter

Sample generator


Copper wire



Bar magnets

3-inch steel nails

Cardboard tubing

Each student will need paper and instructions to record trials.

Guiding Question

What is our relationship to energy?

To understand the connection between fossil fuels (chemical energy) and electricity (electromagnetic energy), it is important to understand the basics of an electrical generator and how it works.

Key Understandings

- An electrical current is produced when a conductive wire is moved through a magnetic field.
- A generator turns mechanical energy into electrical energy.

Objectives

- Students will be able to design a sample electric generator.

Vocabulary

- Alternating current (AC)
- Direct current (DC)
- $E = BLv$ (equation describing the strength of the electrical field)
- Electrons
- Magnetic field
- Velocity

Background Information for the Teacher

A generator is simply a device that moves a magnet near a wire to create a steady flow of **electrons**. The action that forces this movement varies greatly, ranging from hand cranks and steam engines to nuclear fission, but the principle remains the same.

The physics of an electrical generator can be quite complex. However, a few basic concepts are all that are necessary for students to understand how a generator turns mechanical energy into electricity. The first concept is that of a **magnetic field**, as shown in Figure 1. When a conductive wire (copper, aluminum, steel, gold, etc.) is moved perpendicularly to this magnetic field, the electrons in the wire are deflected, causing an electrical current. The second concept involves how the electrons move. If the wire cuts the magnetic field from the top to the bottom, the electrons are deflected away from the field. If the wire cuts the magnetic field from the bottom to the top, the electrons move toward it. This is called an **alternating current** (**AC**) because it moves toward and away from the field. If the current moves in one direction, it is called a **direct current** (**DC**). It is easiest for students to make an AC generator because a DC generator requires a commutator, a device that prevents one half of the electrons from being deflected, allowing the current to flow in only one direction (see Figure 1).

Finally, teachers need to be familiar with the variables affecting the strength of the electrical current. These variables are (a) the strength of the magnetic field, (b) the distance

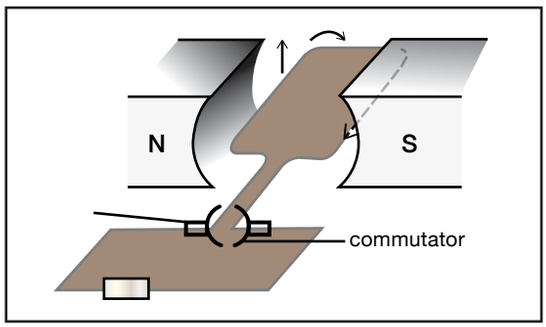


Figure 1. Simple DC generator showing commutator.

the conductive wire cuts through the magnetic field, and (c) the **velocity** at which the conductive wire cuts through the magnetic field. For students to create stronger generators, they will need to (a) have stronger magnets, (b) increase the distance the wire travels through the magnetic field, or (c) increase the velocity with which the wire travels through the magnetic field. The equation explaining this is $\boldsymbol{E = BLv}$. E is the strength of the electric field, B is the strength of the magnetic field, L is the length of the conductor moving through the magnetic field, and v is the velocity of the conductor through the magnetic field.

Watch the building of a simple electrical generator at https://www.youtube.com/watch?v=k7Sz8oT8ou0.

For more information about generators, please refer to http://science.howstuffworks.com/electricity3.htm.

Common Student Misconceptions/ Student Challenges

The physics of an electrical generator can be quite complex for students to understand. To help reduce the complexity, provide students with a chance to try and fail by manipulating just one variable at a time, such as the length of the magnetic field, strength of the magnetic field, or velocity of the conductive wire moving through the magnetic field.

Next Generation Science Standards

Performance Standard.4-PS3-4:

Apply scientific ideas to design, test, and refine a device that converts energy from one form to another.

Disciplinary Core Idea

- Energy can also be transferred from place to place by electrical currents, which can then be used locally to produce motion, sound, heat, or light. The currents may have been produced to begin with by transforming the energy of motion into electrical energy.
- The expression "produce energy" typically refers to the conversion of stored energy into a desired form for practical use.

Science and Engineering Practices

- Apply scientific ideas to solve design problems.

Crosscutting Concepts

- Energy can be transferred in various ways and between objects.
- Engineers improve existing technologies or develop new ones.
- Most scientists and engineers work in teams.
- Science affects everyday life.

Performance Standard.MS-PS2-5:

Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects, exerting forces on each other, even though the objects are not in contact.

Disciplinary Core Idea

- Forces that act at a distance (electrical, magnetic, and gravitational) can be explained by fields that extend through space and can be mapped by their effect on a test object (a charged object, a magnet, or a ball, respectively).

Science and Engineering Practices

- Conduct an investigation and evaluate the experimental design to produce data to serve as the basis for evidence that can meet the goals of the investigation.

Crosscutting Concepts

- Cause and effect may be used to predict phenomena in natural or designed systems.

Materials

- Multimeter (one for teacher demonstration)
- Computer with Internet and projector, PhET generator simulation, *http://phet.colorado.edu/en/simulation/generator*
- Prebuilt sample generator (one for teacher demonstration)
- Bar magnets or neodymium magnets
- 3-inch steel nails
- Spool of copper wire
- 1.75-inch to 2.5-inch-diameter cardboard tube or PVC pipe cut into 5-inch lengths
- Electrical multimeter (several, if possible)
- Optional: Power drill and drill bit large enough for the nail hole

Setup

- Setup varies with the materials used.
- Allow 15 minutes (before class) to set up hot glue guns and sort materials. Precut the PVC pipe or cardboard tube the day before.
- Students can make a predesigned generator if time or material constraints do not allow for individual design.
- Groups of two to three allow all students to contribute while minimizing materials needed.

Length of Lesson

This lesson will take approximately two to three 45-minute class periods. Allow 15 minutes for the introductions, 30 minutes for the research and design, 45 to 90 minutes for building and testing, and 10 minutes for a final assessment. Additional class periods can be used to revise generator design.

Suggested Procedure

1. Demonstrate to the students how the sample generator works by hooking the generator up to the multimeter and turning the magnets to show it produces an alternating current. Students will be able to see the multimeter indicating a small amount of electricty being generated with each turn of the nail/magnets.

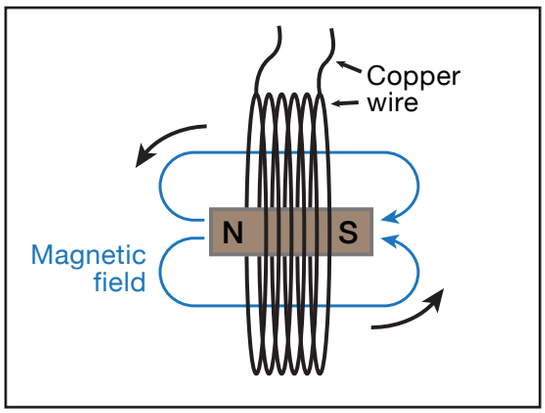


Figure 2. Schematic of a simple AC generator showing the principle concept. The magnets rotate pole over pole inside the copper wire coil, the magnetic field must cross perpendicularly to the conductive wire.

2. Pass the generator around and have students make observations of and predictions for how they think the generator works. Lead a class discussion in which they put forth their ideas. Prior to the following explanation, have students communicate verbally or in writing in their science journals what they feel they know or do not know about electricity generation.

3. Show students the diagram of a magnetic field and explain to them that a conductive wire has to cross the lines of the magnetic field perpendicularly for a current to be generated (Figure 2).

4. Load the simulation from *http://phet.colorado.edu/en/simulation/generator*. First, under the Bar Magnet tab, demonstrate how the magnetic field changes as the magnet is moved around. You may need to demonstrate this several times. Second, under the Pickup Coil tab, demonstrate how the magnetic field creates an electrical current. During this simulation, be sure to change some of the variables, such as the strength of the magnet, the velocity with which it moves, and the number of wire coils. Finally, open the Generator tab and experiment with the speed of the magnet, the number of loops, and the size of the loops. Optional: If students have access to a computer and there is time, have them manipulate these simulations on their own.

5. Next, it is time to have students design their own generators. They should begin by creating a drawing or schematic diagram showing how they will build it and how they think their generator will work. These drawings should include lines representing the magnetic field and how the conductors will move through the magnetic field.

   Students can reference page 4 of *How Electricity Works* by Marshall Brain, William Harris, and Robert Lamb (*https://science.howstuffworks.com/electricity.htm*).

6. When their drawings are complete, have them gather materials and build their generators.

7. As students finish constructing their generators, test them by connecting each end of the copper wire to the multimeter. It is important that students scrape off the enamel coating on the ends of the copper wire so that it makes an adequate connection to the multimeter. Set the multimeter to ~200 mV, and ensure the red cable is plugged into VΩmA and the black

cable is plugged into the COM. Connect the leads from the black and red wires to the terminals on the generator or wrap the wire around them to secure a good connection (see Figure 3).

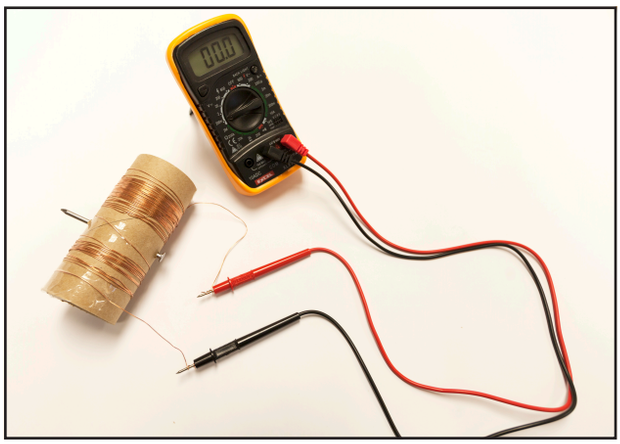


Figure 3. Testing student generator.

After all students have built and tested their generators, remind them about the simulation and have them help identify the variables that will change the amount of electricity their generator produces. (*The variables are the strength of the magnetic field, the amount of copper wire used, and the distances between the interacting objects.*) Help them realize the following: adding more magnets or stronger magnets can change the strength of the magnetic field, and the area of the wire passing through the magnetic field can be changed by using more wire, or wider magnetic field. Turning the magnets faster will increase the speed at which electricity is generated.

Note: *It will be important to discuss at this point that the drill is representing mechanical energy, and in another system, that would be provided by something else, such as moving steam (coal plant), the wind (wind turbine), or moving water (hydropower).*

8. Optional: Show students the equation $E = BLv$. Explain that E is the strength of the electric field, B is the strength of the magnetic field, L is the length of the conductor moving through the magnetic field, and v is the velocity of the conductor through the magnetic field. Tell students they will be competing to create the generator with the highest E or electrical field, as measured with a multimeter. Remind students that to get a higher E, they will need to increase B, L, or v.

9. Now that students have an understanding of the variables necessary to produce electricity, have them redesign their initial generator to be stronger.

10. If time permits, students can make multiple changes to their generators until they find the best design.

11. Have students dismantle the materials for future use.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

Note: *If the Optional portion of the lesson was done, they should address the equation E = BLv and show how they will manipulate each of these variables in their diagrams. In addition, have students explain how mechanical energy is turned into electrical energy by using generators.*

Have students keep track of their initial designs and how their designs changed as they experimented.

Students should describe the variables in their designs and the results of changing these variables. For a final assessment, have students create a diagram showing how they would make the ultimate generator using the knowledge they have gained in their experiments.

Supplemental Materials and References

Marshall, B., William, H., & Lamb, R. (2004). *How electricity works*. Retrieved from https://science.howstuffworks.com/electricity.htm

University of Colorado. (2013). *Generator simulation: PhET interactive simulation*. Retrieved from http://phet.colorado.edu/en/simulation/generator

William, B. (2008). *Simple generator: AC electric generator for science fair.* Retrieved from https://www.youtube.com/watch?v=k7Sz8oT8ou0

Lesson E

Plugged in to CO_2

Quick Overview

Lesson Scope: Connection between electricity generation and CO_2 emissions

Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Kill A Watt® meters, small appliances, surge protectors, Post-it® Note pads, calulators

Materials Needed per Student

Kill A Watt meter

Small appliances

Surge protectors

Post-it Note pads

Calculators

Each student will need paper and instructions to record trials.

This lesson was adapted from the "Plugged in to CO_2" lesson found at Windows to the Universe, *http://www.windows2universe.org/teacher_resources/teach_pluggedCO2.html*.

Guiding Question

What is our relationship to energy?

Students explore their carbon footprint by investigating how much electricity different appliances use and how much **carbon dioxide (CO_2)** is produced in the generation of that electricity.

Key Understanding

- Electricity generation creates CO_2 **emissions**.
- The electricity we use on a daily basis has environmental impacts.

Objectives

- Students will calculate the amount of electricity needed to run various home appliances. Students will use the amount of electricity for home appliances to calculate the amount of CO_2 produced from each appliance.

Vocabulary

- Ampere (amp)
- Carbon dioxide (CO_2)
- Emissions
- Greenhouse effect
- Greenhouse gases
- Kilowatt
- Phantom load
- Voltage (volt)
- Wattage (watt)
- Watt-hour

Background Information for the Teacher

Anything with a plug (appliances, electronics, lamps) draws power through an electrical outlet. Many of these items are left on all day and night. Other items draw some electricity even when they are turned off (called a **phantom load**).

In Part 1 of this activity, a device called a Kill A Watt meter is used to measure the power consumption of appliances and other devices. The meter gives instantaneous measures of volts (voltage) and amps (a measure of current) as well as the watts of power being used (**Note:** watts = volts x amps). The meter also gives a measure of the amount of power used over time in **kilowatt**-hours (kWh) if you leave the device in place for a length of time (from just a couple minutes to weeks or even months). The kWH is a derived unit of energy equal to 3.6 megajoules. If the energy is being transmitted or used at a constant rate (power) over a period of time, the total energy in kWh is the product of the power in kilowatts and the time in hours.

See *Consumer Reports Tests Energy Monitors* (March 2009) for more information. Kill A Watt meters sell for $20 to $25. They are also available at some public libraries.

Notes regarding electrical vocabulary:

Voltage (volt): the electrical potential between two points. For example, a 1.5-volt battery (typical AA battery) has 1.5 volts of electrical potential between the negative and positive battery terminals. A garden hose is often used as an analogue to electricity. In this example, *voltage* would be represented by the pressure of water flowing through the garden hose.

Ampere (amp): the unit of electrical current passing through a point. In the garden hose analogue, an *amp* is represented as the amount of water flowing through the garden hose. The amount of water flowing through the hose can be changed by using a wider or narrower hose. In the same way, amps can be changed by the conductivity of the material the electricity is flowing through. A material with low resistance (high conductivity) will allow more amps with the same voltage when

compared with materials with higher resistance (lower conductivity).

Wattage (watt): a unit of power, measured as the amount of electricity converted to work. In a circuit, when an electrical charge is converted into something else like light, heat, or a magnetic field, it has done work. This work is measured in watts. When measured over time, the watts are multiplied by a time unit. One watt produced over an hour would be one **watt-hour**. Watts are directly related to voltage and amperes in the equation watts = volts × amps. In the garden hose analogue, the *watt* is represented by the water turning a sprinkler. If the pressure or volume goes up, the sprinkler turns faster. In electricity, if the volts or amps go up, so do the watts.

Electricity is generated in different ways in different regions depending, in part, on local natural resources. The pie charts on the map used in Part 2 of this activity show how the mix of electricity sources varies with region of the United States. On average, within the United States, about half of electricity is generated using coal. Natural gas and nuclear power account for about 20% each. Nearly 10% of electricity is from renewable sources, with the bulk of that produced at hydroelectric dams. Petroleum makes up a very small fraction of U.S. electricity. Coal, natural gas, and petroleum are fossil fuels. Burning these fuels generates electricity but also emits CO_2 and other greenhouse gases to the atmosphere, as well as other air pollutants such as sulfur dioxide and volatile organic compounds (VOCs). Electricity generation produces about 40% of the greenhouses gases emitted in the United States.

Greenhouse gas emissions are important to understand because of their environmental impacts. CO_2 and other greenhouse gases trap infrared energy emitted from the Earth's surface and warm the atmosphere. This is a natural process that is being amplified by the additional **greenhouse gases** added to the atmosphere mainly through the burning of fossil fuels. These additional gases cause the **greenhouse effect** to grow stronger, causing global warming. Prior to the Industrial Revolution, CO_2 in our atmosphere was at approximately 280 parts per million (ppm). As of 2009, CO_2 in the atmosphere was present at a rate of approximately 386 ppm and growing steadily upward. In 2013, atmospheric CO_2 levels reached 400 ppm. This increase in atmospheric CO_2 is linked to increased global temperatures, which impact global climate.

Rarely do individuals link the amount of electricity they use on a daily basis, or the accumulation of this use, to greenhouse gas emissions. In addition, most people are unaware of the fact that 40% of electricity in the United States continues to be generated using coal.

In this excercise, it is important that students understand the relationship between using electricity and producing CO_2. Teachers may need some background on electrical units in which electricity is bought and used so they can help students understand quantitatively

how much electricity they use, and how much CO_2 is being produced from that electricity.

Common Student Misconceptions/ Student Challenges

Students may not understand the relationship between electricity and CO_2 being released into the environment. In addition, students rarely understand that the amount of electricity they use each day has a direct impact on the amount of CO_2 released into the environment.

The units used to measure electricity often create a barrier for having conversations related to energy. It is important to review the units prior to moving forward with this activity. These units are a watt, kilowatt, watt-hour, volt, and amp. Please refer to the background information.

Next Generation Science Standards

Performance Standard.MS-ESS3-4:

Construct an argument supported by evidence for how increases in human population and per-capita consumption of natural resources impact Earth's systems.

Disciplinary Core Idea

- Typically, as human population and per-capita consumption of natural resources increase, so do the negative impacts on Earth, unless the activities and technologies involved are engineered otherwise.

Science and Engineering Practices

- Construct an oral and written argument supported by empirical evidence and scientific reasoning to support or refute an explanation or a model for a phenomenon or a solution to a problem.

Crosscutting Concepts

- Cause and effect relationships may be used to predict phenomena in natural or designed systems.
- All human activity draws on natural resources and has both short- and long-term consequences, positive as well as negative, for the health of people and the natural environment.

Materials

- 5 Kill A Watt Meters (if 5 are not available, have groups take turns using the one provided in this kit)
- Assortment of appliances and electronics (a room that has a number of appliances would be good to use—a home economics room or a teacher's lounge might be good places to use if they are available). Small appliances could also be brought into the classroom if needed (things students might use at home: hair dryer, hair straightener, electric toothbrush, phone charger, desk lamp, etc.).

- Power strips/surge protectors (one for each group of students)
- Different Regions of the Country Use Different Fuel Mixes to Generate Electricity graph, Appendix 2.5.1
- Worksheet 1, Appendix 2.5.2
- Worksheet 2, Appendix 2.5.3
- Example Calculations, Appendix 2.5.4
- Trends in Global Carbon Emissions, Appendix 2.5.5
- Paper and writing instrument
- Post-it Note pads
- Calculators

Setup

Plan to spend approximately 15 minutes setting up by plugging in appliances, the power strip, and the kilowatt meter prior to the lesson.

Make copies for each student of

- Worksheet 1, Appendix 2.5.2
- Worksheet 2, Appendix 2.5.3
- Example Calculations, Appendix 2.5.4

Length of Lesson

This lesson will take approximately two 45-minute class periods (longer if groups are sharing only one kilowatt meter)

Introduction

1. Ask students to take out paper and something to write with. Ask the students to record the number of units (per hour) that they think are used by each of the following appliances (this will be their best guess). Remind students that a watt is a unit of power and that they should assume that a 100-watt incandescent light bulb uses 100 units of energy per hour. This means that their guesses should be recorded in watt-hours.

    - A compact-fluorescent light bulb of the same brightness as a 100-watt incandescent light bulb?

      (27 watt-hours)

    - A desk computer?

      (140 watt-hours)

    - A laptop computer?

      (48 watt-hours)

    - A stereo system or iPod speakers?

      (128 watt-hours)

    - A dishwasher?

      (1,800 watt-hours)

    - An electric clothes dryer?

      (2,400 watt-hours)

- A 1,500-watt portable heater?
  (1,500 watt-hours)
- A 500 Btu portable air conditioner?
  (515 watt-hours)

2. Do not provide students with correct answers; instead, ask them to meet with their partner and compare answers. Have student pairs come to a consensus. Once the groups have committed to their answers, have them record these on sticky notes and post them in a common area or board in the classroom. Designate a spot for each appliance. Once all groups have posted, reveal the answers to the classroom.

3. Next, provide students with this brief example of energy units: A 100-watt incandescent light uses 100 watts of electricity when it is on. If that light is on for an hour, it has used 100 watt-hours. If you leave the light all day, from 6:00 am until midnight (18 hours), it will have used (100 x 18) 1,800 watt-hours of electricity. To make the large numbers easier, kilowatts, or units of 1,000 watts, are used to measure electricity usage. In this case, the 1,800 watts is converted to 1.8 kilowatt-hours of electricity.

Activity

Begin by providing some background for this activity:

- Not all electronics use the same amount of electricity. Some use more watts to operate than others.
- In **Part 1** of this activity, you will investigate how much electricity various appliances use, both on and off.
- In **Part 2** of this activity, you will calculate the amount of the greenhouse gas CO_2 that is produced when these appliances are plugged in and using electricity.

Part 1

1. The appliances, power strips, and Kill A Watt meter should already be set up within the room.

2. Introduce students to the Kill A Watt meter. The meter allows us to assess how much electricity an appliance is consuming. It is plugged into the outlet first, then the appliance plugs into it. It measures the flow of electricity coming from the outlet to the appliance and displays the amount of work the appliance is doing in the unit watts. Divide the students into groups of 3 or 4 per Kill A Watt meter. If only one Kill A Watt meter is available, student groups must take turns using the meter.

3. Have the groups pick two appliances each and predict which one of the pair will use the most watts.

4. Have the groups measure both appliances with the Kill A Watt meter and share results with the class. Each student should record the data in the first and second columns of the table on Worksheet 1 (Appendix 2.5.2).

(Note that the watts measured on the meter may not completely stabilize at one value. Help the student making the measurement choose a reasonable number—one way to do this is to check it every 15 seconds for 2 minutes, record the number, then average the numbers.) Students may use calculators.

5. Continue to have student groups measure the watts for each appliance and record their data in the table on Worksheet 1 (Appendix 2.5.2) in the first and second columns, until all of the appliances have been tested and recorded. There is space on the worksheet for nine appliances, but you may decide to test more or fewer with your students. Students may use calculators.

6. As a class, estimate how long each appliance is used each day. Fill in that information in Worksheet 1 in the third column.

7. Calculate how many kWh per day and per year each appliance uses. Students may use calculators. (If time is limited, consider having pairs of students work on this calculation for a single appliance and then sharing data.)

    a. Multiply watts by hours per day to get watt-hours (fourth column).

    b. Divide watt-hours by 1,000 to get kWh per day (fifth column).

    c. Multiply kWh per day times 365 to get kWh per year (sixth column).

8. A supplement to this activity is to check the Kill A Watt meters out to students to take home. At home, students can continue to measure different electrical appliances and bring this information back to the class.

Part 2

Background information: *Make sure students understand that CO_2 is a greenhouse gas that is emitted into the atmosphere when fossil fuels are burned. The release of CO_2 and other greenhouse gases into the atmosphere is of concern due to connection with potential climate changes. In the United States, the burning of fossil fuels, such as coal and natural gas, produces most of our electricity.*

1. Keep students in their previous groups.

2. Tell students that this part of the activity will be about figuring the amount of CO_2 produced by each appliance.

3. Introduce students to the pie charts on the Different Regions of the Country Use Different Fuel Mixes to Generate Electricity in Appendix 2.5.1. These data, from the Edison Electric Institute, show the mix of energy sources found in different parts of the country. Find your region's pie chart and discuss as a class how your energy sources are similar to, or different from, those in other parts of the country.

4. Have students look over the table at the end of Worksheet 2. Explain that each fuel source releases different amounts of CO_2 to produce 1 kWh of electricity. They will need to use this information to complete their calculations.

5. Instruct students to complete Steps 1 and 2 on Worksheet 2.

    - Next, have groups focus on an appliance or electronic device from Part 1 (Step 3 on Worksheet 2). In this step, have each pair/group calculate how many kWh per year of electricity used by their appliance come from each energy source. These values should be recorded in column 2 of the table. The general calculation is:

      [kWh] x [% of energy converted to decimal] = kWh from fuel source.

    - Use your kilowatt-hour calculations from the previous step with the information in column 3 of the table on Worksheet 2 to calculate the amount of CO_2 released by each fossil fuel. The general calculation is:

      [kWh from fuel source] x [lb of CO_2 per kWh] = lb of CO_2 produced.

6. Students should add the three amounts of CO_2 together to get the total pounds of CO_2 sent into the atmosphere by using the appliance or electronic device over a year.

7. Once all groups have completed their work, have them display their results in a way that the rest of the class can see. Facilitate a classroom discussion using the following guiding questions:

    - How well did you do at predicting the energy usage of different appliances? Which ones surprised you? Which ones used about as much as you thought?

    - How could you calculate your own personal usage? What would you have to take into account?

      Usage at home, at school, and various other places you go during the day.

    - What are some hidden usage issues?

      Perhaps appliances that use electricity even when they are not "on."

    - Are the appliances that use the most energy according to the measurements made with the Kill A Watt meter the ones that produce the most CO_2? Why or why not?

    - What are some ways we could reduce the amount of CO_2 emitted into the atmosphere while still using these appliances?

    - What are some questions you have now about appliances, and how could they be explored and answered using the Kill A Watt device?

8. Wrap up the lesson by reviewing the key points. You will want to reinforce the key understanding that the electricity we use on a daily basis has environmental impacts.

Wrap-Up/Student Feedback

What are the assessment opportunities in this lesson?

Students can respond to the questions below as a writing assignment, group discussion, or journal prompt at the end of class:

- How has the population increase and per-capita consumption of natural resources contributed to the rise in CO_2 emissions?
- How has this impacted Earth's systems (water cycle, rock cycle, the atmosphere, oceans, plants, etc.)?
- Use evidence from this activity in your response.

Supplemental Materials and References

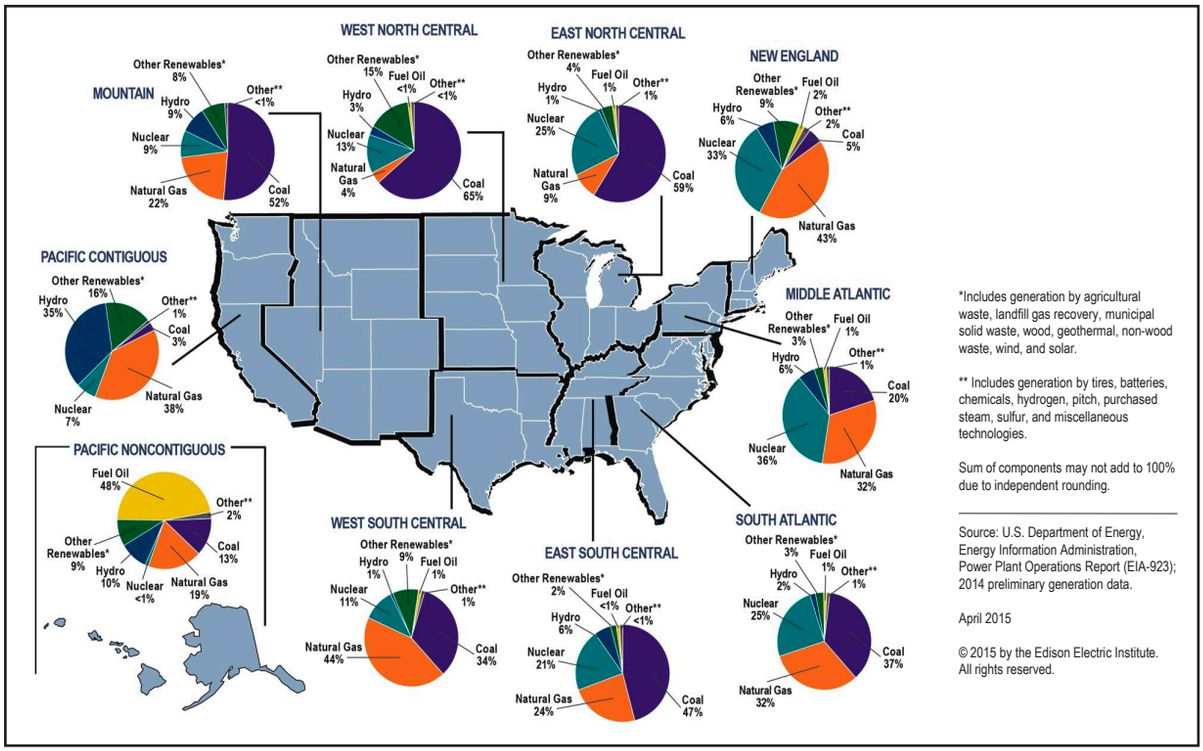
Edison Electrical Institute. (2014). *Different regions of the country use different fuel mixes to generate electricity.* Retrieved from http://www.eei.org/issuesandpolicy/generation/fueldiversity/Documents/map_fuel_diversity.pdf

Gardiner, Lisa. (2009) *Plugged in to CO_2*. Retrieved from http://www.windows2universe.org/teacher_resources/teach_pluggedCO2.html

U.S. Environmental Protection Agency (EPA) (2013) *Global Greenhouse Gas Emission Data* Retrieved from http://www.epa.gov/climatechange/ghgemissions/global.html

http://www.windows2universe.org/physical_science/chemistry/carbon_dioxide.html

Different Regions of the Country Use Different Fuel Mixes to Generate Electricity



Modified from www.eei.org/issuesandpolicy/generation/fueldiversity/Documents/map_fuel_diversity.pdf.

Worksheet 1

Appliance or electronic device	Watts (use Kill A Watt meter)	Hours used per day (estimate)	Watts-hours per day (watts x hours)	KWh per day (divide by 1,000)	KWh per year (multiply by 365)

Source: Windows to the Universe, at http://windows2universe.org/ from the National Earth Science Teachers Association (NESTA). The website was developed in part with the support of UCAR and NCAR, where it resided from 2000–2010.
©2010 National Earth Science Teachers.

NAME: _____

Worksheet 2

Carbon dioxide (CO_2), a greenhouse gas, is released into the atmosphere when fossil fuels are burned. This is a potential cause of recent global warming. The amount of CO_2 released to generate electricity depends on how much of your electricity comes from fossil fuels. In this part of the project, you will figure out how much CO_2 is released to generate the electricity for a particular appliance or electronic device over a year.

How much of your electricity comes from fossil fuels?

1. Look at the U.S. map that shows pie charts for regions that have a particular energy mix. Locate your region and its pie chart.

Fossil fuel	Percent	Decimal
Coal		
Natural gas		
Fuel oil		

2. According to that pie chart, what is the percentage of electricity that comes from coal, natural gas, and fuel oil? Fill in percentages in the table below. Convert to decimal form (for example; 63% = 0.63).

How much CO_2 is formed to make the electricity that powers an appliance or electronic device?

3. Choose one of the appliances or electronics measured in the first part of this project. Record its kWh per year (from Worksheet 1).

   *Appliance/electronic:*_____, *which uses*_____*kWh in a year.*

4. Calculate the number of kWh produced with coal, natural gas, and fuel oil to power the appliance by multiplying the decimal form of the percentage by the total kWh. Fill in these numbers in column B in the table below.

A	B	C	D
Fossil fuel	Step 4: kWh produced with each fuel to power the device over a year	Pounds (kg) of CO_2 per kWh of electricity	Step 5: pounds (kg) of CO_2 produced over a year
Coal		2.095 pounds/kWh (0.95 kg/kWh)	
Natural gas		1.321 pounds/kWh (0.60 kg/kWh)	
Fuel oil		1.969 pounds/kWh (0.89 kg/kWh)	
		Total CO_2	

Source: Windows to the Universe, at http://windows2universe.org/ from the National Earth Science Teachers Association (NESTA). The website was developed in part with the support of UCAR and NCAR, where it resided from 2000–2010. ©2010 National Earth Science Teachers.

5. Multiply the pounds or kilograms of CO_2 per kWh column C) by the number of kWh for each fuel (column B) to get the amount of CO_2 produced by generating electricity for your appliance or electronic. Record your answers in column D.

6. Sum the pounds or kilograms of CO_2 you calculated in column D to get the total pounds of CO_2 sent into the atmosphere by using the appliance or electronic device over a year.

7. Calculate the number of kWh per day and per year each appliance uses.

   a. Multiply watts by hours per day to get watt-hours.
   Example: Fluorescent classroom light fixture 160 watts x *5.5 hours/day = 880 watt-hours per day*

   b. Divide watt-hours by 1000 to get kWh per day (fifth column).
   Example: 880 watt-hours per day ÷ 1,000 = .88 kWh per day

   c. Multiply kWh per day times 365 to get kWh per year (sixth column).
   Example: .88 kWh per day x *365 days per year = 321.2 kWh per yea*r

Example Calculations

Calculating CO_2 from kWh per Year

1. Find the fossil fuel electrical generation for your region.

   Example
   Mountain region: coal 55%, natural gas 19%, and fuel oil 0.5%

2. Multiply your total kWh per year by each percentage.

   Example
   Coal: 321.2 kWh per year x .55 (55%) = 176.66 kWh per year from coal
   Natural gas: 321.2 kWh per year x .19 (19%) = 61.03 kWh per year from natural gas
   Fuel oil: 321.2 kWh per year x .005 (.5%) = 1.6 kWh per year from fuel oil

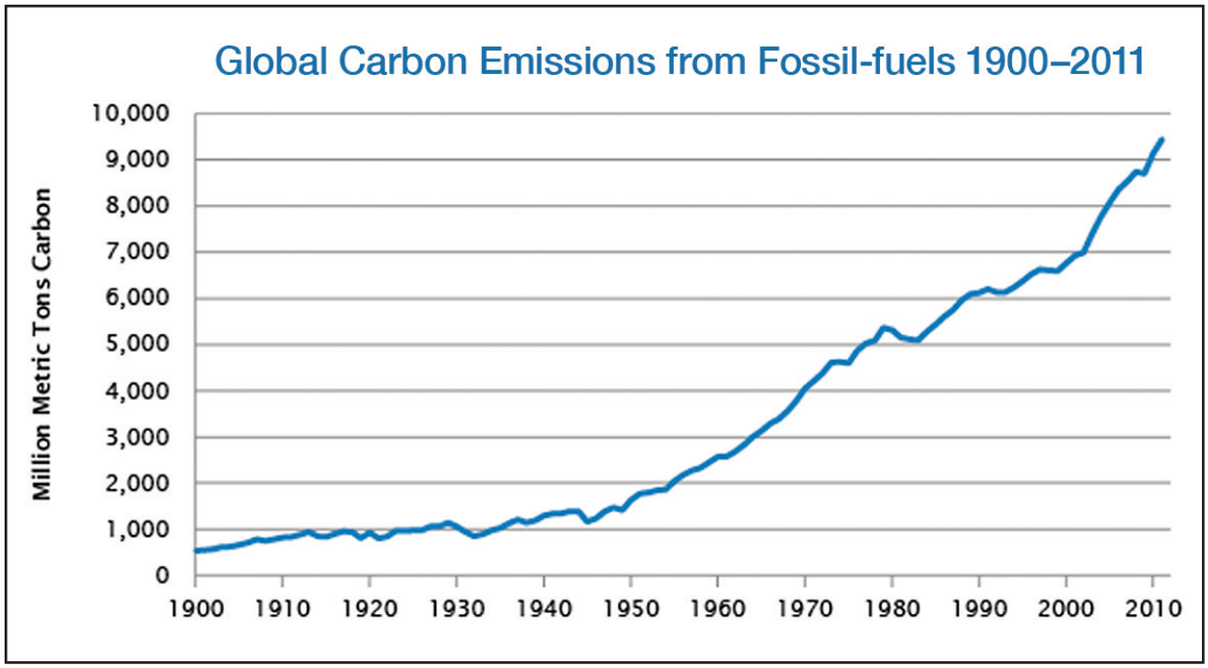
3. Calculate the total amount of CO_2 each fuel source produces yearly by multiplying the total kWh per year for this source by the pounds (lb) of CO_2 per kWh in Worksheet 2.

   Example
   Coal: 176.66 kWh per year x 2.095 lb/kWh = 370.1 lb of CO_2 per year
   Natural gas: 61.03 kWh per year x 1.321 lb/kWh = 80.62 lb of CO_2 per year
   Fuel oil: 1.6 kWh per year x 1.969 lb/kWh = 3.15 lb of CO_2 per year

4. Calculate the sum of CO_2 for a year by adding up the total CO_2 for each fuel type.

   Example
   370.1 lb (coal) + 80.62 lb (natural gas) + 3.15 lb (fuel oil) = 453 lb of CO_2 per year

Trends in Global Carbon Emissions



Source: Boden, T.A., Marland, G., & Andres R.J. (2015). Global, Regional, and National Fossil-Fuel CO2 Emissions. Oak Ridge, TN: U.S. Department of Energy, Oak Ridge National Laboratory, Carbon Dioxide Information Analysis Center, doi 10.3334/CDIAC/00001_V2015. Image retrieved from www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data.

Lesson F

Energy Flows—Trace It to the Source

Quick Overview

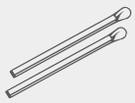
Lesson Scope: Tracing the platforms of energy sources

Length: One to three ~4-minute class periods

Grade Level: 5–8

Materials: Wooden match, battery-powered flashlight, handheld nonbattery flashlight, butcher paper or poster board, markers or colored pencils

Materials Needed per Student



Wooden match

Battery-powered flashlight

Nonbattery flashlight

Butcher paper or poster board

Markers or colored pencils

Each student will need paper and instructions to record trials.

This lesson was adapted from *Energy Flows* by NEED (2018).

Guiding Question

What is our relationship to energy?

This lesson helps students trace the electrical energy they use every day back to the source. Energy flow is a key concept within the unit, helping students understand that electricity is produced from diverse primary energy sources.

- Energy is stored in many different forms.
- Energy is neither created nor destroyed; it is transformed (converted) from one form to another.
- Most of the energy on Earth can be traced back to the sun.
- Energy flows through all dynamic systems on Earth.

Objectives

- Students will be able to trace an energy source from its origin to a wall socket.
- Students will be able to define energy transformation and think of several examples where energy is transformed.

Vocabulary

- Chemical energy
- Conservation of energy
- Energy transformation
- Kinetic energy
- Potential energy

Background Information for the Teacher

When teaching about energy flows, it is helpful to tell a story. There is a story behind the electricity that powers the lights, computers, and cellphones. Several components and routes are involved in understanding this story. The story begins with the energy source and the transformations that occur along the pathways from the original sources.

The sun is an example of a primary source of energy. It emits light energy, which is captured by plants and stored to create food using the process of photosynthesis. After plants die, they become buried in the soil and decompose. After millions of years, the combination of plant material and the *right* conditions can create coal. Coal is mined for use in power plants, where it is burned and transformed to heat energy. This energy can be used to power turbines that generate electricity.

Energy transformations and the concept of energy transformation can be very complex and may be difficult for students to understand. It is important for students to have the opportunity to explore energy conversions and the flow of energy in systems through multiple examples. A good starting place could be with a very simple description. This can be done using strategies such as a whole-group discussion or concept mapping on the board. This initial discussion should be used to uncover what students know, develop a common definition, and build on this definition, increasing its complexity as the lesson progresses.

A very basic definition to begin with might be: *Energy transformation is the process of energy changing from one form to another.* A good resource to learn more about energy types and transformations can be found at *https://www.wiley.com/legacy/Australia/PageProofs/SQ8_AC_VIC/c10_TransferringAndTransformingEnergy_WEB.pdf*.

When discussing **chemical energy** in a battery, it is useful for students to have multiple types of visualizations. The battery circuit kits, flashlights, and pictures are all useful tools for this.

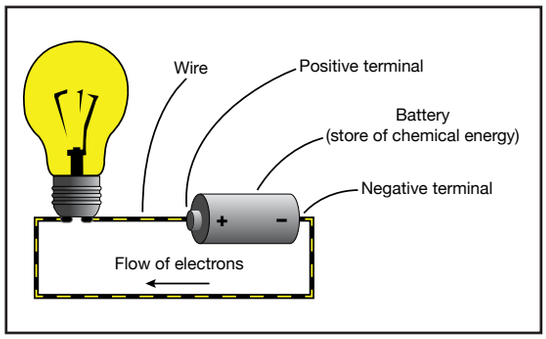


Figure 1. DC circuit with bulb. Source: www.eschooltoday.com/energy/kinds-of-energy/energy-stored-transferred-dissipation.html.

The concepts covered in the lessons about electric currents and electricity generation (Lessons C and D) will be helpful concepts to build on for this lesson. Make sure students understand that the wire needs to make a complete circuit for the electrons to flow and the light bulb to work. This review provides another opportunity to uncover the energy transformations that are occurring in this demonstration. They should observe light energy, heat energy from the light bulb, and chemical energy in the battery.

After making this definition clear, the next step is to make it applicable to students' everyday lives. This step can also take different forms depending on your classroom and the methods you prefer. It might be useful to use a think/pair/share activity at this point. Some thought experiments or questions that could be asked could include:

Think about the activities you did this morning that required energy. Where did this energy come from? What energy transformations occurred to allow you to use this energy?

After students think about these topics, have students discuss their thoughts with a partner, and eventually ask volunteers to share their responses. Further discussion that results from this activity could lead to the development of a compiled list of all of the activities requiring energy, the energy sources used, and the transformations that occurred. It may be necessary to begin the thought process with an example like the following. The teacher can ask students if they ate anything for breakfast.

Did this food give you energy? What kind of energy? Did it require energy to get the food to the grocery store? How did the food grow?

Students will soon understand that there is an "energy story" behind many things we interact with on a daily basis.

An overarching goal of this lesson is for students to discover that energy flows are complex. Many steps are involved in getting energy from a primary form to a form such as electricity that people use.

Common Student Misconceptions/ Student Challenges

Research findings suggest that the idea of **conservation of energy** is more difficult than the ideas of energy transformation and energy transfer (Herrmann-Abell & DeBoer, 2011). Therefore, it is important to provide opportunities to explore the concept through activities to help students identify the diverse forms of energy as they trace the flow from source to service.

Next Generation Science Standards

Performance Standard.4-PS3-2:

Make observations to provide evidence that energy can be transferred from place to place by sound, light, heat, and electric currents.

Disciplinary Core Idea

- Energy can be moved from place to place by moving objects or through sound, light, or electric currents. Light also transfers energy from place to place.
- Energy can also be transferred from place to place by electric currents, which can then be used locally to produce motion, sound, heat, or light. The currents may have been produced to begin with by transforming the energy of motion into electrical energy.

Science and Engineering Practices

- Make observations to produce data to serve as the basis for evidence for an explanation of a phenomenon or to test a design solution.

Crosscutting Concepts

- Energy can be transferred in various ways and between objects.

Materials

- Energy source summary (by state) at website
- Wooden match (for teacher use in a demonstration)
- Battery-powered flashlight
- Handheld nonbattery flashlight
- Butcher paper or poster board
- Drawing materials (marker, colored pencils)

Intermediate Energy Infobook Activites book (NEED, 2018).access to the electronic version or one book per student group (see References)

Series cards 1, 2, and 3 are located on pages 27–35 of the *Intermediate Energy Infobook Activities* guide (one set for each student group)

- Trace It to the Source Prompt (Appendix 1.5.1)
- Trace it to the Source Assessment (Appendix 1.5.2)
- Forms of Energy (Appendix 1.5.3)
- Forms and Sources of Energy (Appendix 1.5.4)

Setup

Make copies of forms or worksheets for student groups.

Length of Lesson

Note: As an assessment, if you did not use the first activity (preassessment), this lesson should start with the Trace It to the Source activity. If you did, skip to Step 2.

This lesson will take one to three 45-minute class periods.

Suggested Procedure

Note: *This assessment was used in Lesson A at the beginning of the unit. It is not necessary to repeat the assessment, but it is attached in Appendix 1.5.2 of this lesson if needed.*

Part I

1. Pretest: Trace It to the Source Assessment. This assessment focuses on electricity and its energy flow from the source to the destination. Ask students to spend a minimum of 10 minutes completing this activity. Depending on the instructor's goals for this assessment, she or he can also give students the rubric developed for the assessment. If the rubric is not used, the teacher will get a better understanding of what students' prior knowledge is because the rubric could lead them to responses they would not otherwise have thought of.

2. Begin the lesson by lighting a wooden match and asking the students to think about what is happening in energy terms. Ask them to think about the energy flow from the sun to the match. This will prompt them to think about energy transformation.

3. Next, write *energy transformation* on the board. Explain the *rules* of this activity, which is called a silent conversation. During the activity, the class will develop a definition of *energy transformation* based on brain-stormed responses that are written on the board. Instruct students to first spend some time thinking and writing down ideas at their seats. When they are ready and have a single idea to contribute, they may walk up to the board and write a word or short sentence that explains what they think it means, all of which is done silently. After several words or phrases are posted, students will be encouraged to contribute to each other's ideas in a constructive manner. The end result should resemble a mind map.

4. After this, the teacher can facilitate a discussion by asking students about certain parts of the mind map. In this way, a definition can be created from the contributions on the board. This can be accomplished by underlining or circling agreed-upon words and phrases. If the discussion is not leading to an accurate definition, the teacher should intervene and guide students to the correct answer when appropriate. Refer to the Background Information section for an example definition.

5. Next, students will need something to write on. Have students create two columns and ask them to think about the events that occurred before they arrived at school. You could prompt them to start with the moment they woke up, and then to brainstorm and record in the first column all the activities they can think of that required energy. In the second column, they should list the energy transformations that occurred while they were doing these activities or the transformations that occurred before the object became usable

to them. Allow students 10–15 minutes to work on this.

6. Have students partner with a classmate. Instruct the pairs that they will have 5 minutes to compare and discuss their lists. After the time is up, they will be sharing three events or objects and the corresponding energy transformations from their combined lists.

7. During this sharing session, the teacher can facilitate discussion. Examples follow: If a student said *I ate a banana for breakfast*, the teacher could respond with: *Where did you get the banana?* (*It came from the store, grew on a plant, Hawaii, etc.*) *Did that require energy?* (*Yes*) *What kind of energy?* (The *sun's energy to grow, gas for transportation, etc.*) *What transformations occurred?* (*The sun's energy was transformed to chemical energy used by the plant to grow; I ate the banana and it was transformed to energy that I can use to grow, walk, run, or dance.*) Students should begin to notice how important the sun is for life to exist. They should also be led to recognize that energy transformations are crucial in this process. This may encourage deeper thinking and awareness of the complex processes that go into making the objects we use daily.

8. Distribute the Forms of Energy handout located in Appendix 1.5.3 (page 8 of *Energy Flows*, NEED, 2018). As a class, discuss and take notes on the forms of energy and place them as either **potential energy** or **kinetic energy**. (*Kinetic energy* is motion energy and temperature is often used to measure the amount of kinetic energy in a system, whereas *potential energy* is the amount of stored energy.) Remind students that energy is spontaneously transferred from warmer regions or objects into colder ones. Students should be able to define potential and kinetic energy and be able to list examples of each.

9. Next, hand out Forms and Sources of Energy located in Appendix 1.5.4 (page 9 of *Energy Flows*, NEED, 2018) and have students work with partners to complete the page. Students will be classifying the source of energy by its form. For example, coal is burned; therefore, the energy is stored as *chemical energy*. The source of energy is coal and the form of energy is chemical. Review the answers (located on page 6 of *Energy Flows*; see References) with the students when they have finished the worksheet.

Part II

Note: *Lessons C and D can be used to deepen students' understanding of electricity and generators. At the teacher's discretion and learning needs, these lessons can be used to supplement this lesson or can be implemented before this lesson. Lesson C develops a basic understanding of electricity and the flow of electrons through materials such as light bulbs, copper wires, and batteries. This lesson also introduces important vocabulary necessary for understanding electricity that is foundational to understanding the generation and transformation of electrical energy. Lesson D helps students understand how a generator creates an electric current and allows students the chance to experiment with building generators.*

Now it is time for students to explore energy transformations with a couple hands-on investigations or activities. Depending on the number of flashlights and hand-generator flashlights you have, this step can be set up in stations or just be conducted as an instructor demonstration.

1. Ask students; "Have you ever considered how energy flows within a battery-powered flashlight?" This is a good opportunity to have students sketch a flashlight and its energy flow or have them discuss energy flow with a partner. After asking a few students to share their thoughts, the teacher reviews **chemical energy**. The teacher can refer to the Background Information for a visual. The purpose here is to make sure students understand that the battery is providing chemical energy, which then creates electricity to power the bulb in the flashlight.

   The next activity involves exploring hand-generated flashlights in an attempt to figure out how they work. Make sure to tell students that these flashlights cannot handle a lot of force or they will break. It would be ideal to have one hand-generated flashlight for a group of 3 or 4 students.

2. After students have observed and used the flashlights, ask them to either sketch the flow of energy within this type of flashlight or discuss this with a partner. In a brief discussion, students should be able to distinguish between mechanical and chemical energy. The hand-generator flashlight is getting its energy from the force the user puts into it. Note the difference between this example and the battery-powered flashlight. Refer to page 10 of *Energy Flows* for the energy transformations in a handheld generator. Students could redraw the correct flow, or the accurate transformations could be discussed together as a class.

3. Now ask students to return to the list of activities they did before school. Ask them how the hand generator or the battery-powered flashlight is related to their list. You could either have a discussion about this or tell students that you do not want a response now and let them ponder this question for now.

Part III

This is a point in the lesson where you could split the lesson in half and resume it the next class period.

4. Each-one-teach-one: Divide the students into several groups and distribute a set of series cards to each group (Series cards 1, 2, and 3 are located on pages 27–36 of *Energy Flows*, NEED, 2018). Ask the students to place the cards on a string or tape them in the correct order. Have the students explain their series order to the class and how they are connected and why. Students should

also use the diagrams and handouts on pages 11–26 for additional resources for this activity.

5. This lesson should conclude with a summative explanation or teacher–student sketch of how energy flows through power plants, turbines, generators, transformers, and distributions lines. Use the diagrams again on pages 11–26 so students can see a visual of the transfers through each system. Finally, revisit the definition of *energy transformation* as a class and relate it back to the preceding activities.

6. Posttest: The next step in this lesson involves the Trace It to the Source activity as the postassessment. Ask students a second time to spend 10 minutes completing this activity. (See Appendix 1.5.2 for the rubric.)

7. After students have completed their assessments, an additional activity can include a collaborative approach to the assessment. Split students into groups of 3 to compare assessments. Encourage students to collaborate and develop a new response based on the teams' collective knowledge. Teachers can give students a larger sheet of paper or poster board to re-create the sketch.

8. The final step is for students to make an informal presentation of each group's sketch. Students could create a large sketch on poster board or similar material. This allows students to see other classmates' work, it gives the instructor a chance to assess the class, and it builds communication skills.

Wrap-Up/Student Feedback

What Are the Assessment Opportunities in This Lesson?

Conceptual Drawing—Trace It to the Source

Depending on the goals of the teacher or the dynamics of the class, this can be done several ways and multiple times. One way is to have students complete the assessment individually. The teacher can grade the assessments with the rubric or a method that best suits her or his grading style. This will give the teacher a perspective on the range of responses and knowledge that the students possess.

Another way is to administer the assessments to groups of students. Making the assessment collaborative can be an effective method for promoting greater conceptual development. The teacher can observe the presentations of the group drawings to identify students' level of understanding.

This lesson includes multiple opportunities for the teacher to evaluate students. Class discussions provide a great opportunity to informally assess students' understanding of these concepts. While students are working in groups, the teacher can roam around the room and listen to students' discussions and ask questions that stimulate deeper thinking.

Supplemental Materials and References

National Energy Education Development Project (NEED). (2018). *Intermediate energy infobook*. Retrieved from http://www.need.org/files/curriculum/guides/Intermediate%20Energy%20Infobook.pdf

- This book is an excellent resource for adding content to lessons for students as well as background information for the teacher. This handbook outlines the various types of energy sources, the use and generation of electricity, and a brief history of energy.

National Energy Education Development Project (NEED). (2018). *Energy Flows*. Retrieved from http://www.need.org/files/curriculum/guides/Energy%20Flows.pdf

- This resource contains handouts and diagrams essential for this lesson on energy flows and transformation.

U.S. Energy Information Administration: http://www.eia.gov/state/

Trace It to the Source!

Prompt:

You charge your phone, tablet, or other electronic device by plugging it into a wall socket. Where does this electricity come from? Take a minute to think about the original source(s) of electricity and the steps that occur before it powers your electronics.

Step 1

Start by brainstorming and organizing your thoughts on a piece of paper. What are the types of sources you have learned thus far that produce electricity? Think about renewable resources and nonrenewable resources and the processes that each source under-goes as it is transformed into electricity and travels to a wall socket. Identify with words or in a sketch the primary energy sources learned so far.

Step 2

Choose one of the sources above and create a series of drawings to represent the steps that trace the source from its origin to your home.

Step 3

Make sure your sketch is organized and is as detailed as you can make it. Include descriptive words, phrases, and sentences to help someone understand your drawings and what you are thinking. Use the following steps below to extend this as a group activity:

Step 4

In your group, evaluate each of your classmate's drawings. Give each classmate an opportunity to share his or her sketch. Collectively gather new information from each person's drawing and create a new sketch that includes the additional information. Were there other sources traced? Are there any additional steps in the energy's path to the socket that were not included in your drawing? Prepare the best drawing possible to present to the class.

Step 5

Once the new drawing is completed, present it to the entire class. Each person must share something about the drawing. Finally, respectfully listen and take note of other student's presentations. This is your chance to collectively learn from your peers to expand your knowledge of electricity and the many ways it travels to get to your electronics.

Trace It to the Source! Assessment Rubric

Category	4	3	2	1
Energy Sources (coal, uranium, natural gas, geothermal, solar, wind, hydro, petroleum)	Drawings clearly and accurately identify 5 of the primary energy sources.	Drawings identify 4 of the primary energy sources.	Minimal drawings identify 2 of the primary energy sources.	Drawings do not fully present or identify a primary energy source.
Transportation (train, rail, barge, pipeline, trucks)	Sketch represents at least 3 accurate transportation modes for delivering a primary energy source to the generation facility.	Sketch represents at least 2 accurate transportation modes for delivering a primary energy source to the generation facility.	Sketch represents at least 1 transportation mode for delivering a primary energy source to the generation facility.	Sketch does not identify a transportation mode.
Electricity Generation (coal fired, hydroelectric dams, nuclear power plant, natural gas fired, wind turbines, solar panels)	An advanced drawing with words or phrases that accurately describe how and at what point in the process the energy source is transformed into electricity.	A proficient drawing with words or phrases that accurately describe how and at what point in the process the energy source is transformed into electricity.	A basic or partially accurate drawing with words or phrases describe how and at what point in the process the energy source is transformed into electricity.	A vague or inaccurate drawing with words or phrases that describe how and at what point in the process the energy source is transformed into electricity.
Electricity Transmission (transmission lines, substation, distribution lines, fuse box, electrical lines through house)	At least 5 accurate representations of the transmission process from the energy source's origin of its final destination.	At least 4 accurate representations of the transmission process from the energy source's origin of its final destination.	At least 3 accurate representations of the transmission process from the energy source's origin of its final destination.	Two or fewer representations of the transmission process from the energy source's origin of its final destination.

Note: Rubric is designed to assess individual student drawings and does not assess a group presentation or a final poster created by the group.

Forms of Energy

All forms of energy fall under two categories:

POTENTIAL

Stored energy and the energy of position (gravitational).

CHEMICAL ENERGY is the energy stored in the bonds of atoms and molecules. Gasoline and a piece of pizza are examples.

NUCLEAR ENERGY is the energy stored in the nucleus of an atom – the energy that holds the nucleus together. The energy in the nucleus of a plutonium atom is an example.

ELASTIC ENERGY is energy stored in objects by the application of force. Compressed springs and stretched rubber bands are examples.

GRAVITATIONAL POTENTIAL ENERGY is the energy of place or position. A child at the top of a slide is an example.

KINETIC

The motion of waves, electrons, atoms, molecules, and substances.

RADIANT ENERGY is electromagnetic energy that travels in transverse waves. Light and x-rays are examples.

THERMAL ENERGY or heat is the internal energy in substances – the vibration or movement of atoms and molecules in substances. The heat from a fire is an example.

MOTION is the movement of a substance from one place to another. Wind and moving water are examples.

SOUND is the movement of energy through substances in longitudinal waves. Echoes and music are examples.

ELECTRICAL ENERGY is the movement of electrons. Lightning and electricity are examples.

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Forms and Sources of Energy

In the United States we use a variety of resources to meet our energy needs. Use the information below to analyze how each energy source is stored and delivered.

1 Using the information from the *Forms of Energy* chart, and the graphic below, determine how energy is stored or delivered in each of the sources of energy. Remember, if the source of energy must be burned, the energy is stored as chemical energy.

NONRENEWABLE		RENEWABLE	
Petroleum	_____	Biomass	_____
Coal	_____	Hydropower	_____
Natural Gas	_____	Wind	_____
Uranium	_____	Solar	_____
Propane	_____	Geothermal	_____

2 Look at the U.S. Energy Consumption by Source graphic below and calculate the percentage of the nation's energy use that each form of energy provides.

What percentage of the nation's energy is provided by each form of energy?

Chemical _____

Nuclear _____

Motion_____

Thermal _____

Radiant _____

What percentage of the nation's energy is provided by nonrenewables? _____

by renewables? _____

U.S. Energy Consumption by Source, 2013

NONRENEWABLE

PETROLEUM 35.2%
Uses: transportation, manufacturing

NATURAL GAS 26.6%
Uses: heating, manufacturing, electricity

COAL 18.5%
Uses: electricity, manufacturing

URANIUM 8.5%
Uses: electricity

PROPANE 1.7%
Uses: heating, manufacturing

RENEWABLE

BIOMASS 4.7%
Uses: heating, electricity, transportation

HYDROPOWER 2.6%
Uses: electricity

WIND 1.6%
Uses: electricity

SOLAR 0.3%
Uses: heating, electricity

GEOTHERMAL 0.2%
Uses: heating, electricity

*Total does not add to 100% due to independent rounding.
Data: Energy Information Administration

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Activity Set 2

CO_2 AND CLIMATE CHANGE

A Supplemental Curriculum
for Middle School Learners

A Midwest Geological Sequestration Consortium Program Funded by the U.S. Department of Energy

Activity Set 2

CO_2 and Climate Change

Lessons G–K

- **G** The Carbon Cycle
- **H** What is CO_2?
- **I** The Greenhouse Effect
- **J-1** Evaluating the Evidence of Climate Change
- **J-2** Evaluating the Evidence of Climate Change; Addressing Misconceptions
- **K** Examining Your Carbon Footprint, Human Impact Through Transportation

Lesson G

The Carbon Cycle

Quick Overview

Lesson Scope: Following a carbon atom through the carbon cycle

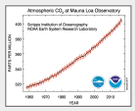
Length: One to two ~45-minute class periods

Grade Level: 5–8

Materials: Posters, position cards, atmospheric CO_2 graph

Materials Needed per Student

Position cards



Atmospheric CO_2 graph

Each student will need paper and instructions to record trials.

This lesson has been adapted from the National Center for Atmospheric Research at *https://scied.ucar.edu*.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect temperature?

This lesson explores how **carbon** cycles through the different Earth systems. Students play the role of carbon atoms traveling through the carbon cycle to gain understanding of carbon's varied pathways and the relevance of carbon to living and non-living things. Students will also understand the increasing levels of CO_2 in the atmosphere over the past half-century.

Key Understandings

- Carbon cycles through Earth's systems.
- Carbon makes up all living things and parts of nonliving systems such as the ocean, air, rocks, etc.
- Atmospheric levels of CO_2 are on the rise.

Objectives

- Students will follow the journey of a carbon atom through the carbon cycle.
- Students will observe evidence and answer questions about the rise in atmospheric levels of CO_2

Vocabulary

- Carbon
- Carbon cycle
- Carbon dioxide (CO_2)
- Greenhouse gas
- Greenhouse effect
- Carbon sinks

Background Information for the Teacher

Teachers can explore the following websites to become familiar with or refresh their knowledge of the **carbon cycle**.

http://www.windows2universe.org/earth/Water/co2_cycle.html

http://eo.ucar.edu/kids/green/cycles6.htm

http://www.physicalgeography.net/fundamentals/9r.html

Carbon is an element found all over the world and in every living organism. Oxygen is another element that is found in the air. When carbon and oxygen bond, they form a colorless, odorless gas called **carbon dioxide (CO_2)**. The CO_2 in the Earth's atmosphere is considered a **greenhouse gas**, which means it traps heat. This **greenhouse effect** naturally helps keep the Earth's temperature at a level that can support life on the planet. (See Lesson I on the Greenhouse Effect for more information about the Earth's atmosphere.)

Other places on Earth also store carbon. The oceans store large amounts of carbon; plants and soil store carbon. Carbon constantly moves and cycles from one part of the Earth to another, and this is known as the carbon cycle. This cycle helps to maintain a balanced level of CO_2 in the atmosphere.

Humans are influencing the natural balance by adding more CO_2 to the atmosphere. Human activities, such as heating our homes, driving cars, using electricity, and making products, add CO_2 to the atmosphere. This CO_2 is being added to the atmosphere faster than natural processes (plants photosynthesizing, ocean sinks, etc.) can remove it, causing the atmosphere to trap more heat and the Earth's temperature to rise. Scientists have found that recent levels of CO_2 in the atmosphere are abnormally high (see graph in Appendix 2.1.2) compared with the long-term historical trend, and these levels are continuing to increase at an unprecedented rate.

Common Student Misconceptions/ Student Challenges

Carbon dioxide or CO_2 is a term used frequently in the news and in that forum is mostly associated with the atmosphere. Many students do not realize that the molecule CO_2 is a very important element to life and that the carbon in CO_2 is constantly cycled through organisms and places on Earth.

Students are often able to insert definitions and scientific names into the parts of the carbon cycle without being able to explain how the carbon cycle actually works (Mohan et al. 2006).

Next Generation Science Standards

Performance Standard.MS-ESS2-1:

Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.

Disciplinary Core Idea

All Earth's processes are the result of energy flowing and matter cycling within and among the planet's systems. This energy is derived from the sun and Earth's hot interior. The energy that flows and the matter that cycles produce chemical and physical changes in Earth's materials and living organisms.

Science and Engineering Practices

Develop and use a model to describe phenomena.

Crosscutting Concepts

Explanations of stability and change in natural or designed systems can be constructed by examining the changes over time and processes at different scales, including the atomic scale.

Materials

Posters and position cards from the Carbon Cycle game, *https://climatechangelive.org/img/fck/file/carbon_cycle_game.pdf*.

- Atmospheric CO_2 graph, Appendix 2.1.1

Setup

It will take about 30 minutes to cut out the game cards. (You may want to laminate these.) Estimate about 5 minutes to print (in color) and hang posters (place the various carbon stations and each station's tickets in various locations about the room).

Length of Lesson

Plan to spend 65 to 70 minutes on this lesson. The lesson may be conducted over one or two 45-minute class periods.

- Introduction – 10 minutes
- Playing the game – 20 minutes
- Reflection – 15 minutes
- Analyzing graph – 15 minutes

Suggested Procedure

http://scied.ucar.edu/events/climate-change-connections-2013

http://scied.ucar.edu/carbon-cycle

Activity Set 2 | CO_2 and Climate Change

Activity

1. Determine what students know and do not know about carbon and the carbon cycle. A series of formative assessment questions can create a good picture of students understanding of carbon. You can use these as a pre-assessment or a discussion tool.

   a. Where do trees get the mass (atoms) to build their leaves, trunks, bark, and other matter?

   Trees respire or breathe in carbon dioxide from the air and use the carbon to form molecular bonds in their tissues.

   b. Where does carbon dioxide released from burning fossil fuels come from?

   The carbon is stored in the chemical bonds of the fossil fuel while the oxygen comes from the atmosphere.

   c. Most of the molecular bonds in the human body are carbon based. Where do humans get the carbon in their bodies?

   Animals cannot consume carbon from carbon dioxide in the atmosphere, so they must ingest it by eating plants that respire carbon dioxide or by eating animals that have eaten plants.

2. Tell students that they will be exploring carbon sources and **carbon sinks** in today's lesson through the cycling of a carbon atom.

3. Explain that the various carbon stations and each station's tickets represent the places where carbon can be found within the carbon cycle.

4. Discuss with the students where carbon might be stored within each of the following locations in the carbon cycle.

   a. **The Atmosphere**
   Carbon exists as a gas in the form of carbon dioxide.

   b. **Surface Water**
   Carbon dioxide from the atmosphere is dissolved into surface waters and oceans.

   c. **Animal Waste**
   These wastes contain carbon left over from the plants or animals the animals consumed.

   d. **Soils**
   Microorganisms and plant roots bring in carbon dioxide from the atmosphere then respire the carbon dioxide into the soil.

   e. **Rocks**
   Abiotic elements such as rocks also change states from one type to another (i.e. metamorphic to sedimentary through deposition, time, and erosion). This is known as the rock cycle and is driven by the heat from the Earth's core. Heat moves through the mantle of the Earth causing faulting,

volcanism, earthquakes, and movement of materials in the rock cycle.

f. **Living Animal Consumers**
Consumer animals ingest carbon from the plants they eat and store it in their tissues.

g. **Living Animal Predators**
Predators ingest carbon into their tissues by eating other animals.

h. **Living Plants**
During photosynthesis and respiration, plants bring in carbon dioxide from the atmosphere and store the carbon in their plant tissue.

i. **Dead Plants and Animals**
When living organisms die, carbon stays in their tissues until they are eaten by something else.

j. **Decomposers and Parasites**
These breakdown the tissues of dead organisms for energy and in the process ingest the carbon from the dead organisms into their tissues.

5. Tell students that they will be playing the role of a carbon atom. They will be given a passport paper and be assigned to start at one of the stations. At each station, they will draw a ticket to determine where to go next. They will record the various stations they visit by marking their travels on the passport worksheet. Students should write notes on the passport to explain how they went from one station to the next (i.e. decomposition, animal respiration, etc.) and the energy source that is driving their movement (either the Sun or the heat from the Earth's core).

6. Randomly assign students to the various stations and have them begin their journeys.

7. Students will take 10 trips through the carbon cycle, moving through the stations until their passports are filled.

8. Have students reflect on their journeys—they could do this by writing a story, drawing a cartoon series, or writing a poem. Students should be encouraged to share their reflections with the class, a classmate, or someone else. Have students incorporate the reasons they moved from one carbon sink to another.

9. Discuss these questions with your students following the activity. They can respond in their journals or discuss with partners.

   a. In the real world, does the cycling of carbon stop?

   b. How could energy source(s) change the carbon cycle?

   c. How were your journeys the same? Different? Why?

d. What might cause an increase or decrease of carbon dioxide in a given location, such as the atmosphere or under ground?

e. Where do you think most of our planet's carbon is located?

f. In this lesson you examined the carbon cycling in a forest ecosystem. How might this same carbon cycle look different in your town or city? What types of cards would you need for that system?

g. Where are fossil fuels located in this system?

For a review, consider showing the carbon dioxide video at https://science360.gov/obj/video/54d4fb2c-4e7e-4c92-9ebf-f8a50d6106d6/chemistry-co2-carbon-dioxide or https://www.youtube.com/watch?v=VTbxS9evlkQ.

10. Finally, distribute the CO_2 graph in Appendix 2.1.2 to students or place it on the board for the students to see. Explain that this graph was created from data collected over many years by scientists studying atmospheric CO_2 concentrations. The Mauna Loa graph plots CO_2 data (red curve), measured as the mole fraction in dry air, on Mauna Loa and constitutes the longest record of direct measurements of CO_2 in the atmosphere. This data was initiated by C. David Keeling of the Scripps Institution of Oceanography in March of 1958 at a facility of the National Oceanic and Atmospheric Administration. The black curve represents the seasonally corrected data. Explain to students that because CO_2 spreads throughout the world's atmosphere, measurements from Mauna Loa are a good representative of the global average CO_2 levels. The following questions can be asked as a group, discussed with partners, or used as an individual assessment.

a. What patterns do you notice in the graph?

An increase in CO_2, followed by a decrease in CO_2, creating a repeating pattern of peaks and troughs much like a wave. Another pattern is an increase in heights of both peaks and troughs over time.

b. What does the increase in the height of the peaks and troughs mean?

The overall amount of CO_2 in the atmosphere is increasing.

c. Extension: Have students explain how the brief increases and decreases in CO_2 are related to the seasonal changes of CO_2 concentration in the atmosphere and the growing season for plants. CO_2 in the atmosphere decreases during the growing season and increases during the rest of the year, which leads to a maximum buildup in April and May before photosynthesis begins to take over again. Photosynthesis, where plants take up CO_2 from the atmosphere and release oxygen, dominates during the growing season (the warmer

part of the year). Respiration, by which plants and animals take up oxygen and release CO_2, occurs all the time but dominates during the colder months of the year. However, the seasons are opposite in the Northern and Southern Hemispheres, causing CO_2 concentrations in the Northern Hemisphere to increase and CO_2 concentrations in the Southern Hemisphere to decrease and vice versa when the seasons change. However, there is more land in the Northern Hemisphere, causing the carbon cycle to be more pronounced in the north. These reasons explain the slight undulation in the graph.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

Student reflections in Step 8 should provide evidence of their learning about the carbon cycle. The following question can be posed to further students' understanding of the carbon cycle or as an assessment.

The human activities of burning fossil fuels for energy have released a tremendous amount of carbon that was once stored in the ground back into the atmosphere. Explain the relationship the atmosphere has with other parts of the carbon cycle. How might extra carbon in the atmosphere affect every other part of the carbon cycle?

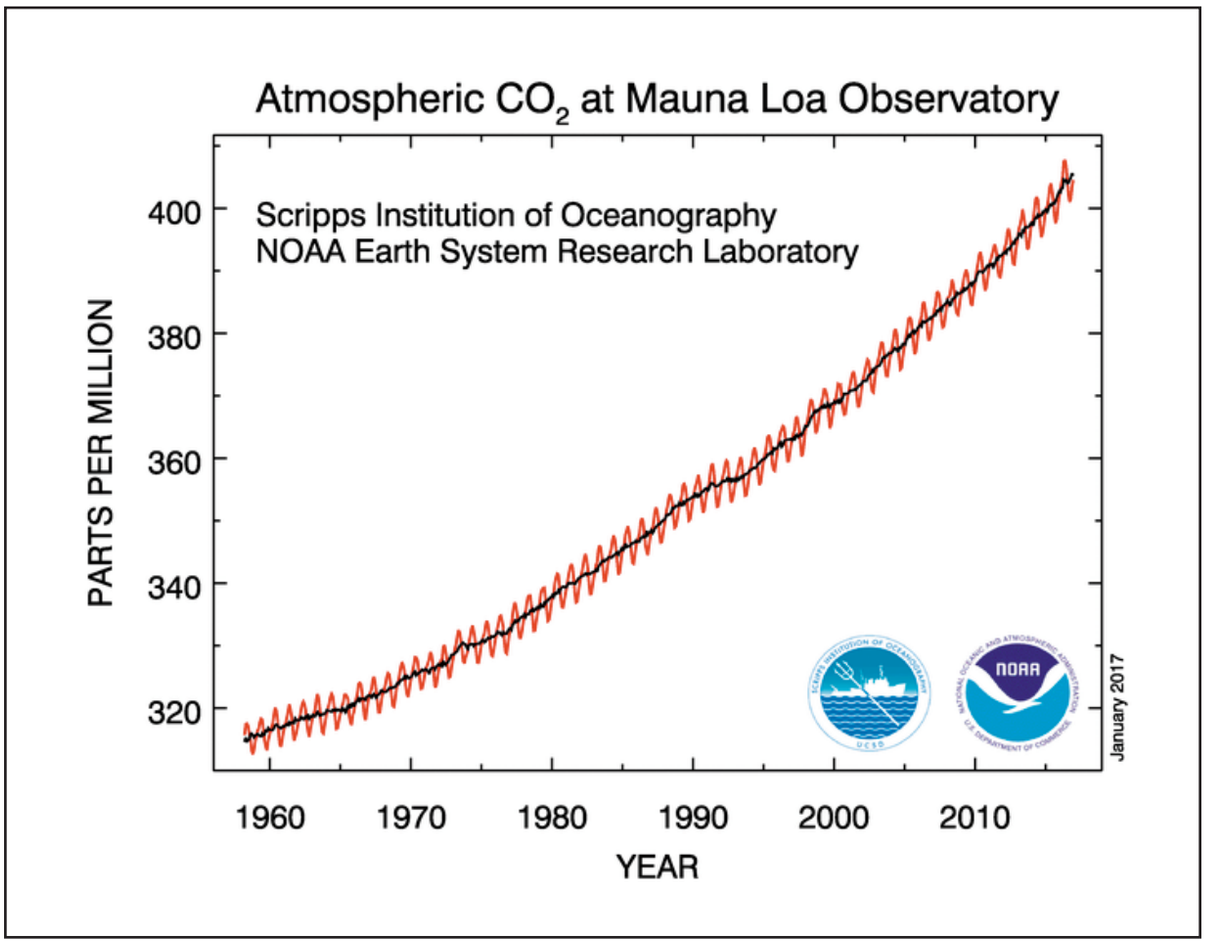
Supplemental Materials and References

Keeling, C. D., Bacastow, R. B., Bainbridge, A. E., Ekdahl, C. A., Guenther, P. R., Waterman, L. S., & Chin, J. F. (1976). Atmospheric carbon dioxide variations at Mauna Loa observatory, Hawaii. *Tellus*, 28(6), 538–551.

Keeling, C. D., & Whorf, T. P. (2005). Atmospheric CO_2 records from sites in the SIO air sampling network. *Trends: a compendium of data on global change, 2009.*

Mohan, L., Sharma, A., Jin, H., Cho, I., & Anderson, C.W. (2006). *Developing a carbon cycling learning progression for K–12*. Paper presented at the annual meeting of the National Association for Research on Science Teaching (San Francisco, CA).

National Ocean & Atmospheric Administration. (n.d.). *Earth system research laboratory: Global monitoring division*. Retrieved from http://www.esrl.noaa.gov/gmd/ccgg/trends/



Source: www.esrl.noaa.gov/gmd/ccgg/trends/full.html

Lesson H

What is CO_2?

Quick Overview

Lesson Scope: CO_2 in the atmosphere and the affects on global temperature

Length: One ~45-minute class period

Grade Level: 5–8

Materials: 10-gallon aquarium, plastic tray, bubbles with blowing wand, candle, matches, balloons, hot pads, 1-pint water bottles, ice pick, plastic bag, 5 to 12-pound block of dry ice, gloves, Styrofoam ice chest, safety goggles

Materials Needed per Student

10-gallon aquarium

Plastic tray

Bubbles & blowing wand

Candle

Matches

Balloons

Each student will need paper and instructions to record trials.

This lesson has been adapted from The Bureau of Economic Geology, The University of Texas at Austin, *http://www.beg.utexas.edu/gccc/forum/codexdownloadpdf.php?ID=38* for STEP.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect global temperature?

In order to understand **greenhouse gases**, in particular **carbon dioxide (CO_2)**, students need to understand the properties of this gas. This lesson will demonstrate CO_2 in solid and gas form. As students observe the **sublimation** of dry ice, certain characteristics can be demonstrated visually in the colorless, odorless gas that is produced.

Key Understandings

- CO_2 is a colorless, odorless gas.
- CO_2 is necessary for life on Earth.
- Increased levels CO_2 in the atmosphere make it hard for radiant heat to return to space. Increasing the amount of CO_2 in our atmosphere is like adding extra glass to a greenhouse and trapping the radiant heat.

Activity Set 2 | CO_2 and Climate Change

Objectives

- Students will identify basic properties of CO_2.
- Students will list ways CO_2 is used in our everyday lives.

Vocabulary

- Greenhouse gases
- Carbon dioxide (CO_2)
- Sublimation
- Photosynthesis
- Greenhouse effect
- Global climate change

Background Information for the Teacher

CO_2 is a colorless, odorless gas that is among the most prevalent gaseous materials in the world. It is found nearly everywhere in the Earth's atmosphere and plays an important part in many biological processes. Humans release CO_2 during natural metabolic processes (exhaling), and CO_2 is also released into the atmosphere when fossil fuels are burned. Plants use CO_2 during **photosynthesis,** and it is also used in many popular items, including soda, yeast, baked goods, and "dry" fire extinguishers.

The physical properties of CO_2 can vary by temperature. While this material is a gas under most surface conditions, CO_2 forms a solid dry ice at temperatures below -109°F (-78°C). It may also transform into a liquid when it is dissolved in water under constant pressure.

Increased concentration of CO_2 is trapping heat in the atmosphere; this is commonly referred to as a **greenhouse effect**. As concerns over **global climate change** have increased, the impacts of CO_2 concentrations have become a major source of debate for politicians and environmental activists.

Common Student Misconceptions/ Student Challenges

Students may think CO_2 exists only as a gas. Actually, CO_2 can be found in three phases: solid (when at temperatures below -109°F (-78°C), a liquid when dissolved in water under constant pressure, or as a gas (at atmospheric pressure).

Very few people realize how common CO_2 is in materials we use every day. The bubbles in a carbonated drink are CO_2 and dry fire extinguishers often use CO_2. Other uses:

- Food and beverages
- Pulp and paper
- Water and wastewater treatment
- Welding and metal fabrication
- Pharmaceutical and biotechnology
- Oil and gas

- Health care
- Heat removal/cooling

CO_2 is a non-combustible gas that *can* be toxic to people and animals exposed to large quantities of accumulated CO_2 gas in a confined area. The CO_2 gas is heavier than air; therefore, it can fill low spaces; breathing high concentrations of the gas can cause asphyxiation. Due to the high levels of CO_2 required to cause harm, concentrations of CO_2 are often expressed as a percentage of the gas in air by volume—parts per million (ppm).

Next Generation Science Standards

Performance Standard.5-PS1-3:

Make observations and measurements to identify materials based on their properties.

Disciplinary Core Idea

Measurements of a variety of properties can be used to identify materials.

Science and Engineering Practices

Make observations and measurements to produce data to serve as the basis for evidence for an explanation of a phenomenon.

Crosscutting Concepts

Standard units are used to measure and describe physical quantities such as weight, time, temperature, and volume.

Materials

- 10-gallon aquarium (fish tank) or similar container that will not crack with very cold temperature (plastic containers work well and can be easily transported, but they need to be transparent)
- Towel for under plastic tray
- Plastic tray to fit inside aquarium (dry ice placed directly on glass may crack it; a container provides insulation)
- Bubbles and blowing wand
- Candle
- Matches or lighter
- Two hot pads (gloves also work) for safe handling of dry ice
- Several 1-pint water bottles filled with drinking water
- Balloons
- Ice pick
- Plastic bag to cover work surface
- 5 to 12-lb block of dry ice (refer to safe handling guidelines: *https://www.continentalcarbonic.com/dry-ice-safety.html*)

- Small Styrofoam ice chest for transporting dry ice
- Safety goggles (to be used when breaking dry ice)

Setup

1. Freeze caution: Warn students not to touch dry ice with bare skin.
2. Wearing safety goggles and gloves, use the ice pick to break off several ice cube size chunks of dry ice. Place a towel at the bottom of the aquarium or other container. Place the plastic tray on top of the towel, which will provide insulation so that the cold from the dry ice will not crack the glass.
3. Place the dry ice in the plastic tray.
4. Conduct demonstration in a setting that is not turbulent or breezy to allow the CO_2 to build up. Wait at least 15 minutes to allow for adequate CO_2 to accumulate. It may be helpful to break the dry-ice block into more pieces to increase the rate of sublimation. This experiment is not suitable for outside demonstrations.

Length of Lesson

This lesson will take one 45-minute class period.

Suggested Procedure

- Distribute the worksheet (Appendix 2.2.1); If you prefer, you can distribute the worksheet at the end of the demonstration.
- Explain to the students that dry ice is frozen CO_2. Ask: What happens when an ice cube (water) melts?

  It turns to water—or returns to its original state.

  Ask: Does anyone know where the CO_2 goes as the dry ice sits in the room and warms up? Do you see any liquid CO_2 drips?

  No, liquid CO_2 does not exist at atmospheric pressure. Frozen CO_2 sublimates directly to gas or returns to its original state.

  Can anyone see the CO_2?

  No, because it is a colorless, odorless gas.
- Explain that the pieces of dry ice have been sitting in the fish tank since before class began. Begin a discovery session with students by asking them: What is happening to the dry ice?

  *It is warming up and through the process of **sublimation** it is changing from a solid to a gas.*

The dry ice has been sublimating for at least 15 minutes so the tank is full of CO_2 gas. Can the students see it?

No, because CO_2 is a colorless gas.

But even though CO_2 gas is invisible (transparent to light), is there any way we can test to see if it is there?

Remember, CO_2 gas is heavier than air (air is mostly nitrogen and oxygen).

Ask the students what they think will happen if we blow bubbles full of regular air into the tank?

Common answers include, it will burst, it will sink, it will go up. The correct answer is that they will "float" on the interface between the room air and the CO_2 in the tank.

- Ask for a volunteer to gently blow bubbles into the tank. Avoid blowing down (or directly into the tank) as this will displace some of the CO_2 and minimize the effect. Bubbles will naturally try to sink but will gently bounce on the interface of room air and CO_2.
- Allow students to test the process by gently blowing bubbles into the tank.
- Another way to demonstrate properties of CO_2 is to place a small piece of dry ice into a clear water bottle half filled with water. (Use safety glasses and gloves when breaking the dry ice into small pieces.) Immediately after dropping the dry ice into the water bottle place a medium size balloon over the top. The CO_2 gas will blow up the balloon.
- A third activity to demonstrate properties of CO_2 is to light two small candles. Leave one on the desk in front of the fish tank and gently lower the other one into the fish tank with the dry ice. Ask students what they think will happen?

  The one on the desk will continue to burn because there is plenty of oxygen available to feed the fire. The one in the fish tank with the CO_2 will quickly extinguish because there is not enough oxygen for the candle to burn.

  Is CO_2 explosive?

  As we just demonstrated it is not explosive. In fact it is used in some commercially available fire extinguishers.

- Have students pour a transparent cup half full of drinking water. Using gloves, add a small cube of frozen CO_2. Ask students what will happen to the water after the dry ice sublimates?

  The water warms the frozen CO_2 causing it to form gas, which fizzes (CO_2 makes the fizz in carbonated beverages). As some of the CO_2 dissolves in the water, it forms a weak acid. If you drink the water, it tastes slightly tangy (like lemon), which is the taste of acid.

Teaching Strategy

Silent conversation, mind map, demonstration

Questions to Pose to Students:

The following questions are designed to encourage students to recall information or assumptions they have about CO_2.

- How important is CO_2 to your everyday life? Think of ways CO_2 might be used and list them on your worksheet.
- Is CO_2 dangerous or explosive?

  In low and moderate concentrations, CO_2 is not dangerous to people. It is a naturally occurring gas necessary to live on earth, and it is present in this room right now. However, CO_2 collects in low places and displaces lighter oxygen.
- What properties of CO_2 make it one of the greenhouse gases?

Activity

- Encourage students to observe and record carefully. Appendix 2.2.1
- Teacher performs demonstration
- Encourage students to list properties of CO_2 as they recognize or become aware of them.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

- Participation in the activity.
- Completed *"What is Carbon Dioxide?"* workshee, Appendix 2.2.1

Extensions

Have students research uses for CO_2 as homework. Encourage students to find ways CO_2 is used in residential, industry, and medical settings. Students should make a list and bring the list to class.

Properties of Carbon Dioxide Demonstration

List CO_2 properties observed:	**Drawing or illustrations:**
Questions I have:	

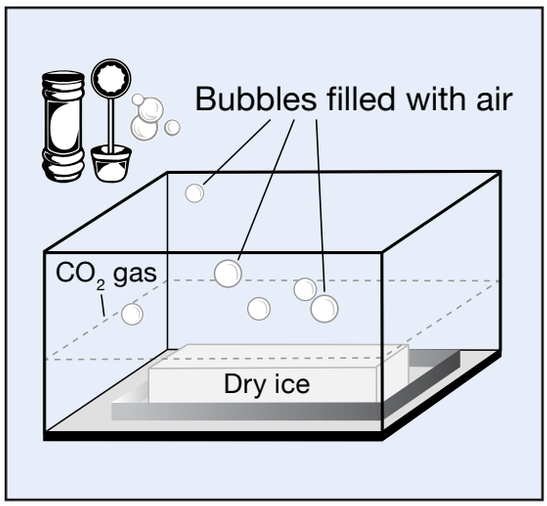
Uses of CO_2:

Properties of Carbon Dioxide Demonstration – Teacher Guide Answer Sheet

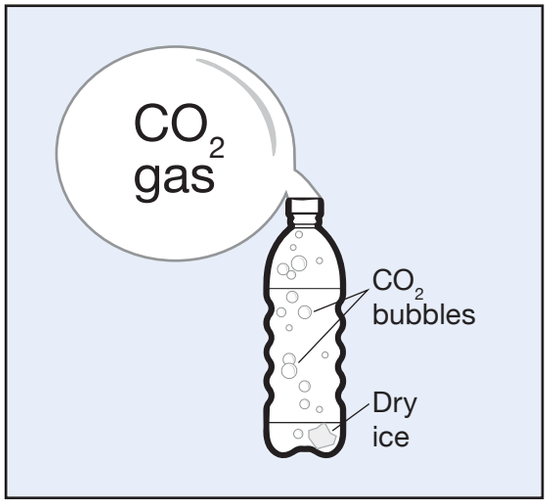
List CO_2 properties observed:	**Drawing or illustrations:**
Colorless Odorless Takes up space Can be gas or solid Really cold as a solid Sublimates Heavier than air	
Questions I have:	

Uses of CO_2:

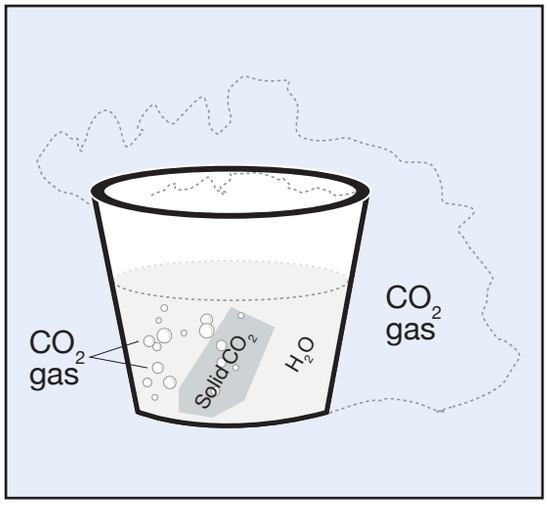
- Food and beverages
- Pulp and paper
- Water and wastewater treatment
- Welding and metal fabrication
- Pharmaceutical and biotechnology
- Oil and gas
- Health care
- Heat removal/cooling



Carbon dioxide gas collects at the bottom of the container and bubbles rest on the interface between room air and CO_2.



As dry ice sublimes, gas expands and fills the balloon.



Making carbonated water with dry ice.

Lesson 1

The Greenhouse Effect

Quick Overview

Lesson Scope: Greenhouse gases and their impact on the atmosphere

Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Meter sticks, Fahrenheit/Celsius thermometers, stakes, duct tape, mason jars, flasks

Materials Needed per Student

Meter (yard) sticks

Thermo-meters

Stakes

Duct tape

Mason jars

Flasks

Each student will need paper and instructions to record trials.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect global temperature?

Students will gain an understanding of how greenhouse gases interact in the **atmosphere** by investigating how both the atmosphere and the Earth's surface are affected by sunlight and **infrared radiation**.

This Lesson Is Composed of Three Activities

Activity 1: How the Atmosphere Is Heated

Activity 2: The **Greenhouse Effect**

Activity 3: How Greenhouse Gases Work—It's All About the Cooling

Key Understandings

- The Earth's atmosphere is a layer of gases surrounding the Earth that are retained by gravity.
- The atmosphere protects and allows for life on Earth by absorbing **ultraviolet solar radiation**, warming the surface through the greenhouse effect, and reducing temperature extremes between day and night.
- The greenhouse effect is a process by which **thermal radiation** (heat) from a planetary surface is absorbed by atmospheric greenhouse gases and is re-radiated in all directions. Part of this re-radiation is directed back toward the surface of the Earth,

which results in an elevation of the average surface temperature above what it would be in the absence of gases.

- Models (such as the **greenhouse gas** model) are helpful in explaining how the atmosphere works but can only explain part of the story—it is important to understand that these models break down in their explanatory effectiveness. (See Next Generation Science Standards [NGSS] standard and common misconceptions below).
- The rate at which different greenhouse gases cool is a key factor in the greenhouse effect.

Objectives

- Students will compare temperature readings on the surface and above the ground (in the air).
- Students will make predictions and write an explanation based on a model of the greenhouse effect.
- Students will be able to determine how greenhouse gases affect cooling and heating rates.

Vocabulary

- Atmosphere
- Infrared radiation
- Greenhouse effect
- Ultraviolet solar radiation
- Thermal radiation
- Greenhouse gas
- Radiation
- Convection
- Conduction
- Metadata
- Stratosphere
- Troposphere

Background Information for the Teacher

The Earth's Atmosphere

Whether or not a planet retains an atmosphere over geologic time depends on the mass of the planet, the radius of the planet, the temperature of the atmosphere, and the composition of the atmosphere. The atmospheres of the Earth-like planets (Mercury, Venus, Earth, and Mars) originated from the gases associated with volcanism. On Earth, any hydrogen or helium that comes from volcanoes can *easily* escape the Earth. Hence, we have very little hydrogen and helium in our atmosphere. On the other hand, any atom or molecule with an atomic mass greater than 20 units (mass = 20 units) (i.e., neon), will be retained by the Earth because the mean speed of the atom is less than the escape velocity. Gravity is a major determinant of the stability of a planet's atmosphere.

For massive planets, gravity overcomes the effects of thermal agitation and a molecule cannot escape the gravity field entirely. On the other hand, for a small planet like Mercury or the moon, the gravity is small and the random thermal velocity that an atom has will send it zinging into interplanetary space.

- To summarize, Earth has more atmosphere than most of the other inner planets for two reasons: (1) gravity and (2) volcanism. Being the most massive of all of the inner planets, Earth can hold more atmosphere with its gravity alone. Furthermore, volcanoes continually release gases into the Earth's atmosphere at a rate faster than they escape into space. Unlike Venus, however, which has an even thicker atmosphere, Earth also has oceans where gases in the atmosphere are stored, and gases can also be precipitated (or crystallized) into rocks. *https://scijinks.gov/atmosphere-formation*.

The Greenhouse Effect

The Earth's surfaces and atmosphere get most of their energy from the sun. Energy from the sun travels to the Earth in the form of electromagnetic **radiation**, primarily visible light. The Earth's atmosphere is mostly transparent to visible light, so the light travels through the atmosphere without heating it up. Visible light hits the Earth's surface where some of it is absorbed, resulting in heating of the surface.

The warmer surface then transfers energy into the cooler atmosphere in the form of infrared radiation. Greenhouse gases in the atmosphere absorb some of the infrared radiation and transfer some of the heat back onto the Earth's surface after it has cooled down during the night. This causes the Earth's surface and atmosphere to have a slower cooling rate than if there were no greenhouse gases present. Over time, the addition of more greenhouse gases in the atmosphere results in a higher average temperature.

Since the start of the Industrial Revolution, people have been adding substantial amounts of CO_2 and other greenhouse gases (such as methane, nitrous oxide, and fluorinated gases) to the atmosphere. Greenhouse gases trap heat in the atmosphere, which makes the Earth warmer (EPA, 2018).

Different types of greenhouse gases have different impacts on the climate, depending on such factors as how much of the gas is produced, how long it stays in the atmosphere, and how much heat it traps. "Carbon dioxide equivalent" (or CO_2e) is a unit of measurement that allows the effects of different greenhouse gases to be compared using CO_2 as a standard unit for reference (EPA, 2018).

The primary sources of greenhouse gas emissions in the United States (EPA, 2013) are:

- **Electricity production**
  Electricity production generates the largest share of greenhouse gas emissions in

the United States. Around 65% of our electricity comes from burning fossil fuels, mostly coal (30%) and natural gas (34%).

- **Transportation**
  Greenhouse gas emissions from transportation primarily come from burning fossil fuels for cars, trucks, ships, trains, and airplanes.

- **Industry**
  Greenhouse gas emissions from industry primarily come from burning fossil fuels for energy, as well as from certain chemical reactions necessary to produce goods from raw materials.

- **Commercial and residential**
  Greenhouse gas emissions from businesses and homes arise primarily from fossil fuels burned for heat, the use of certain products that contain greenhouse gases, and the handling of waste.

- **Agriculture**
  Greenhouse gas emissions from agriculture come from livestock (e.g., cows), agricultural soils, and crop production (including transport, production, harvest, etc.).

- **Land use and forestry**
  Plants and soil can act as an overall "carbon sink" (absorbing CO_2 from the atmosphere) or as a source of greenhouse gas emissions during cellular respiration or when harvested and burned. In some countries, large amounts of forest are being burned or cut down, which adds CO_2 emissions to the atmosphere.

Note: *See the Gas Emissions pie charts and graphs (Appendix 2.3.2) for comparisons and trends.*

Common Student Misconceptions/ Student Challenges

General misconceptions about the greenhouse effect include:

- Linking global climate change and greenhouse gases to stratospheric ozone depletion.
- Confusing air pollutants with greenhouse gases.
- Not knowing that greenhouse gases are made up of water vapor, carbon dioxide, methane, nitrous oxide, and ozone.
- Recognizing that solar rays are involved in the greenhouse gas effect but not realizing terrestrial radiation is also a factor.

One of the biggest misconceptions is that students may think that the greenhouse effect works in the same way a greenhouse does. In other words, students may not understand how the model of a greenhouse demonstrates only a part of this system. For example, glass in a greenhouse does not allow infrared radiation to escape, while greenhouse gases in the atmosphere absorb infrared radiation, then re-emit some of the infrared radiation back to the Earth's surface and some into space. A greenhouse for plants and the

greenhouse effect for the Earth work in very different ways. Using the same term for phenomena that are actually quite different could be confusing.

A common misconception is that the actual CO_2 molecules are adding more heat to the Earth's surface. Most of the energy added to the Earth's surface comes from the sun. CO_2 is merely slowing the energy released from the Earth's surface back into space at night, which prevents the necessary cooling of the planet. Solar radiation passes through the Earth's atmosphere, hits the Earth, is converted to heat energy, and is radiated back to the atmosphere as infrared radiation. Some of that radiation is trapped by greenhouse gases and reradiated back down to the Earth, making the planet warm.

Next Generation Science Standards

Performance Standard.HS-ESS2-4:

Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate.

Disciplinary Core Idea

The foundation for Earth's global climate systems is the electromagnetic radiation from the sun, as well as its reflection, absorption, storage, and redistribution among the atmosphere, ocean, and land systems, and this energy's re-radiation into space. Changes in the atmosphere caused by human activity have increased carbon dioxide concentrations and thus affect the climate.

Science and Engineering Practices

Use a model to provide mechanistic accounts of phenomena.

Crosscutting Concepts

Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

General Setup

Plan to spend close to four hours on this lesson.

Full Length of Lessons

The lesson may be conducted over five to six 45-minute class periods.

Safety Materials

- Safety goggles
- Gloves
- Tongs

Activity 1 – How the Atmosphere Is Heated

Materials

- 8 meter (yard) sticks
- 8 Fahrenheit/Celsius thermometers
- 8 stakes (at least 4 feet long)
- Duct tape
- Activity 1 Student Worksheet (Appendix 2.3.1)
- Activity 1 Data Collection Worksheet (Appendix 2.3.1)
- "How the Earth's Atmosphere Heats Up" PowerPoint presentation

Activity 1 – Setup

Copy enough Activity 1 data collection and student worksheets (Appendix 2.3.1) for each student

Activity 1 – Length of Lesson

This activity will require approximately two 45-minute class periods.

Activity 1 - Suggested Procedure

Introduction (formative assessment): Have students formulate answers to the following questions and write their answers in a journal, discuss their answers in small groups, or discuss as a class.

Imagine you are traveling from sea level into the mountains. As you increase in elevation does the temperature increase or decrease? Is it colder in the mountains or at sea level?

As you move closer to a hot object, what happens in terms of the amount of heat you feel? Think about sitting closer or further away from a campfire. However, we also know that the air temperature at 30,000 feet (9,144 m) above ground (a common altitude for a commercial jet) is about –29°F (–34°C). How is this possible?

Activity 1 – Day 1

- Tape two meter (yard) sticks together so that they are equivalent to 2

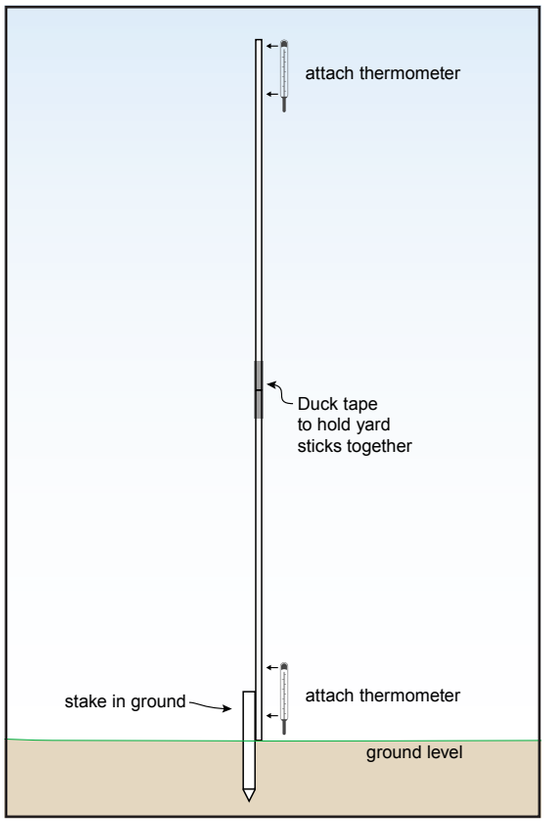


Figure 1. Activity 1 setup.

meters in length. This may require reinforcement in the form of a ruler or other piece of rigid material to support the end to end yard sticks.

- Tape a thermometer at the top and bottom of the meter sticks.
- Pick four outdoor locations around your school. Two of the locations should be in the shade most of the day. The other two should be exposed to sunlight most of the day.
- Drive stakes into the ground and tape the meter stick setup to the stakes so that students can still read the thermometers.

1. Explain to your students that you will be setting up an investigation to demonstrate how the sun warms the Earth. To do this, students will be taking temperature readings at four locations (Appendix 2.3.1 Data Collection Worksheet). At each location, there will be a thermometer at ground level and another thermometer two meters above the ground. Temperature readings will be taken every hour to gather data on how temperatures change throughout the day.

   Note: *It is important to get temperature readings every hour throughout the day. There are a variety of ways data can be collected. Depending on class needs, each class could collect the data, a select group of students could gather data, or the teacher could collect the data when students are not available.*

   On the reverse side of the data collection worksheet, have students predict which thermometer they think will record a warmer temperature—the thermometer on the ground or the one 2 meters above the ground. Ask students to define the type of energy transfer occurring on the ground and in the air (**convection**, radiation, or **conduction**). After they have answered this, ask them if they think the time of day will make a difference. Discuss why they think this.

2. Lead a class discussion in which students share their thoughts about the questions they just answered. Be sure to acknowledge all responses and have students explain their reasoning. At this point, answers are not to be evaluated as correct or incorrect but by the thoroughness of their reasoning.

   Note: *This question set can be used as a pre-assessment to determine what your students understand about sun, Earth, and atmosphere interactions. This also serves as a prediction for the investigation.*

3. Make sure students are well-versed in reading thermometers. If they are not, it would be worth the time to review how to read a thermometer before continuing to the next step. Refer to *How to Read a Thermometer* in the supplemental materials for further explanation.

4. At the designated data collection times, accompany the students as they record the temperatures on both thermometers. Students should also record the **metadata** (data about the data), including whether or not the setup is exposed to sunlight

and other conditions such as wind, precipitation, and cloud cover.

5. After students have returned to the classroom, have them reevaluate their hypothesis. Students can write a new hypothesis on the data worksheet.

6. Have students collect temperature data for each location throughout the day.

Activity 1 – Day 2

Students should be able to explain that ground temperatures in the morning warmed up first and the air above them warmed second. This was because the lower atmosphere is not warmed by the sun but by the Earth's surface. Later in the day, air temperatures and ground temperatures should be approximately the same. Temperature readings may not fit exactly into this model. Ground temperatures and air temperatures will be approximately the same in areas that are shaded or cloud covered. Winds have a tendency to mix air. In some cases a warm pocket of air can move over the cool ground, making it seem like the atmosphere was warmed before the surface.

1. Setup: "How the Earth's Atmosphere Heats Up" PowerPoint.

2. Display temperature data recorded from every hour on day 1. If students do not have these data recorded in their tables they should do so now.

3. Have students evaluate their hypothesis. Was their hypothesis supported by their data?

4. Present the PowerPoint "How the Earth's Atmosphere is Heated."

5. Have students develop explanations regarding the data collected based on the information they gained in the PowerPoint.

Activity 2 – The Greenhouse Effect

Activity 2 – Materials

(Supplies needed for each student group)

- 2 mason jars or flasks and 1 lid per student group
- 2 Fahrenheit/Celsius thermometers that fit inside the mason jars
- What is the greenhouse effect website at http://climatekids.nasa.gov/greenhouse-effect/

Activity 2 – Setup

Find a sunny spot outside or near a window to set up the jars. If there is no direct sunlight, an incandescent lamp can be used as a substitute.

Activity 2 – Length of Lesson

This activity will require approximately one 45-minute class period.

Activity 2 – Sugested Procedure

Now that students understand how the atmosphere is heated, they will explore an important aspect of the atmosphere, the greenhouse effect.

1. Review the main points from Activity 1:

   - The upper atmosphere, or **stratosphere,** heats up as the sun's radiation is absorbed by ozone gas.

     The Earth's surface is heated by the sun's radiation. Then energy from the Earth's heated surface transfers heat to the lower atmosphere through radiation, conduction, and convection.The lower atmosphere, or **troposphere**, where humans live and all weather processes take place, does not heat up very much from the sun's radiation.

   - This means the troposphere is heated from the bottom to the top.

2. Once students have a good understanding of how the atmosphere is heated, pose the following formative assessment question:

   Imagine you are stuck inside a car on a hot, sunny day. All the windows are rolled up, and the car is sitting directly in the sun. After a couple of hours, what will the temperature be like in the car compared to the temperature outside? Explain the reasoning behind your answer.

   The temperature in the car would be hotter than the outside temperature. This is because radiant energy from the sun is transmitted through the windows of the car, but the infrared radiation from the solid objects in the car is not easily transmitted out the windows.

   Note: *Students can discuss this question as a class, in pairs, or write it down on a piece of paper. The important part is that the teacher identifies the students' preconceptions and addresses any misconceptions by the end of the unit.*

3. Arrange students in groups of two to three. Give each student group two flasks, one lid, and two thermometers.

4. Tell students they will be placing the jars with thermometers out in the sunlight. However, only one jar will have a lid.

5. Have students predict which jar will get warmer in the sunlight. Students should also be able to show the reasoning behind their prediction.

   Note: *There are various ways to have students conduct this formative assessment; one idea would be to use think-pair-share where pairs of students discuss their reasoning behind the prediction, then share with the rest of the class.*

6. Have students place their jars under a light or in direct sunlight for 20 minutes.

7. During this time lead the students through the website "What is the greenhouse effect?" at http://climatekids.nasa.gov/greenhouse-effect/.

   Note: *There are multiple strategies to having students decipher the information from the "What is the greenhouse effect?" website. This website breaks down the greenhouse effect into the following questions.*

What is a greenhouse?

How is the Earth a greenhouse?

Is it warm in here or is it just me?

Why can't we just plant more trees?

Don't clouds keep the Earth cooler?

Or do clouds make Earth warmer?

Note: *One strategy would be to read the information as a class or have students read the information independently. Then have students formulate answers to the six questions above. Students can share these answers in a discussion or write them down in science journals.*

8. After 20 minutes, students should measure the temperatures of each jar and record the temperature on a piece of paper.

9. Embedded assessment—Have students raise their hand if their jar without a lid had a higher temperature than the one with a lid. Most students should not raise their hands. If some did raise their hands, ask the class what might have been different with the student's investigation that resulted in different data. Some factors students might come up with are human error in reading the thermometer, mixing up the data, faulty thermometer, or the jar with the lid being in the shade.

10. Have students write explanations regarding why they think the temperature in the jar with a lid is higher than the one without a lid. Have them use evidence to back up their claims.

11. Tell students that greenhouse gases in the lower atmosphere allow solar radiation to pass through them to heat up the Earth's surface. The heated Earth's surface then emits infrared radiation back to the atmosphere. Infrared radiation is absorbed by greenhouse gases, which are heating up the atmosphere. Based on the investigation with the jars, have students compare and contrast how the atmosphere would heat up without greenhouse gases in it.

Possible Answer: *Without greenhouse gases in the Earth's atmosphere, solar radiation would still heat the Earth's surface. The heated surface would still emit infrared radiation back into the atmosphere. Without greenhouse gases to absorb the infrared radiation the atmosphere would quickly cool off, making air temperatures on Earth colder.*

Activity 3 – How Greenhouse Gases Work—It's All About Cooling

Activity 3 – Materials

(Supplies needed for each student group)

- 3 Erlenmeyer flasks
- Beakers big enough to hold the flasks
- Hot plate
- 10 ml water
- 3 Fahrenheit/Celsius thermometers
- Three rubber stoppers with a hole for the thermometer

- Stopwatch
- ~50 g dry ice

Activity 3 – Setup

- Print enough Activity 3 worksheets for each student to have one (Appendix 2.3.3).
- Set up as many stations as supplies allow. Students working in groups of two are ideal.
- Each station should have a hot plate, a beaker, three flasks, three stoppers to fit in the flask, and three thermometers.
- The first flask should be empty, the second flask should be filled with 10 ml of water, and the third flask should have 5 g of dry ice in it.
- Insert a stopper into each flask and then insert a thermometer into the stopper about 10 cm so that the bottom of the thermometer sits in the middle of the flask.
- After inserting the stopper, let the dry ice completely sublimate before starting. The dry ice should be gone and the flask should be clear.
- Ensure the temperatures in the flasks are approximately the same before allowing students to begin.
- At each station, fill a beaker about half full of water and put it on the hot plate. Students will use the beaker as a water bath to regulate the temperature when heating their flask.

Activity 3 – Length of Lesson

This activity will require approximately two 45-minute class periods.

Activity 3 – Suggeted Procedure

Safety note: *Students should be wearing safety glasses. Students should also use gloves or tongs to move the hot flasks, while taking extra caution with the hotplates. Remind students that hot gases expand. If the flask remains on the burner longer than a minute, the stopper could shoot out or cause the flask to explode.*

Students will be heating flasks containing different types of greenhouse gases to determine how greenhouse gases affect cooling and heating rates.

1. Teachers should set up the materials for Activity 3 before class. Start heating the water baths at the beginning of class. Have flasks full of air, water vapor, and carbon dioxide, with a stopper and thermometer inserted beforehand.

- Students should be familiar with the greenhouse effect from Activity 2. Show students the one or more of these videos to introduce them to Activity 3. https://illinois.pbslearningmedia.org/resource/phy03.sci.phys.matter.greenhouse2/global-warming-the-physics-of-the-greenhouse-effect/#.WzOpXC2ZMUG or https://www.c2es.org/content/climate-basics-for-kids/

2. Have students record their hypotheses regarding which flask will become the hottest: the one with air, water vapor, or carbon dioxide. Students can record their hypotheses on their Activity 3 – Student Worksheet.

3. The teacher should circulate and verify that all flasks are set up properly and safety equipment is being used.

4. Students should follow the procedure in the Activity 3 student procedure worksheet (Appendix 2.3.3). Once students have completed the investigation they should analyze and graph their data on the data table and graphs worksheet (Appendix 2.3.3).

5. Lead a group discussion in which the students share their data. Did it match their hypotheses? Why or why not?

6. If you need, you can tell students that the greenhouse gases absorb the heat in the flask, then release it later, causing those flasks to cool at a slower rate. In the atmosphere, the greenhouse gases slow the cooling of the Earth's surface. This creates warmer nights and leftover heat the next day when the sun starts heating the Earth's surface again. More greenhouse gases results in higher average temperatures.

7. Additionally, students can read "The Greenhouse Effect" excerpt in Appendix 2.3.3 to help them understand the reason the various gases in the flask reacted the way they did.

Final Assessment

Have students make sense of the greenhouse effect by posing (and having them answer in writing) the following scenario:

The planet Mars has very little atmosphere. The highest temperatures there reach about 70°F (21°C), while the lowest temperatures reach –225°F (–143°C). The planet Venus has a very thick atmosphere composed of a lot of greenhouse gases. The day and night temperatures on Venus stay relatively the same at 864°F (462°C). Based on your experiences in Activity 3, use your results to explain why the temperatures on Mars are colder and vary a lot, while the temperatures on Venus are hotter and do not vary. How does this relate to what you have learned about Earth's atmosphere and why we are concerned with the rise in carbon dioxide? Make at least three to four claims and back them up with evidence from these activities. Be sure to address the concern for the rise of carbon dioxide in the atmosphere. (See Appendix 2.3.4 for the CO_2 Assessment Rubric.)

Supplemental Materials and References

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Appendix 2.3.1 | Activity 1 – Data Collection Worksheet

Day 1

Time	Location			
	1	2	3	4
8 am				
Metadata				
9 am				
Metadata				
10 am				
Metadata				
11 am				
Metadata				
12 am				
Metadata				
1 pm				
Metadata				
2 pm				
Metadata				
3 pm				
Metadata				

Activity 1

1. What was your hypothesis from day 1 as to which thermometer would record a warmer temperature — the thermometer on the ground or the one two meters above the ground.

2. Did you think the time of day would make a difference? Why?

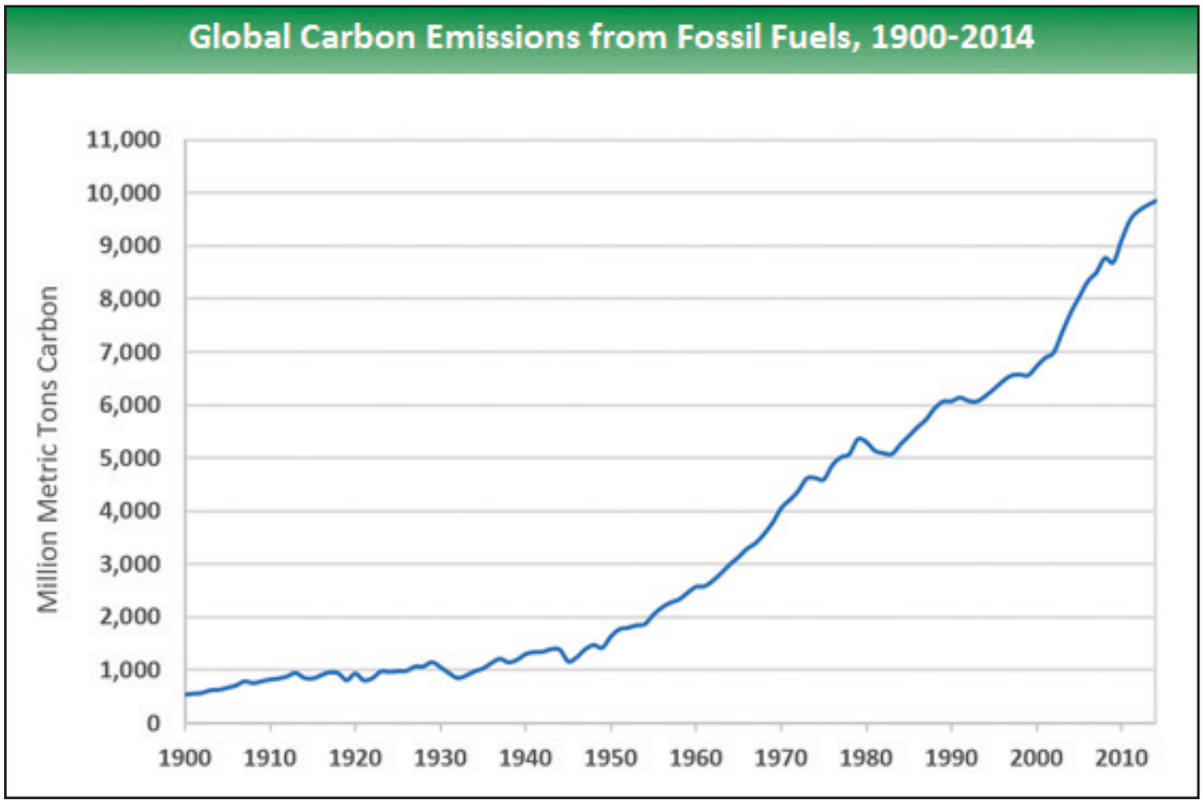
3. If your collected data changes your prediction, develop a new hypothesis.

4. How is thermal (heat) energy being transferred at each thermometer?

5. After your class records the first temperature readings, reevaluate your hypothesis. Do you still stand by it? If so, state why. If not, develop a new hypothesis.

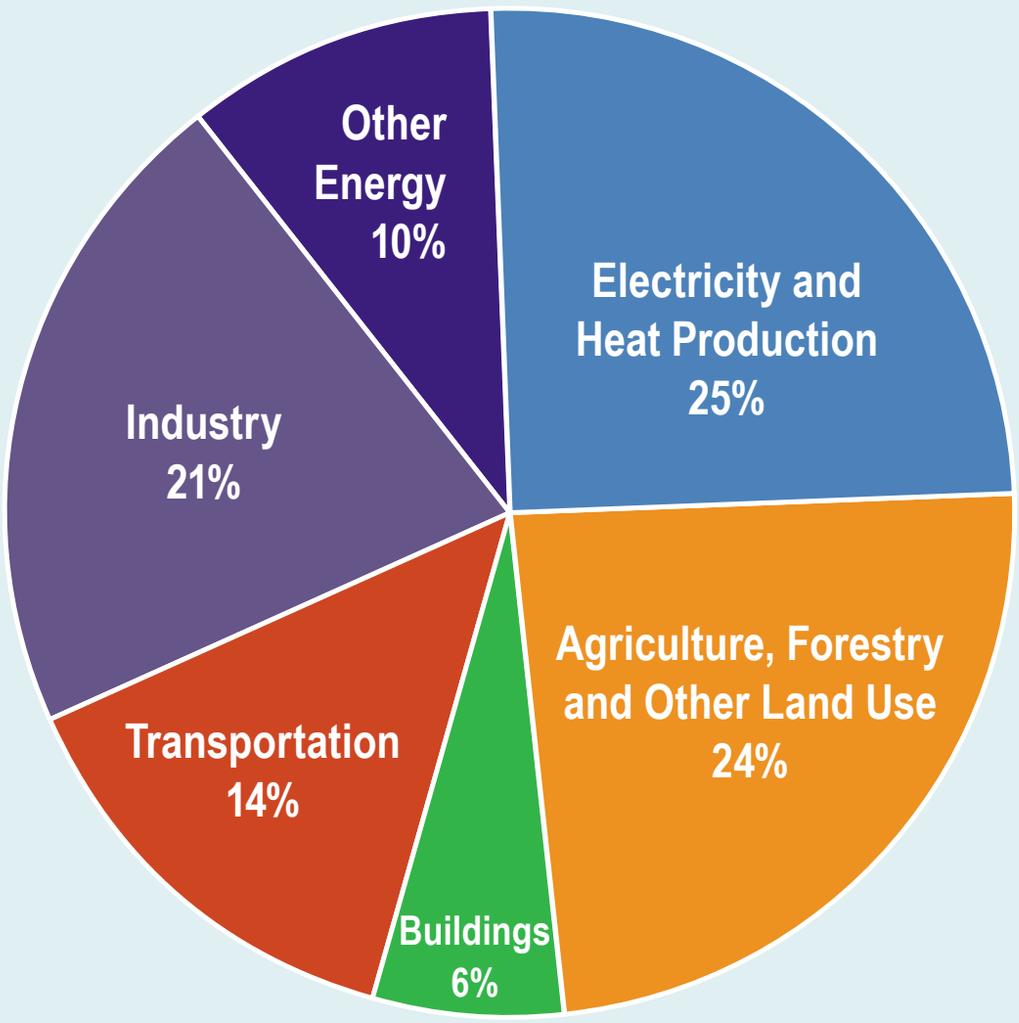
6. After examining a full day's worth of temperature data, determine if your hypothesis was correct (was supported). Explain why or why not.

7. Explain the temperature data collected on Day 1 based on the information you have gained from the PowerPoint. Use your data to explain how the atmosphere around your school was heated on the day you took the temperature reading.

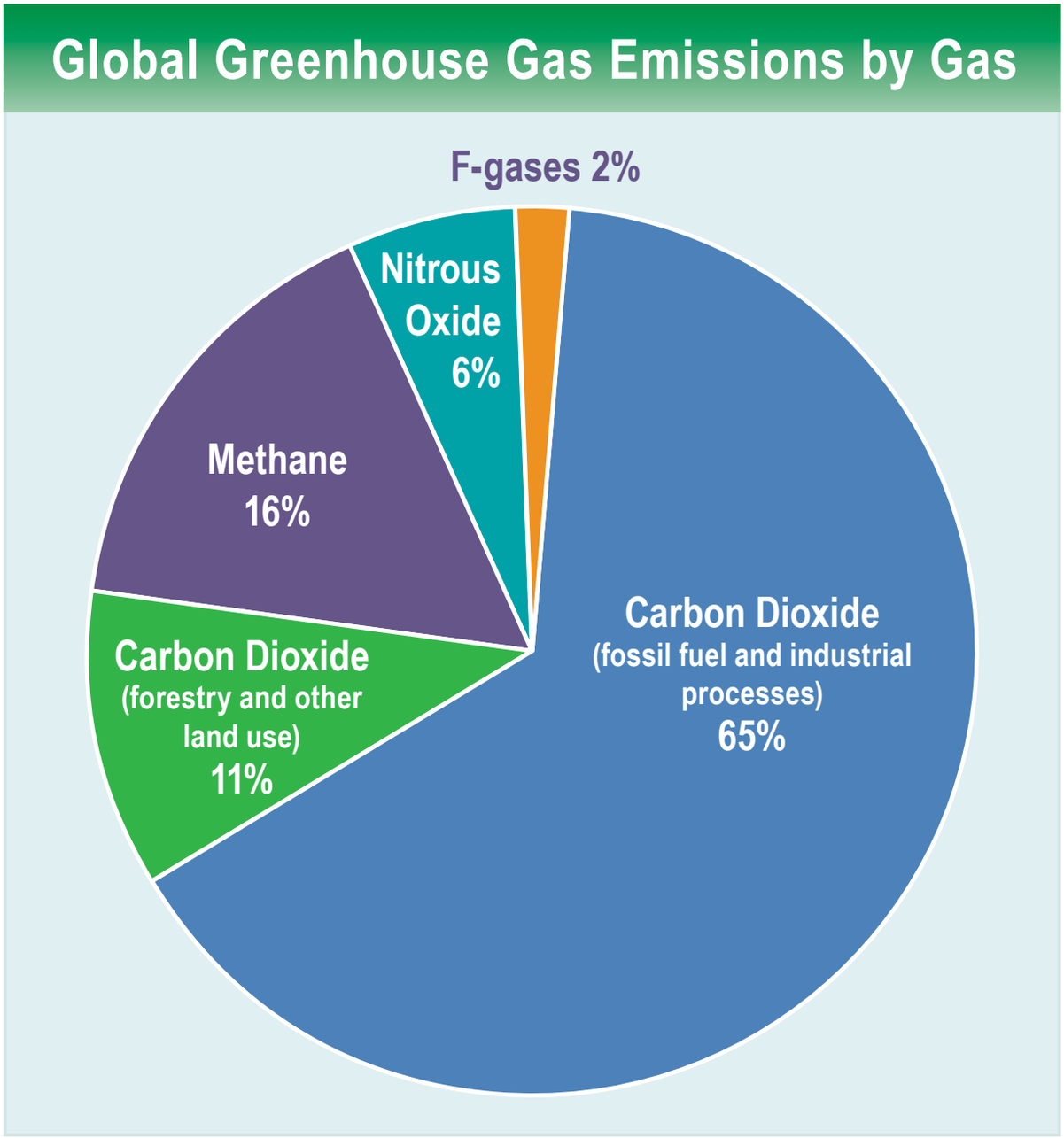


Source: Boden, T.A., Marland, G., and Andres R.J. (2015). Global, Regional, and National Fossil-Fuel CO2 Emissions. Carbon Dioxide Information Analysis Center, Oak Ridge National Laboratory, U.S. Department of Energy, doi 10.3334/CDIAC/00001_V2015. Host Website: www.epa.gov. Source data link: www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data

Global Greenhouse Gas Emissions by Economic Sector



Source: IPCC (2014); based on global emissions from 2010. Details about the sources included in these estimates can be found in the Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.Host Website: www.epa.gov. Source data link: https://www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data#Trends



Source: IPCC (2014); based on global emissions from 2010. Details about the sources included in these estimates can be found in the Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.Host Website: www.epa.gov. Source data link: https://www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data#Trends

(F-gases = flourinated gases)

Student Procedure

1. Ensure the experiment is set up properly. Water in the water bath should be at a low boil. Three flasks labeled air, water vapor, and carbon dioxide should be set up with a stopper and thermometer inserted in each of them. All flasks should be close to the same temperature. If they are not, give them more time to become equalized.
2. Once the flasks are equalized, record the initial temperature for each in the data table (Appendix 2.3.3 - Data Table).
3. Place the flask of air into the water bath and start a stopwatch.
4. After one minute, remove the flask with air and record the temperature.
5. At the same time as Step 4, insert the flask of water vapor into the water bath.
6. After one more minute, remove the flask with water vapor from the water bath, and record the temperatures of both flasks in the data table.
7. At the same time as Step 6, insert the flask of CO_2 into the water bath.
8. After one more minute, remove the flask marked CO_2 and record the temperature of all three flasks in the data table.
9. At this point, you can turn off the hotplate. Be sure not to put any flask back into the water bath.
10. Continue to record the temperatures of each flask every minute until all 15 minutes are filled in on the data table.

Data Analysis

11. Create a graph showing the cooling for each flask under the data table.

Calculate the cooling rates for each flask. The cooling rate is calculated by dividing the change in temperature by the change in time.

Example — Cooling Rates for Air

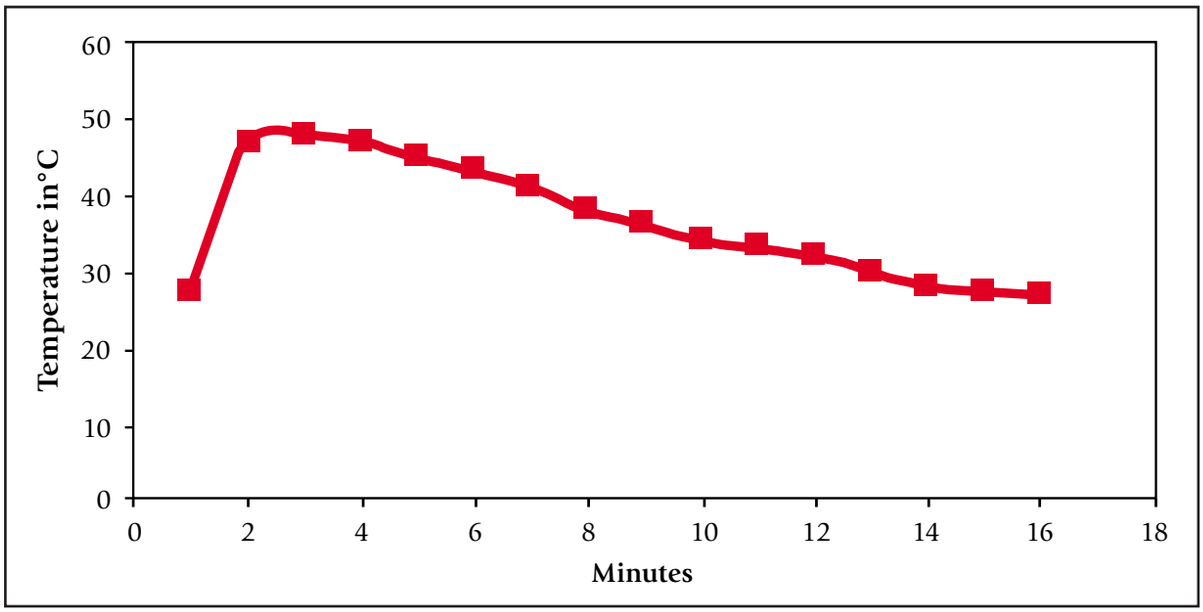
Note that the beginning of the graph shows an initial warming trend. Next, look for a segment in the graph where the line is fairly straight, showing cooling. This means the cooling is happening at a fairly steady rate. In this example, you could start at 4 minutes and go to 8 minutes.

Calculating the cooling rate.

4 minutes at 45°C, and 8 minutes at 37°C. Equation for cooling rate temperature change divided by change in time.

(45°C – 37°C) ÷ (8 minutes – 4 minutes) = cooling rate

8°C ÷ 4 minutes = 2°C per minute



Example chart

12. Calculate the cooling rates for each of your flasks based on the above example.

Air _____ **Water Vapor** _____ **CO_2** _____

13. How might the cooling rates of various gases affect the atmosphere?

14. Calculate the average temperatures for each flask by adding up all of the temperatures except the initial temperature. Divide the sum of the temperatures by 15.

Air _____ **Water Vapor** _____ **CO_2** _____

15. Predict what would happen to the average temperature if each flask were only allowed to cool for 5 minutes before being inserted into the water bath for another minute.

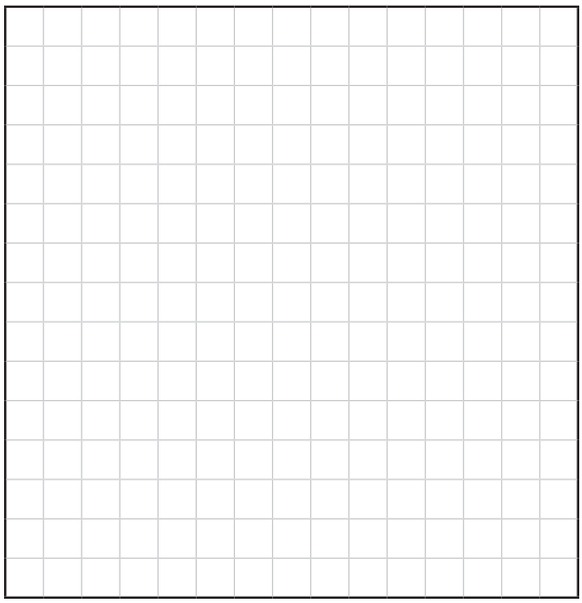
16. Imagine the same scenario as in question 14 or 15 only the cycle of heating for 1 minute and cooling for 5 minute continues 10 times. What would happen to the average temperatures for each flask?

Activity 3 Data Table

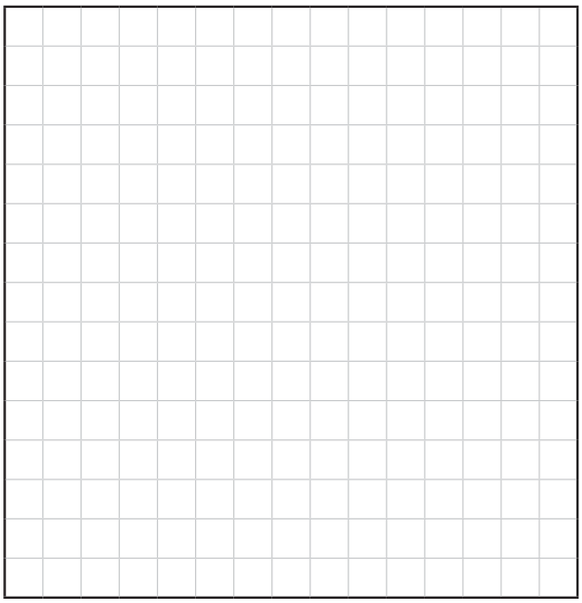
Air	Temp	Water Vapor	Temp	CO_2	Temp
Initial Temperature		Initial Temperature		Initial Temperature	
1 Min		1 Min		1 Min	
2 Min		2 Min		2 Min	
3 Min		3 Min		3 Min	
4 Min		4 Min		4 Min	
5 Min		5 Min		5 Min	
6 Min		6 Min		6 Min	
7 Min		7 Min		7 Min	
8 Min		8 Min		8 Min	
9 Min		9 Min		9 Min	
10 Min		10 Min		10 Min	
11 Min		11 Min		11 Min	
12 Min		12 Min		12 Min	
13 Min		13 Min		13 Min	
14 Min		14 Min		14 Min	
15 Min		15 Min		15 Min	

Activity 3 Data Table and Graphs

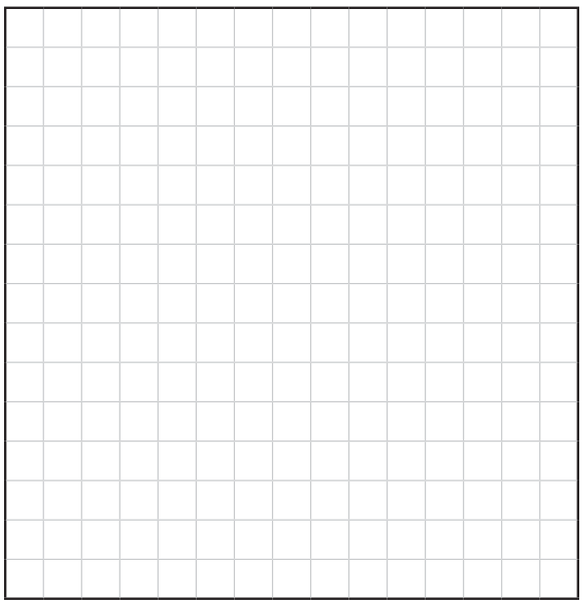
Air Cooling Trend



Water Vapor Cooling Trend



CO_2 Cooling Trend



Activity 3 – Excerpt from The Greenhouse Effect from the University Corporation for Atmospheric Research (UCAR)

Greenhouse Gases

Carbon dioxide (CO_2) is one of the greenhouse gases. It consists of one carbon atom with an oxygen atom bonded to each side. When its atoms are bonded tightly together, the carbon dioxide molecule can absorb infrared radiation and the molecule starts to vibrate. Eventually, the vibrating molecule will emit the radiation again, and it will likely be absorbed by yet another greenhouse gas molecule. This absorption-emission-absorption cycle serves to keep the heat near the surface, effectively insulating the surface from the cold of space.

Carbon dioxide, water vapor (H_2O), methane (CH_4), nitrous oxide (N_20) and a few other gases are greenhouse gases. They all are molecules composed of more than two component atoms, bound loosely enough together to be able to vibrate with the absorption of heat. The major components of the atmosphere (N_2 and O_2) are two-atom molecules too tightly bound together to vibrate and thus they do not absorb heat and contribute to the greenhouse effect.

Greenhouse Effect

Atmospheric scientists first used the term 'greenhouse effect' in the early 1800s. At that time, it was used to describe the naturally occurring functions of trace gases in the atmosphere and did not have any negative connotations. It was not until the mid-1950s that the term greenhouse effect was coupled with concern over climate change. In recent decades, we often hear the term greenhouse effect used in somewhat negative ways. The negative concerns are related to the possible impacts of an enhanced greenhouse effect. It is important to remember that without the greenhouse effect, life on earth as we know it would not be possible.

While the Earth's temperature is dependent upon the greenhouse-like action of the atmosphere, the amount of heating and cooling are strongly influenced by several factors just as greenhouses are affected by various factors.

In the atmospheric greenhouse effect, the type of surface that sunlight first encounters is the most important factor. Forests, grasslands, ocean surfaces, ice caps, deserts, and cities all absorb, reflect, and radiate radiation differently. Sunlight falling on a white glacier surface strongly reflects back into space, resulting in minimal heating of the surface and lower atmosphere. Sunlight falling on a dark desert soil is strongly absorbed, on the other hand,

and contributes to significant heating of the surface and lower atmosphere. Cloud cover also affects greenhouse warming by both reducing the amount of solar radiation reaching the earth's surface and by reducing the amount of radiation energy emitted into space.

Scientists use the term albedo to define the percentage of solar energy reflected back by a surface. Understanding local, regional, and global albedo effects is critical to predicting global climate change.

CO_2 Assessment Rubric

Category	4	3	2	1
Depth of Understanding	Explanation uses at least 4 accurate scientific claims or ideas tied to the activity that are thoughtfully explained and linked to the descriptions and explanations of the patterns for the greenhouse effect.	Explanation uses at least 3 accurate scientific claims or ideas tied to the activity that are thoughtfully explained and linked to the descriptions and explanations of the patterns for the greenhouse effect.	Scientific information is simplified and basic. Scientific connections of patterns for the greenhouse effect are implied.	Scientific information has inaccuracies and scientific connections are unclear or absent.
Addressing Carbon Dioxide	Explanation accurately and thoughtfully addresses the concern for the rise in carbon dioxide.	Explanation proficiently addresses the concern for the rise in carbon dioxide.	Explanation basically addresses the concern for the rise in carbon dioxide.	Explanation has inaccuracies or very minimally addresses the concern for the rise in carbon dioxide.

*Note: Rubric is designed to assess individual student drawings and does not assess group presentation or final poster created with the group.

Lesson J-1

Evaluating the Evidence of Climate Change

Quick Overview

Lesson Scope: CO_2 in the atmosphere and the impact on global temperature

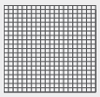
Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Colored pencils and graph paper

Materials Needed per Student

Colored pencils



Graph paper

Each student will need paper and instructions to record trials.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect global temperature?

This lesson is designed to initiate student understanding of **climate** change through assessing evidence on the link between CO_2 and temperature.

Key Understandings

- There is a link between temperature and CO_2 concentrations in the atmosphere.
- Earth's temperature affects climate.
- Analyzing patterns of data from the past and present can help humans understand climate cycles and predict changes in the climate.
- Analyzing and critiquing scientific evidence can help increase understanding around a topic.
- Information can be represented and interpreted in multiple ways.

Objectives

- Students will compare and contrast different types of climate change evidence.
- Students will analyze the link between atmospheric temperature and CO_2 concentrations (by graphing ice core data spanning hundreds of thousands of years).
- Students will analyze photos of glacier melts in Alaska.
- Students will begin to formulate an understanding of climate change.

Activity Set 2 | CO_2 and Climate Change

Vocabulary

- Climate
- Anthropogenic
- Weather
- Regional climate
- Global climate
- Climate change
- Temperature anomaly

Background Information for the Teacher

Today, more than ever before, print and electronic media shape public perception. Helping students develop the skills necessary to filter, assess, and critique incoming information as evidence or opinion is an important skill. When first trying to understand this topic, information should come from peer-reviewed sources, experts, and scientists. In later lessons, students will be provided with an opportunity to critique popular media and assess the validity of statements.

Climate change is both controversial and complicated. There is significant evidence that Earth's climate is warming, yet in the popular media climate change often is portrayed as an ongoing debate. To compound this perception, scientists are still unclear on how to quantify the magnitude of **anthropogenic** climate change. The point of this lesson is not to convince students that climate change is anthropogenic or to what extent humans are changing climate, but rather to have them analyze current research to initiate thinking about this subject and build their understanding of climate change through scientific evidence.

This lesson provides an opportunity for students to draw conclusions based on scientific evidence. The evidence will be drawn from a variety of peer-reviewed resources about some of the methods used to measure climate change. The role that CO_2 plays in the Earth's system is fundamental to understanding if the climate is changing. Students will analyze numerical data displaying changes over a long period of time, correlating trends between CO_2 and temperature. Students will also view historical and current photographs of glacier melts in Alaska, further demonstrating that climate is not in equilibrium. Finally, students will assess these different pieces of evidence and extrapolate the conclusions that can be drawn from each.

It is also important to distinguish between **weather** and climate. Weather is a specific event or condition that happens over a period of hours or days. For example, a thunderstorm, a snowstorm, and today's temperature all describe the weather. Weather is highly variable from day to day and from one year to the next. For example, Minneapolis might have a warm winter one year and a much colder winter the next. This kind of change is typical. But when the average pattern over hundreds

of years (which is climate) changes, it is considered a change in the climate.

Climate in one area, like the Midwest or Hawaii, is called a **regional climate**. The average climate around the world is called **global climate**. When scientists talk about global **climate change**, they're referring to patterns of change happening around the world over many years.

Rising global temperatures are thought to be causing other changes around the world, such as stronger hurricanes, melting glaciers, and the loss of wildlife habitat, which can all be empirically observed and measured. By examining trends in weather and temperature data, CO_2 emissions, and atmospheric CO_2, scientists can collect evidence of current climate change.

There are several ways scientists can measure climate history and current climate trends. A few of them include ice core sampling and measuring tree rings, sea levels, and ocean acidity (pH). The method that provides the longest record of past climate trends is measuring the ice cores collected from glaciers because the ice has been there for hundreds of thousands of years.

The thick layers of ice that have accumulated in places like Greenland and Antarctica have recorded clues about past temperatures, precipitation, atmospheric gases, and other aspects of the environment. To reveal these clues, researchers drill into glaciers and ice sheets and remove cylinder-shaped samples of ice called ice cores. Scientists can use chemical sampling techniques to determine the age of each layer of ice and the concentrations of different gases trapped in tiny air bubbles within the ice, which reveals the composition of the atmosphere in the past. They can also examine the water molecules in the ice itself to get information about historical temperatures. Trapped pollen and dust provide additional clues about the climate. Ice core records can go back hundreds of thousands of years, and they help scientists figure out whether the rapid increase in CO_2 levels and temperature we are currently observing fits a natural pattern or not (EPA, 2013).

By the end of this lesson, students should begin to develop an understanding of climate change rooted in science. The link between CO_2 and temperature can easily be observed from the empirical data provided.

Common Student Misconceptions/ Student Challenges

Students may have several misconceptions about climate change. These misconceptions may have to do with understanding how Earth's systems operate, including the atmosphere and the greenhouse effect, the sun's rays, and glacial periods. Misconceptions could also stem from the way the scientific community and popular media discuss change. There is an abundance of research that identifies misconceptions surrounding this topic. A few of the major misconceptions are addressed in the table below.

- Common Misconceptions about Climate and Climate Change *https://c2es.org/site/assets/uploads/2017/03/misconceptions-realities-climate-science-06-2012.pdf*.
- Common Misconceptions about Polar Weather and Climate *https://nsdl.oercommons.org/courses/common-misconceptions-about-polar-weather-and-climate*
- Climate Misconceptions: A Top 10 List. *https://nsdl.oercommons.org/courses/climate-misconceptions-a-top-10-list*
- Greenhouse Gas Overview *http://www.epa.gov/ghgemissions/overview-greenhouse-gases*

Misconception	Reality
The sun is the cause for recent global warming.	The output of energy from the sun has been monitored by satellites for 30 years and has not increased during this period of recent global warming.
Climate has changed many times in the distant past, before humans began burning coal and oil, so the current warming cannot be caused by humans burning coal, oil, and natural gas.	There are several drivers that cause climate to change, and some of the key drivers have both natural and human sources. It is believed that increases in global temperatures result mostly from higher levels of heat-trapping gases in the atmosphere, like CO_2. In the past, CO_2 has been released naturally. Today, it is released both naturally and from human activity.
The last few years have been cooler, so global warming can not be real; or, global warming stopped in 1998; or, The world has been cooling for the past decade.	The climate is defined by long-term averages in global temperatures and other climate metrics, and those are still increasing.
There is no scientific consensus on the existence or causes of global climate change.	Most earth scientists believe that there is strong agreement that emissions of heat-trapping gases from the burning of fossil fuels make a significant contribution to global warming.
Scientists predicted global cooling in the 1970s.	When the next ice age might occur became a topic of debate during the 1970s, but there was no consensus on the topic and most of the debate was already focused on global warming.
Atmospheric water vapor is the heat-trapping gas that is primarily responsible for global warming.	Water vapor is increasing in the atmosphere in response to rising CO_2 concentrations, amplifying the warming effect of manmade CO_2 emissions.

Table obtained from https://c2es.org/site/assets/uploads/2017/03/misconceptions-realities-climate-science-06-2012.pdf

Next Generation Science Standards

Performance Standard.MS-ESS3-5:

Ask questions to clarify evidence of the factors that have caused the rise in global temperature over the past century.

Further resources that summarize student misconception can be accessed at:

Disciplinary Core Idea

Human activities, such as the release of greenhouse gases from burning fossil fuels, are major factors in the current rise in Earth's mean surface temperature. Reducing the level of climate change and reducing human vulnerability to whatever climate changes do occur depend on the understanding of climate science, engineering capabilities, and other kinds of knowledge, such as understanding human behavior and applying that knowledge wisely in decisions and activities.

Science and Engineering Practices

Ask questions to identify and clarify evidence of an argument.

Crosscutting Concepts

Stability might be disturbed either by sudden events or gradual changes that accumulate over time.

Materials

- A copy of *Vostok Data Instructions* for each student (Appendix 2.4.1)
- A copy of *Ice Core Data* for each student (Appendix 2.4.2)
- A copy of *CO_2 & Temperature Tables* for each student (Appendix 2.4.3)
- A copy of *Ice Core Data Answer Key* for each student (Appendix 2.4.4)
- A copy of the *Ice Core Graphs* for each student (Appendix 2.4.5)
- A copy of *CO_2 & Temperature Table Answer Key* for each student (Appendix 2.4.6)
- A copy of Student Worksheet (Appendix 2.4.6)
- *Ice Core Data Questions* (Appendix 2.4.7)
- *Ice Core Data Answers* (Appendix 2.4.8)
- Glacier photos (Appendix 2.4.9)
- Colored pencils
- Graph paper

Length of Lesson

This lesson will take one approximately 90 minute class period or two 45-minute class periods.

Suggested Procedure

Lesson Adapted from EPA's Climate Change Lesson on "Getting to the Core" (EPA, 2013).

Activity

1. Introduce the lesson by initiating a conversation about the current climate change debate. Start probing students about the types of statements that they have heard. Ask: Who was the source? A scientist? A politician? Are they backing up their statements with evidence? What constitutes evidence? What is their interest on the subject? For example, is a scientist employed by a coal company presenting impartial evidence? Elaborate on how people are often confronted with the challenge of distinguishing between claims that are evidence-based and those that are merely opinions. Furthermore, tell the students that assessing the credibility of scientific research on Earth's climate is even more complicated. Tell students they will be looking at different types of evidence on climate change and clarify the conclusions that can be accurately drawn from each. The data will come from a research station called Vostok Station in Antarctica and from photographs taken over the last century.

2. Next, engage the students by asking them the difference between climate and weather. This can be done via a silent conversation, creating a t-chart on the board and having students fill in each column, or by having them classify examples as climate or weather. At the end of the discussion students should know that weather is the condition of the atmosphere over a short period of time (daily, weekly, monthly) while climate is the course of weather over a relatively long period of time (decades, centuries, millennia).

Part I - Empirical/Scientific Evidence

3. First, you may want to show students the location of the Vostok Research station on a map or globe. Explain to students that scientists at this remote research station gather ice core samples that provide clues about Earth's past from hundreds of thousands of years ago. You may also want to show pictures of ice coring from the web.

4. Discuss with students what ice core samples are and how they are used to gather information about CO_2, temperature, pollen, and dust. (See the background and resources for more information. It may also be a good idea to review the greenhouse effect from previous lessons.)

5. Explain that the temperature data in the tables and graphs are represented as temperature anomalies (deviations from the norm or reference point). An analogy using human body temperatures might help to explain why anomalies are useful. (The "average" human body temperature is about 98.6°F (37°C), but some people have lower normal temperatures than others. For someone with a normal body temperature of

98.0°F (36.6°C), a temperature reading of 98.9°F (37.2°C) would indicate a fever. The use of 98.9°F (37.2°C) alone would not indicate this condition to most people but the use of an anomaly or departure from the normal body temperature for this person (+0.9°F, -17°C) would indicate the feverish condition.) In summary, a **temperature anomaly** means a departure from a reference value or long-term average. A positive anomaly indicates that the observed temperature was warmer than the reference value, while a negative indicates that the observed temperature was cooler that the reference value. For this data set, the reference value or long-term average is -69°F (-56°C), and the temperatures listed are deviations from this value.

6. Depending on the grade and level of your students, this lesson can be adjusted by having students graph and plot the data points obtained from the Vostok Antarctica ice core data or by having students analyze the graphs provided in the Appendix 2.4.5. The directions for either delivery are given below.

Note: *For lower level middle school students and/or greater time constraints:*

   a. Distribute the graphs and tables in Appendix 2.4.5 and Appendix 2.4.6 and initiate questions to help students interpret the graphs and data tables. Direct student learning by proceeding through the questions in Appendix 2.4.7. This can be done as a class or in groups.

Note: *For higher grade level students and/or less time constraints:*

   b. For more advanced students, hand out copies of the "Vostok, Antarctica, Ice Core Data" worksheets and directions (Appendix 2.4.1, 2.4.2, and 2.4.3) two sheets of graph paper per student, and colored pencils.

   c. Have students follow the directions in Appendix 2.4.1 about how to create a graph. They may need a bit of direction for this part. Consider plotting a few of the points on a graph to help students get started.

   d. As a class, in small groups, or individually have students complete and answer the question in Appendix 2.4.7.

7. Explain to students that scientific evidence consists of observation and experimental results that serve to support, refute, or modify a scientific hypothesis or theory, when collected and interrelated in accordance with the scientific method. Begin to ask students to think about the data to determine what type of information this study provides. Empirical?

   Yes, this data provides numerical measurements.

   Was the information rooted in opinion or biases?

   Because this data is peer reviewed and has been cited in many other places, its most likely

unbiased. But one can never be sure until they read the paper and decided for themselves.

Can we think of any limitations or errors that could have influenced the results?

This is a very complex paper and there could be errors in their calculations, but errors would not significantly change the whole picture.

What is the validly of this study in relation to climate change?

This research is valid.

Part II - Photos as Evidence and Comparing Evidence

8. Next, distribute or place on the board the photographs in Appendix 2.4.8. Ask students what they notice about these photos. What is the time span of the photos?

   About a hundred years

   What can be inferred from these photos?

   Temperatures are warming because glaciers have melted.

9. Devise a statement that can be made from the photos such as: *Glaciers have been receding the past century, therefore climate is warming.*

10. Again, ask questions similiar to those from Step 7. Prompt students to think about the data to determine what type of information this study provides. Empirical?

    These photos provide a visual observation.

    Was the information rooted in opinion or biases?

    The photos do not appear to be biased.

    Can we think of any limitations of these photos?

    The photos provide a non-numerical observation and do not rely on statistics to prove a result. This type of data is more unstructured.

    To what extent do these photos prove that Earth's climate is changing?

    Again, these photos tell us only part of the story and do not provide any measureable or numerical data, yet still display a significant change in the climate of a particular region, in this case, Alaska. The photos only go back a hundred years.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

1. Have students develop a set of statements that summarize each of the two pieces of evidence (the graphs from Vostok research station and from the glacier pictures).

   What is the difference between these types of evidence when addressing climate change?

   One contains specific numerical measured data on a global perspective from 400,000 years into the past while the other looks at pictures from a region a hundred years back.

   Is one more useful than the other?

   Because the research study is grounded in the scientific method, dates back 400,000 years, and is peer reviewed, it is

most likely more useful in answering the question of climate change.

What are the drawbacks and benefits of each?

Price, time, regional representation, peer-reviewed, measurable observations, etc.

2. Compare or contrast these methods to the methods used by the media.
3. What is your thinking about climate change? Has your thinking changed from what it was prior to this lesson? Why or why not?

Supplemental Materials and References

Center for Climate and Energy Solutions (C2ES). (2017). *Realities vs. misconceptions about climate change science.* Arlington, VA: Author. Retrieved from https://c2es.org/site/assets/uploads/2017/03/misconceptions-realities-climate-science-06-2012.pdf.

McCaffrey, M. S., & Buhr, S. M. (2008). Clarifying climate confusion: addressing systemic holes, cognitive gaps, and misconceptions through climate literacy, *Physical Geography,* 29(6), 512–528.

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Petit, J. R., Jouzel, J., Raynaud, D., Barkov, N. I., Barnola, J. M., Basile, I., & Stievenard, M. (1999). Climate and atmospheric history of the past 420,000 years from the Vostok ice core, Antarctica. *Nature, 399* (6735), 429–436.

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

VOSTOK DATA INSTRUCTIONS

Instructions for Filling Out the "Vostok, Antarctica, Ice Core Data" Worksheet

1. In the space provided in column three, round the carbon dioxide (CO_2) concentration to the nearest whole number.

2. In the space provided in column five, round the temperature anomaly to the nearest tenth of a degree.

Instructions for Plotting the Graphs

1. You will create two graphs: one for CO_2 concentration and one for temperature anomaly.

2. On both graphs, your x-axis will represent years. Start with 400,000 BC on the left and number as far as the the year 0 on the right, counting by intervals of 10,000 years. Label the axis.

3. On the first graph, the y-axis on the left side of the paper will represent the CO_2 concentration using units of parts per million (ppm). Begin with 100 ppm at the lower end, and number up to 400 ppm, counting by intervals of 10 ppm. Label the axis.

4. On the second graph, the y-axis on the left side of the paper will represent the temperature anomaly in degrees Celsius (°C). Begin with -10.0 °C at the lower end and number up to 2.0 °C, counting by intervals of 0.5 °C. Label the axis.

5. Using different colored pencils, plot the points for CO_2 concentration and temperature anomaly.

6. Write a title on each graph.

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

VOSTOK, ANTARCTICA, ICE CORE DATA—ANSWER KEY

Carbon Dioxide Concentration and Temperature Anomaly Data
(398,000 BC to 400 BC)

Year (BC)	CO_2 concentration (ppm)	CO_2 concentration rounded to nearest whole number	Temperature anomaly (°C)	Temperature anomaly (°C) rounded to nearest tenth of a degree
398,000	278	**278**	-1.64	**-1.6**
388,000	255.2	**255**	-5.34	**-5.3**
378,000	245.9	**246**	-4.88	**-4.9**
368,000	229.7	**230**	-5.42	**-5.4**
358,000	206.4	**206**	-5.8	**-5.8**
348,000	193	**193**	-7.64	**-7.6**
338,000	220.4	**220**	-7.44	**-7.4**
328,000	234.2	**234**	-4.9	**-4.9**
318,000	271.8	**272**	-0.12	**-0.1**
308,000	256.3	**256**	-3.32	**-3.3**
298,000	241.9	**242**	-3.08	**-3.1**
288,000	240.2	**240**	-6	**-6**
278,000	207.7	**208**	-6.17	**-6.2**
268,000	231.4	**231**	-5.95	**-6**
258,000	184.7	**185**	-8.3	**-8.3**
248,000	203.9	**204**	-6.52	**-6.5**
238,000	230.4	**230**	-2.12	**-2.1**
228,000	245.2	**245**	-6.15	**-6.2**
218,000	212.2	**216**	-4.31	**-4.3**
208,000	244.6	**245**	-3.07	**-3.1**
198,000	242.6	**243**	-2.68	**-2.7**
188,000	231.4	**231**	-6.49	**-6.5**
178,000	213.2	**213**	-6.34	**-6.3**
168,000	197.9	**198**	-7.01	**-7**
158,000	204.4	**204**	-6.25	**-6.3**
148,000	191.9	**192**	-7.34	**-7.3**
138,000	192.3	**192**	-8.99	**-9**
128,000	263.4	**263**	1.47	**1.5**

https://www3.epa.gov/climatechange//kids/documents/temp-and-co2.pdf

EPA United States Environmental Protection Agency

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

CARBON DIOXIDE CONCENTRATION AND TEMPERATURE RATE OF CHANGE

NAME: _____ **DATE:** _____

48,000 BC to 400 BC
Length of time: ____ years

Variable	Value in 48,000 BC	Value in 400 BC	Change	Rate of change per year
CO_2 concentration (ppm)				
Temperature anomaly (°C)				

1901 to 2011
Length of time: ____ years

Variable	Value in 1901	Value in 2011	Change	Rate of change per year
CO_2 concentration (ppm)	296.1 ppm	391.6 ppm		
Temperature anomaly (°C)	-0.16 °C	0.51 °C		

Data source: U.S. EPA, Climate Change Indicators in the United States:
Original host Web site: https://19january2017snapshot.epa.gov/climatechange_.html

CO_2 concentrations are from Antarctica (1901) and Hawaii (2011).
Temperature anomaly is a global average.

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

Carbon Dioxide Concentration and Temperature Anomaly Data
(398,000 BC to 400 BC)

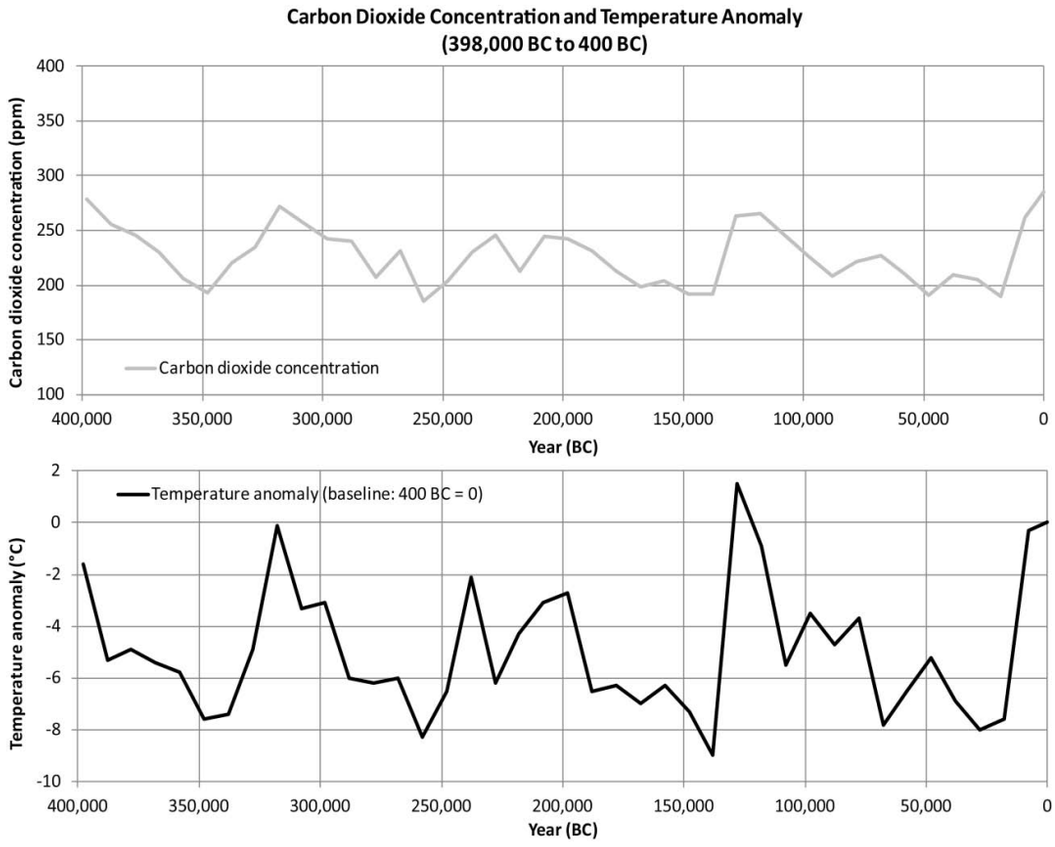
Year (BC)	CO_2 concentration (ppm)	CO_2 concentration rounded to nearest whole number	Temperature anomaly (°C)	Temperature anomaly (°C) rounded to nearest tenth of a degree
118,000	265.2	**265**	-0.86	**-0.9**
108,000	245.7	**246**	-5.53	**-5.5**
98,000	225.9	**226**	-3.45	**-3.5**
88,000	208	**208**	-4.69	**-4.7**
78,000	221.8	**222**	-3.66	**-3.7**
68,000	227.4	**227**	-7.84	**-7.8**
58,000	210.4	**210**	-6.53	**-6.5**
48,000	190.4	**190**	-5.18	**-5.2**
38,000	209.1	**209**	-6.91	**-6.9**
28,000	205.4	**205**	-7.95	**-8**
18,000	189.2	**189**	-7.62	**-7.6**
8,000	261.6	**262**	-0.28	**-0.3**
400	284.7	**285**	0	**0**

Data source: National Oceanic and Atmospheric Administration (NOAA):
Host Web site: www.esrl.noaa.gov.

https://www3.epa.gov/climatechange//kids/documents/temp-and-co2.pdf

EPA United States Environmental Protection Agency

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE



Data source: National Oceanic and Atmospheric Administration (NOAA)
Host Web site: www.esrl.noaa.gov.

GETTING TO THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

CARBON DIOXIDE CONCENTRATION AND TEMPERATURE RATE OF CHANGE—ANSWER KEY

48,000 BC to 400 BC
Length of time: 47,600 years

Variable	Value in 48,000 BC	Value in 400 BC	Change	Rate of change per year
CO_2 concentration (ppm)	190.4 ppm	284.7 ppm	+94.3 ppm	94.3 ppm / 47,600 years = 0.0020 ppm per year
Temperature anomaly (°C)	-5.18 °C	0 °C	+5.18 °C	5.18 °C / 47,600 = 0.00011 °C per year

1901 to 2011
Length of time: 110 years

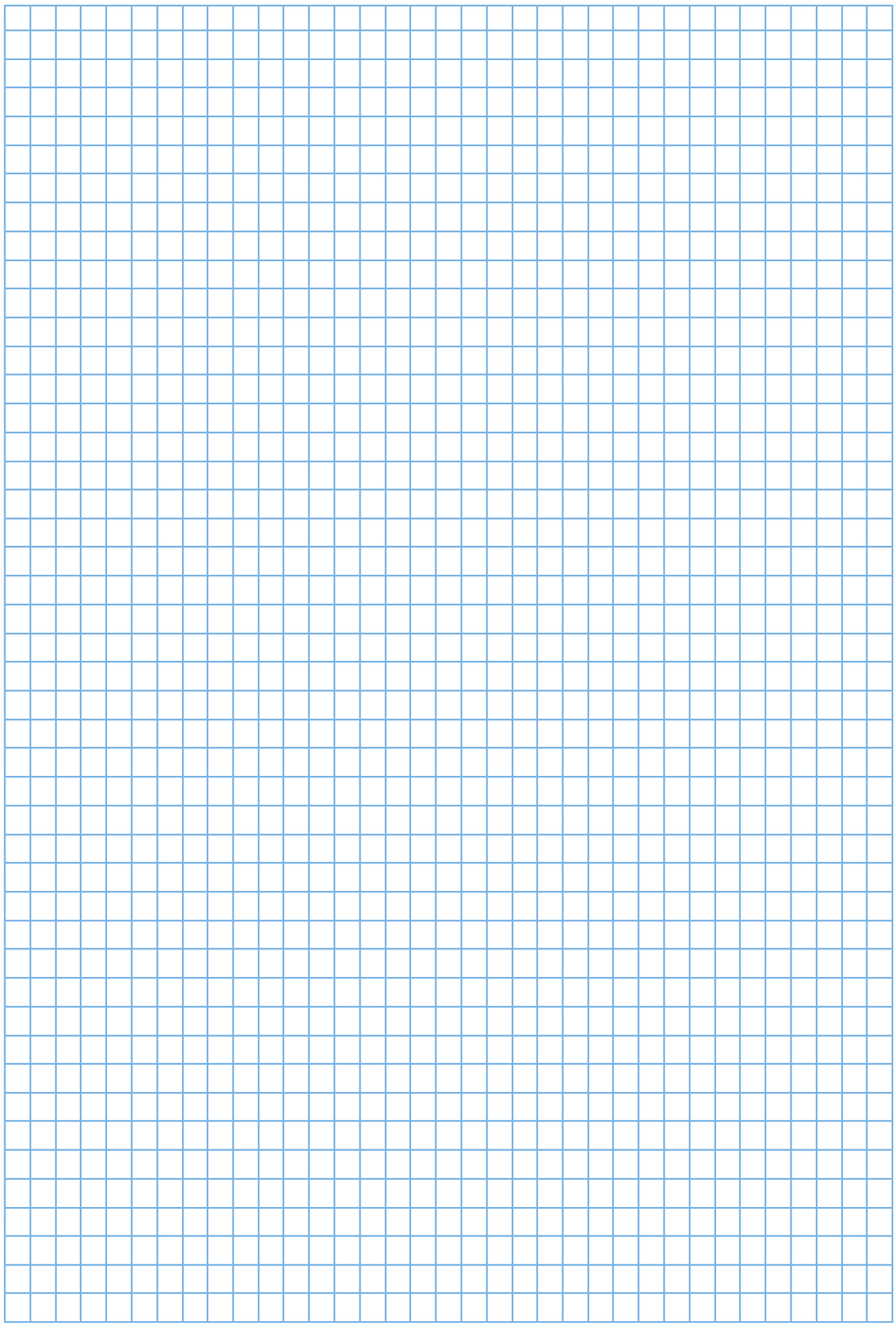
Variable	Value in 1901	Value in 2011	Change	Rate of change per year
CO_2 concentration (ppm)	296.1 ppm	391.6 ppm	+95.5 ppm	95.5 ppm / 110 years = 0.868 ppm per year
Temperature anomaly (°C)	-0.16 °C	0.51 °C	+0.67 °C	0.67 °C / 110 years = 0.0061 °C per year

Data source: U.S. EPA, Climage Change Indicators in the United States:
Original host Web site: https://19january2017snapshot.epa.gov/climatechange .html

CO_2 concentrations are from Antarctica (1901) and Hawaii (2011).
Temperature anomaly is a global average.

GETTING AT THE CORE: THE LINK BETWEEN TEMPERATURE AND CARBON DIOXIDE

NAME: TITLE: DATE:



https://www3.epa.gov/climatechange//kids/documents/temp-and-co2.pdf

EPA United States Environmental Protection Agency

1. What variable is on the x-axis (independent variable) and what variable is on the y-axis (dependent variable) for each graph?

2. What pattern(s) do you notice on the graphs?

3. How many peaks (top) and troughs (bottom) can you identify?

4. What is the approximate number of years in one complete cycle?

5. Do the peaks represent glacial (cold) periods or do the troughs?

6. Now look at the table, how many years of data are shown in the 48,000 BC to 400 BC table?

7. How many years of data are shown in the "1901 to 2011" table?

8. Which table(s) show a warming trend?

9. What trend, upward or downward, are we currently experiencing?

10. What is the change of temperature anomaly (globally) between 1901 and 2011?

11. In 1971, the globally averaged CO_2 concentration was approximately 330 ppm. If the CO_2 concentration in 2000 was about 384 ppm, calculate the average rate increase per year.

12. What do these graphs tell us about the correlation between CO_2 and temperature?

Answers:

1. What variable is on the x-axis (independent variable) and what variable is on the y-axis (dependent variable) for each graph?

   x-axis: temperature, y-axis: CO_2 and temperature

2. What pattern(s) do you notice on the graphs?

   A repeating cycle. When the carbon dioxide concentration goes up, temperature goes up. When the carbon dioxide concentration goes down, temperature goes down.

3. How many peaks (top) and troughs (bottom) can you identify?

   Five peaks and four troughs

4. What is the approximate number of years in one complete cycle? (A cycle is the time between two peaks or between two troughs.)

   100,000 years

5. Do the peaks represent glacial (cold) periods or do the troughs?

   Troughs, because the temperature is at its lowest

6. Now look at the table, how many years of data are shown in the 48,000 BC to 400 BC table?

   About 47,600 years

7. How many years of data are shown in the "1901 to 2011" table?

   110 years

8. Which table(s) show a warming trend?

   Both. CO_2 concentrations increase and temperature anomaly increases. Both increase at a greater rate more recently.

9. What trend, upward or downward, are we currently experiencing?

   Upward for both CO_2 concentration and temperature anomaly

10. What is the change of temperature anomaly (globally) between 1901 and 2011?

    Approximately 0.68°C

11. In 1971, the globally averaged CO_2 concentration was approximately 330 ppm. If the CO_2 concentration in 2000 was about 384 ppm, calculate the average rate increase per year.

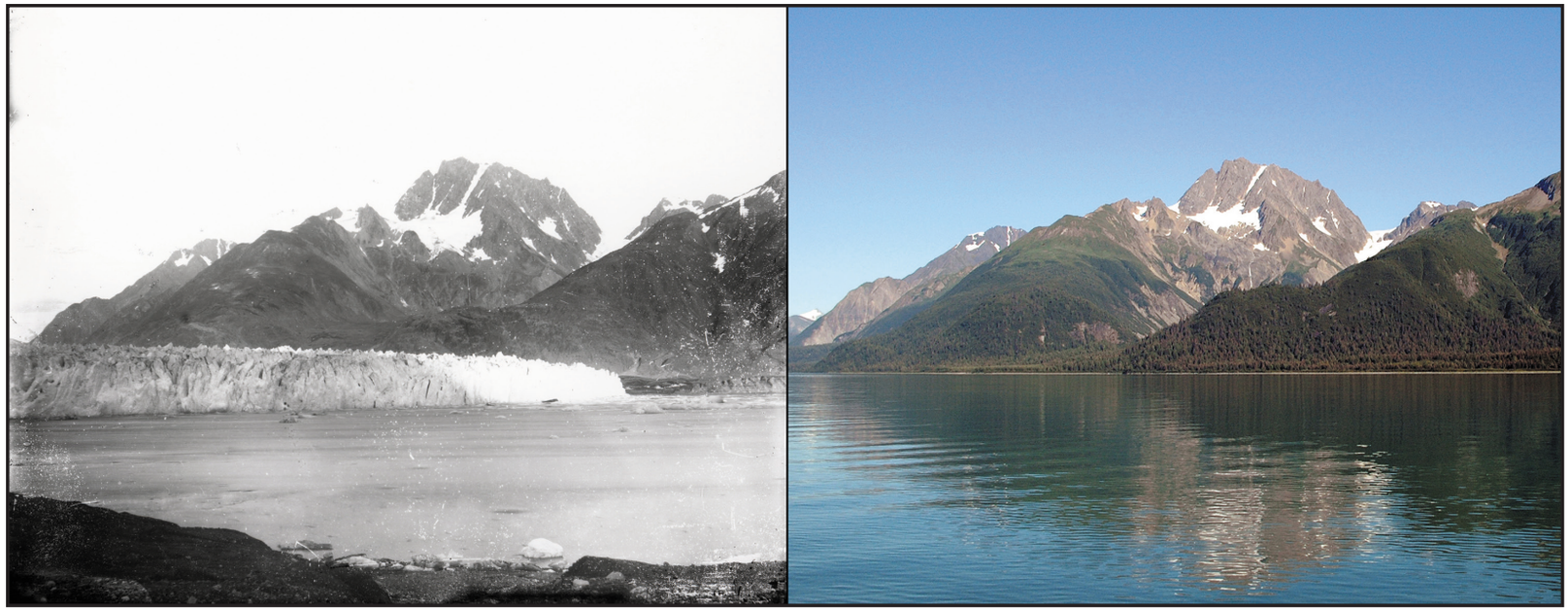
    Approximately 1.5 ppm per year

12. What do these graph tell us about the correlation between CO_2 and temperature?

    As CO_2 increases, so does temperature. There is also a short lag from the time when CO_2 increases and when temperature increases.

Muir Glacier, Alaska. Left: September 2, 1892. Right: August 8, 2005.

1892 photo taken by Harry Fielding; 2005 photo taken by Bruce F. Molnia. Courtesy of The Glacier Photograph Collection, National Snow and Ice Data Center/World Data Center for Glaciology.



Source of Image: https://svs.gsfc.nasa.gov/10892. Host Web site: http://climate.nasa.gov

Okpilak Glacier, Alaska. Left: 1907. Right: 2004.

1907 photo taken by Ernest Leffingwell; 2004 photo taken by Matt Nolan. Courtesy of the Glacier Photograph Collection, National Snow and Ice Data Center/World Data Center for Glaciology.

Source of Image: http://climate.nasa.gov/system/gallery_images/large/Icemelt_Alaska.jpg. Host Web site: http://climate.nasa.gov

Lesson J-1

Evaluating the Evidence of Climate Change; Addressing Misconceptions

Quick Overview

Lesson Scope: CO_2 in the atmosphere and the effects on global temperature

Length: Two ~45-minute class periods

Grade Level: 5–8

Materials: Computer with internet access

Materials Needed per Student

Computer with internet access

Each student will need paper and instructions to record trials.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect global temperature?

CO_2 in the atmosphere is playing a role in Earth's surface temperature increase, which is resulting in accelerated global warming. There are many arguments and misconceptions about climate change. This lesson will target some of these climate change arguments and help students think about using scientific research to counter the claims and develop a more accurate understanding of climate change.

Key Understandings

- There are many misconceptions and arguments found in popular media and among ordinary citizens regarding climate change.
- Scientists have been researching climate change since the 1980s and have collected evidence that supports the claim that part of Earth's warming is due to human activities.
- When forming an accurate opinion on climate change, information should come from knowledgeable professionals, such as scientists and experts.
- Information can be represented and interpreted in multiple ways.

Objectives

- Students will clarify and ask questions about climate change evidence.
- Students will research various claims made about climate change.
- Students will present their findings to the class.
- Students will build and formulate an understanding of climate change.

Vocabulary

- Peer-reviewed science

Background Information for the Teacher

Climate science can be complicated, as there are many feedback mechanisms and processes that contribute to our understanding of climate. Because of this, the popular media often characterizes climate change and global warming as a debate even though a large body of scientific evidence supports the claim that part of Earth's warming is anthropogenic. *Skeptical Science* is a blog and information resource created by Australian author and solar physicist John Cook. This website uses primarily peer-reviewed articles to negate popular arguments against climate change.

This lesson is designed to address misconceptions and misrepresented information about climate change. Students should leave this lesson with an understanding that there is no debate about climate change among climate scientists, and that many of the arguments against climate change stem from economic and political interests, which are further popularized by the media. Students benefit by becoming aware of the media's heavy influence on opinion and public perception and that media's portrayal of information is often unbalanced, biased, or incorrect. When forming an understanding about a topic as complicated as climate change, information and knowledge should come from unbiased, knowledgeable professionals.

Common Student Misconceptions/ Student Challenges

Most students have preconceived notions about climate change based on what they have heard from popular media sources or at home. The media have characterized climate change as a scientific debate, even though climate scientists agree that the Earth's climate is getting warmer and human activity is one of the causes. The challenge is for students to understand how **peer-reviewed science** is different from media-interpreted science reporting and to build an understanding of climate change based on scientific evidence and not from media or other sources that may be biased.

Next Generation Science Standards

Performance Standard.MS-ESS3-5:

Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

Disciplinary Core Idea

Human activities, such as the release of greenhouse gases from burning fossil fuels, are major factors in the current rise in Earth's mean surface temperature (global warming). Reducing the level of climate change and reducing human vulnerability to whatever climate changes do occur depend on the understanding of climate science, engineering capabilities, and other kinds of knowledge, such as understanding of human behavior and on applying that knowledge wisely in decision and activities.

Science and Engineering Practices

Ask questions to identify and clarify evidence of an argument.

Crosscutting Concepts

Stability might be disturbed either by sudden events or gradual changes that accumulate over time.

Materials

- Computer with internet access for each student (or student group)
- Skeptic arguments against climate change website: *http://www.skepticalscience.com/argument.php?f=percentage*

Setup

Note: *If computers are not available for every student/group, this lesson can be modified as a classroom discussion.*

You will want to become very familiar with the website listed above. Allow yourself a couple of hours to become familiar with it prior to teaching this lesson for the first time.

Length of Lesson

This lesson will take 90 minutes (or two 45-minute class periods).

Suggested Procedure

Introduction

1. Read the following quote from *www.skepticalscience.com* to your students.

   "Scientific skepticism is healthy. Scientists should always challenge themselves to improve their understanding. Yet this isn't what happens with climate change denial. Skeptics vigorously criticize any evidence that supports [Anthropomorphic or human-made] caused global warming and yet embrace any argument, op-ed, blog or study that purports to refute global

warming. This website gets skeptical about global warming skepticism. Do their arguments have any scientific basis? What does the peer-reviewed scientific literature say?"

2. Post and present one example of each of the following (these can be easily found by doing a search of each example):
   - A scientific journal article on global warming that is peer reviewed by other scientists.
   - An opinion paper by a climate scientist on global warming.
   - An opinion paper by a general scientist on global warming.
   - A news article by a journalist on global warming.
   - A speech by a politician on global warming.
   - An OpEd by a political pundit on global warming.
   - Website about global warming.
3. Ask students to rank these articles based on how likely they are going to be accurate. Then have students rank each of them based on the ability of the articles to persuade the reader. Discuss with students how it is important to be skeptical of any article they read or phrases they hear, especially ones from any type of media or politicians. Students should do their own research before assuming anything they hear or read is true. The best scientific literature is peer reviewed by other scientists. Explain the peer-review process:
   a. Peer review is the process by which an author's peers, recognized researchers in the field, read and evaluate a paper (article) submitted for publication and recommend whether the paper should be published, revised, or rejected.
   b. Peer review is a widely accepted indicator of quality scholarship in a discipline or field. Articles accepted for publication through a peer-review process meet the discipline's expected standards of expertise.
   c. Peer-reviewed (or refereed) journals are scholarly journals that only publish articles that have passed through this review process.
   d. While not perfect, peer review increases the credibility of scientific research.

Activity

1. Ask students to identify statements they have heard about why climate change is not happening. This could be done as a "think-pair-share" activity to eliminate having the spotlight on a single student or having several students dominate the

discussion. During the sharing part, compose a class list of the top five to 10 reasons students have heard.

2. Have each student (or a pair of students) choose an argument from *The Scientific Guide to Global Warming Skepticism* (http://www.skepticalscience.com/The-Scientific-Guide-to-Global-Warming-Skepticism.html) that best matches one of the reasons listed on the board. (Each group should have a different statement to research.) Encourage students to find an argument similar to one of the reasons brainstormed at the beginning of the activity, or another argument they have heard frequently.

3. Have students read the evidence against their argument then write three to five bulleted statements summarizing their article.

4. Skeptic argument example: It can't be warming because it is cold!

   "Austria is today seeing its earliest snowfall in history with 30 to 40 centimeters already predicted in the mountains. Such dramatic falls in temperature provide superficial evidence for those who doubt that the world is threatened by climate change."

   Example of student bulleted summary:

   a. *It's normal for weather to have its ups (warm weather) and downs (cold weather), but this is not climate, it is weather.*

   b. *Climate is the pattern of weather over a period of years.*

   c. *New records for cold weather are still being set every year, but with climate change the number of record cold days is decreasing.*

   d. *New high temperature records occur almost twice as often as low temperature records.*

5. Have students present the information by reading their argument to the class, then presenting their summary findings.

6. Classroom discussion: Ask students if they still believe the reasons brainstormed earlier are valid. If so, why were they not convinced? Which group or set of arguments convinced them the most? Why? If not, what convinced them? What other information would they want to be convinced? Would what they learned help them to convince someone else (say a family member)?

7. Discuss reasons why someone might want to hold onto a belief, in spite of evidence to the contrary? What might be some of the reasons that the media might not report events about climate change accurately?

   Funding sources may include those whose financial interests would be harmed if climate change issues cause a decrease in the use of their products—such as fossil fuels.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

Note: *It would be good to emphasize that while climate change is resulting in a broad array of problems, because of this we will need critical thinking citizens to help come up with solutions to these problems. Being able to assess bias and interpret data will be critical skills for this process.*

Return to the Introduction Activity and re-rank the list of sources. Did your opinion of any of them change? Next, review their original climate change statements; have students determine if they still believe the statement from before, and if the statement is still plausible. Using a strategy such as "I used to think_____, but now I know_____ because _____ might be a good way to get students to reflect on this lesson.

Supplemental Materials and References

Cook, J. (2014). Skeptical Science: Getting Skeptical about Global Warming Skepticism. Retrieved from *http://www.skepticalscience.com/argument.php?f=percentage*

Lesson K

Examining Your Carbon Footprint, Human Impact Through Transportation

Quick Overview

Lesson Scope: CO_2 in the atmosphere and the effects on global temperature

Length: One ~45-minute class period

Grade Level: 5–8

Materials: Plastic bags, charcoal briquettes, gloves, plastic tablecloths

Materials Needed per Student

Plastic bags

Charcoal briquettes

Gloves

Plastic tablecloths

Each student will need paper and instructions to record trials.

Part of this lesson was adapted from *"Audience-Pleasing Physical Models to Support CO_2 Outreach"* at http://www.beg.utexas.edu/gccc/forum/codexdetails.php?itemID=39.

Guiding Question

How does carbon dioxide (CO_2) in the atmosphere affect global temperature?

This lesson helps students understand their own carbon footprint. Specifically, students are asked to identify their "dream" car and then determine the carbon output based on a short trip. Students are asked to reflect on their car selection based on their findings.

Key Understandings

- The transportation sector contributes 13% of total **carbon dioxide** (CO_2) emissions.
- Changes in human behavior and policy regulations could significantly reduce CO_2 emissions in the transportation sector.

Objectives

- Students will observe the amount of carbon in one gallon of gasoline.
- Students will calculate the amount of CO_2 produced from one gallon of gasoline.

Activity Set 2 | CO_2 and Climate Change

- Students will be able to compare and calculate the carbon emissions produced from different automobiles.
- Students will observe the evidence of fossil fuels' contribution to CO_2 emissions by examining and using scientific graphs.
- Students will brainstorm solutions to reducing CO_2 emissions.

Vocabulary

- Carbon dioxide (CO_2)
- Combustion
- Matter

Background Information for the Teacher

Combustion (burning) is an exothermic reaction between a fuel (i.e. gasoline) and an oxidant (oxygen) that produces CO_2. As a result of increased use of fossil fuels since the Industrial Revolution, CO_2 levels in Earth's atmosphere have been increasing and are now at levels that are causing concern.

The equation for combustion of octane in gasoline: $C_8H_{18} + 25\ O_2 = 18\ H_2O + 16\ CO_2$ + energy

For every 6 pounds (1 gallon) of gasoline that is burned, 19 pounds (8.6 kg) of CO_2 is released into the atmosphere. The extra mass originates from atmospheric oxygen that is combined with carbon during combustion.

Teachers should take time before the lesson to familiarize themselves with the U.S. Department of Energy's fuel economy website (*http://www.fueleconomy.gov/*). The website provides a wealth of information on fuel mileage, and students will be using the "Find and Compare Cars" search to access EPA mileage to calculate the mileage and amount of gasoline needed for a road trip.

Calculating the distances can be done using a website such as MapQuest (*http://www.mapquest.com*) or Google Maps (*https://maps.google.com*).

The transportation sector is one of the main contributors to CO_2 emissions. It is difficult to capture emissions from moving vehicles. One way to lower emission rates from vehicles is to build better emissions systems on new vehicles. However, that does not change emission for vehicles already on the road. Several countries, such as France, Germany, and Sweden, have already adopted emission policies by changing their vehicle taxation systems. These policies set standards for CO_2 emission rates or for fuel economy, and manufacturers must meet the standards or pay fines for noncompliance. The European Union sets greenhouse gas emissions rate standards and mandated a decrease in emissions rates of about 20 percent from 2009 to 2015; as in the United States, further decreases are expected through 2020. In addition to the increasing stringency of the standards, there has been a recent trend toward using taxes on vehicle purchase and ownership to reduce CO_2 emission

rates. Historically, many European countries subsidized or taxed vehicle purchase or ownership depending on vehicle characteristics correlated with CO_2 emissions—for example, Germany taxed vehicle ownership largely according to engine size. Since the mid-2000s, many of these countries have reformed their tax systems to link taxes directly to the CO_2 emission rates of the vehicles. Policy regulations have the potential for influencing both the types of cars we drive and the incentive to produce less carbon emissions (Klier, 2012).

Students should be aware of both the need and benefit for these types of policies in the United States as we look for ways to reduce global greenhouse gas emissions. Human behavior is heavily influenced by enforced laws, and policies that aim to tax CO_2 will inevitably be a part of future solutions. This lesson introduces an understanding of the amount of CO_2 released from transportation activities and should conclude with conversations and brainstorming of the many possible ways to reduce emissions through changing human behavior, creating new and better technologies, and implementing carbon emission policies.

Teachers should review the graphs outlining CO_2 emissions at the end of this lesson. The graphs can be used to summarize the effect that automobile use has on current CO_2 emissions. A conversation about this impact as a class or in small groups is recommended to tie up the lesson.

Common Student Misconceptions/ Student Challenges

Students may have difficulty making the connection between the burning of fossil fuels (through driving a car, using electricity, etc.) and the actual CO_2 particles that are emitted into the atmosphere. Because CO_2 is a colorless, odorless gas, it is difficult to conceive of CO_2 as a damaging substance. CO_2 has mass and takes up space, giving it the property of matter. Viewing CO_2 as **matter** is essential to realizing that this substance is in high amounts in the atmosphere. These connections become clearer when students calculate the actual amount of CO_2 released from driving and when they touch and see the amount of carbon in physical briquettes.

Next Generation Science Standards

Performance Standard.MS-ESS3-5:

Ask questions to clarify evidence of the factors that have caused the rise in global temperatures over the past century.

Disciplinary Core Idea

Human activities, such as the release of greenhouse gases from burning fossil fuels, are major factors in the current rise in Earth's mean surface temperature (global warming). Reducing the level of climate change and reducing human vulnerability to whatever climate changes do occur depend on the understanding of climate

science, engineering capabilities, and other kinds of knowledge, such as understanding of human behavior and on applying that knowledge wisely in decisions and activities.

Science and Engineering Practices

Ask questions to identify and clarify evidence of an argument.

Crosscutting Concepts

Stability might be disturbed either by sudden events or gradual changes that accumulate over time.

Materials

- Access to the Internet and the following websites:

  http://www.fueleconomy.gov/feg/findacar.shtml

  http://www.mapquest.com

  http://maps.google.com

- One worksheet (Appendix 2.6.1) for each student or group of students
- *Calculating Cost and CO_2 Emissions for Different Vehicles* (Appendix 2.6.2). One set for each student or group of students
- CO_2 emission charts (Appendices 2.6.3 and 2.6.4) One for each student or group of students
- Plastic bags for charcoal briquettes (4 per bag: 4 to 5 briquettes = 1 mile (1.6 kilometers)
- Gloves for handling briquettes
- Plastic tablecloths or butcher paper for desks on which briquettes will be placed

Setup

Note: *Students will need to pick a petroleum-based car, as an all-electric car is difficult to compare with a petroleum car.*

The teacher should become familiar with the websites and try the activity. This preparation will take approximately an hour.

It is important that students have access to the Internet so that they can determine the miles/gallon of their chosen vehicle.

Length of Lesson

This lesson will take one 45-minute class period.

Procedure

1. Pour charcoal briquettes on a table as a visual aid. This lets participants see the amount of carbon going into the atmosphere when fuel is burned. Say, "this is a type of fossil fuel which eventually runs our cars."
2. Write or pose the question: How much CO_2 is produced during combustion of fossil fuel? Ask students to write down their ideas or talk with partners.

Explain that because CO_2 is a colorless gas transparent to light, people think that it must be unimportant and harmless. One gallon (3.8 l) of gasoline has about 5.2 pounds (2.4 kg) of carbon. Charcoal briquettes are almost all carbon. A 5-pound (2.26 kg) bag of charcoal holds about 100 briquettes. At 26 miles/gallon (42 km/3.8 l), that's 0.2 pounds (0.9 kg) of carbon (four to five charcoal briquettes)/ mile. Participants can count out briquettes to equal the amount of carbon that will be released to the atmosphere during a normal drive.

3. Write the equation for combustion on the board ($C_8H_{18} + 25\ O_2 = 18\ H_2O + 16\ CO_2 + energy$) and explain how combustion (the burning of fuel) is the combining of hydrocarbons (fuel) with oxygen, producing water, CO_2, and energy. This is the reaction that occurs in our cars and produces emissions. Reading the article at the website below as a class or in small groups will help students understand how gasoline becomes CO_2 emissions. *http://www.slate.com/articles/news_and_politics/explainer/2006/11/how_gasoline_becomes_co2.html*

4. Conclusion: A standard U.S. car throws a charcoal briquette (or more) of carbon from its tailpipe about every ¼ mile. If people could see the carbon that was being released, would it change people's behaviors? Ask students to respond in writing or discuss with partners.

5. Next introduce the day's activity: Today we will be comparing and calculating the carbon emissions produced from different automobiles. Ask students to identify their dream car, including the year, make, and model. Next, ask them to briefly explain why they selected that vehicle. (Most likely you will need to provide several examples to students regarding make and model).

6. Ask: How many of you know how many miles per gallon (MPG) your car uses? Instruct students to spend several minutes figuring this out by using the following site: *http://www.fueleconomy.gov/feg/findacar.shtml*

   Note: *Some students might not be familiar with miles per gallon (MPG). Explore students' background knowledge by discussing what MPG means. Have students do simple calculations, figuring out how many gallons of gas are burned during different scenarios. For example if a car gets 25 miles (40 km) to the gallon and drives 100 miles (161 km) how many gallons of gas does it use? Or, if a car has driven 400 miles (644 km) and the amount of gas needed to fill the tank is 10 gallons (38 l), what was the average MPG for that tank of gasoline?*

   To do this, they will need to enter the year, make, and model of their dream car. Instruct them to record the combined miles per gallon on the worksheet provided in Appendix 2.6.1. Information on the most current year of car is not always available. Students might need to select the previous year. According to *fueleconomy.gov*, *"Combined fuel economy is a weighted average of City and Highway MPG values that is*

calculated by weighting the City value by 55% and the Highway value by 45%."

7. Students can also explore the difference between highway and city MPG. Ask students to brainstorm: Why do most vehicles get better mileage on the highway than in the city? What kind of vehicle might get better city mileage than highway?

   Most vehicles get worse mileage in the city because they are stopping and going, and not driving at a constant rate. A hybrid car usually gets better mileage for city driving because they run off their electric motors. At a stoplight the electric motor shuts off, then starts again when the vehicle needs to accelerate. This makes it more efficient.

8. Now have students figure how many gallons they will use to take a road trip in their dream car.

   Note: *Teachers can pick a destination or allow students to pick their own. Calculate the one-way distance with a website like maps.google.com or mapquest.com. Example: from Laramie, WY to Denver, CO is approximately 145 miles (233 km). The equation is as follows: gasoline consumed = miles one way x 2 ÷ combined miles/gallon*

9. Once students have calculated their gasoline consumption, provide them with instructions to figure out the carbon footprint of the trip. They can do this by multiplying the number of gallons of gasoline consumed by 19.4. Burning one gallon of gasoline emits 19.4 pounds (8.8 kg) of CO_2 into the atmosphere.

   Note: *Ask students how a gallon of gas weighing about 6 pounds (2.7 kg) can release 19 pounds (8.6 kg) of CO_2. The answer is due to the combining of atmospheric oxygen with the carbon during combustion.*

10. Have students figure out how much it costs to drive 25 miles (40 km) using the *fueleconomy.gov* website. Now calculate the cost of fuel for the road trip. They can also use the application on the MapQuest website. (See above equation to calculate the cost of fuel for the road trip.)

Embedded Assessment

Ask students if they feel the cost of their road trip is expensive or cheap. Which students would be willing to get a more efficient car in order to save money? For advanced students, have them calculate what their costs would be if the price of gasoline increased by $1/gallon.

11. Now, let's imagine that in any average week you drive about 290 miles (467 km) (the same distance as your one trip to Denver). How much CO_2 are you emitting to the atmosphere in a month? How much will it cost? How much CO_2 in a year? What are the yearly costs?

    Note: *Select a large city a few hundred miles from your location.*

    Please refer to Appendix 2.6.2 for a supplement table to calculate the cost and CO_2 emissions for different vehicles driven over various distances in a year.

a. Discuss with students if they are surprised about how much CO_2 is produced from their trip. Students can refer to the tables in Appendix 2.6.2 to see how gas mileage will affect the cost and CO_2 produced for cars driven different distances.

b. Finally, distribute or display on the board the graphs in Appendix 2.6.3 and Appendix 2.6.4. Discuss the values regarding the amount of CO_2 emissions produced from fossil fuels. Ask students to contemplate how automobiles contribute to global CO_2 emissions and the changes that need to occur to curb emissions. Solutions could include policy changes such as a CO_2 tax, better transportation systems, new technologies, and behavioral change.

Wrap-Up/Student Feedback

(Including Assessment Opportunities)

Ask students the following questions:

1. If a car were not very fast or luxurious, would I still purchase it if it used less gasoline? Provide the following five options: (1) I would definitely do it, (2) I would almost certainly do it, (3) I would probably do it, (4) perhaps I would do it, and (5) I would probably not do it. Ask students to choose an option and then write about why they selected this option. It is important for them to include evidence from the activity in their answers.

2. Distribute or display on the board the graphs in Appendix 2.6.3 and Appendix 2.6.4. Transportation is a major contributor of CO_2 to the atmosphere. Based on what you have learned in this lesson, how can personal behaviors significantly reduce the CO_2 emissions? What are other ways to reduce CO_2 emissions?

   Note: *Consider using these graphs in the beginning of the class to show how automobiles contribute to CO_2 emissions.*

Supplemental Materials and References

Bureau of Economic Geology (n.d.). *Audience-pleasing physical models to support co_2 outreach*. Retrieved from http://www.beg.utexas.edu/gccc/forum/codexdetails.php?itemID=39.

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U.S. Environmental Protection Agency. (2014). *Clean Energy.* Retrieved from http://www.epa.gov/cleanenergy/energy-resources/refs.html.

U.S. Environmental Protection Agency. (2013). *Global greenhouse gas emission data.* Retrieved from https://www.epa.gov/climate-indicators/climate-change-indicators-global-greenhouse-gas-emissions.

Student Worksheet

1. Pick out your dream car. You need to include the year, make, and model. Record that here:

2. Briefly explain why you selected this vehicle. What do you like about it? What factors did you consider? (i.e. size, type, power, utility)

3. Next, you are going to figure out how many miles per gallon (MPG) this car uses. Go to the following site: *http://www.fueleconomy.gov/feg/findacar.shtml* and enter the year, make, and model of your dream car*. Record the city MPG here: _____ record the highway MPG here: _____ record the combined MPG here: _____

*Information on the most current year of car is not always available. You might need to select the previous year's model.

4. Why do you think that most vehicles get better mileage on the highway than in the city?

5. What kind of vehicle might get better city mileage than highway?

6. Now figure how many gallons you will use to take a road trip in your dream car. First, calculate the distance using Google Maps or MapQuest. Record that here: _____

7. Next, take that number and divide it by the MPG you found for your vehicle.

   Distance _____ /MPG _____ = _____ gallons

8. You will take this number and use it to calculate both:
   a. The cost of the trip (Gallons X current price per gallon = Cost of the trip $_____) and
   b. The amount of carbon emitted (Gallons X 19.4= _____ pounds of CO_2)

9. How is it when a gallon of gas weighs only about 6 pounds, your car can release 19 pounds of carbon dioxide? Hint: How does your car engine work? What is needed for the gasoline to power the car?

10. Do you feel like the cost of your road trip is expensive or inexpensive? Why?

11. Would you be willing to get a more efficient car in order to save money? Why or why not?

12. Bonus—Calculate the difference for this same trip if gasoline costs an additional $1.00 per gallon. Would that change your answer to question #11?

13. Now, let's say that on any average week you drive about 290 miles. How much CO_2 are you emitting to the atmosphere in a month? How much will it cost? How much CO_2 in a year? What are the yearly costs?

Calculating Cost and CO_2 Emissions for Different Vehicles

	Average MPG	Cost of driving 20,000 miles/ year	Amount of CO_2 emitted	Cost of driving 10,000 miles/ year	Amount of CO_2 emitted	Cost of driving 5,000 miles/ year	Amount of CO_2 emitted	Amount of $ saved by driving 10,000-15,000 miles less per year	Difference in amount of CO_2 emitted by driving 10,000-15,000 miles less per year
Car #1									
Car #2									
Car #3									
Car #4									
Car #5									

What other sorts of questions could be asked using this data?

For example:

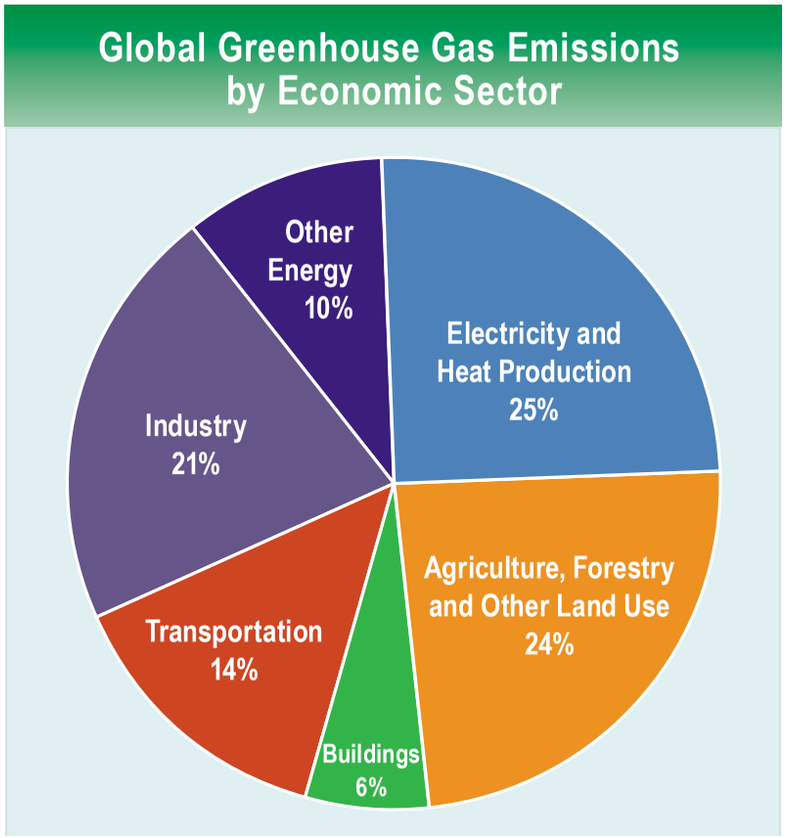
- What if you compare a town like Laramie (30,800 in 2010) and a city like Denver (620,100 in 2011) in terms of carbon load, assuming one car for every three or four people? What happens when you remove X number of cars (due to public transportation or use of bicycles)?
- What if we taxed people on carbon load in addition to having them pay for gasoline?

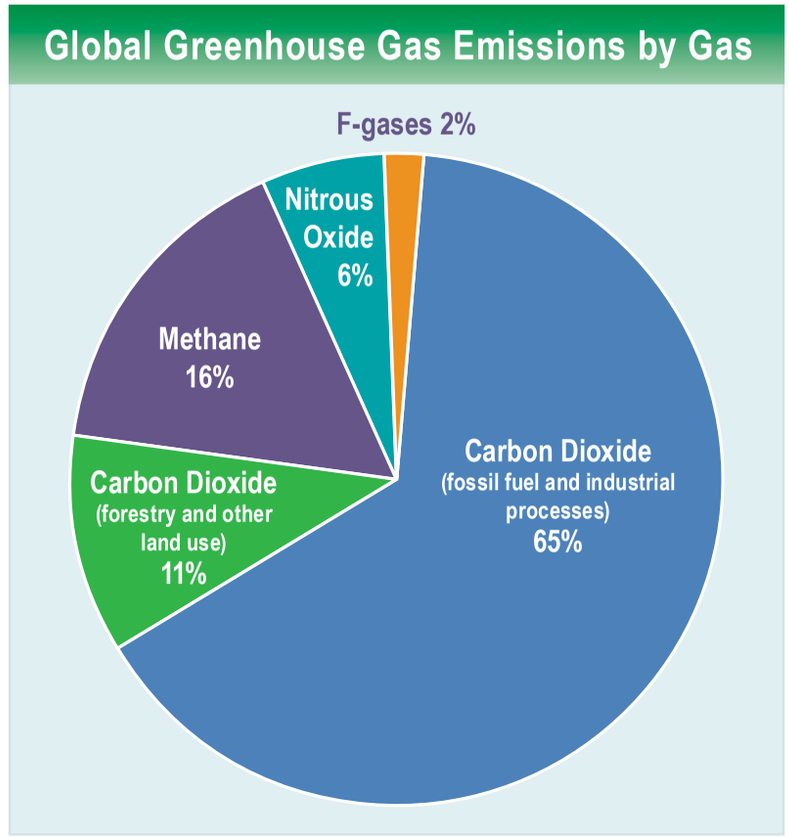
More background information for the teacher

Note: There are many ways this data could be looked at—many questions that could be asked and potentially answered. See what your students think of and follow their lead!

- It takes approximately 15 years to turn over "the fleet" of cars on the road. Programs like Cash for Clunkers, a government incentive program from the early 2000s, gave people large rebates for trading in high mileage vehicles for lower mileage ones. This helped to hasten a changeover. High gasoline prices can also hasten a changeover—something we have seen in the past decade.
- Public transportation, while emitting its own CO_2 load, could still be calculated and considered in terms of decreasing emissions.

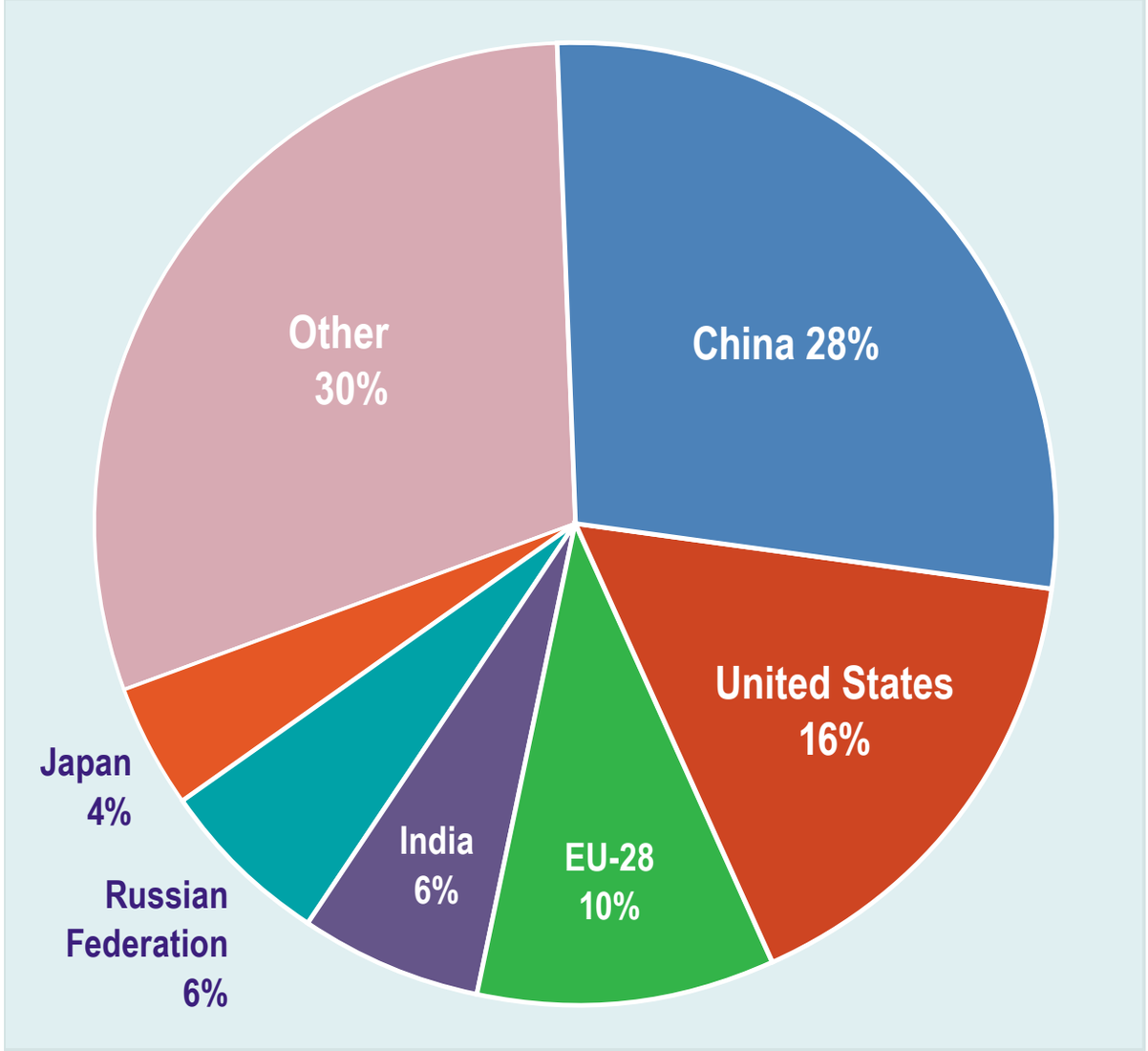
Emissions Charts





Source: IPCC (2014); Based on global emissions from 2010. Details about the sources included in these estimates can be found in the Contribution of Working Group III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change. Host Website: www.epa.gov. Source data link: https://www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data#Trends

2011 Global CO_2 Emissions from Fossil Fuel Combustion and Some Industrial Processes



Source: Boden, T.A., Marland, G., and Andres, R.J. (2015). National CO_2 Emissions from Fossil-Fuel Burning, Cement Manufacture, and Gas Flaring: 1751-2011, Carbon Dioxide Information Analysis Center, Oak Ridge National Laboratory, U.S. Department of Energy, doi 10.3334/CDIAC/00001_V2015. Host Website: www.epa.gov. Source data link: https://www.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data#Trends

Activity Set 3

THE SCIENCE OF CCS: ONE OPTION IN A BALANCED ENERGY PORTFOLIO

ACTIVITY SET 3

The Science of CCS: One Option in a Balanced Energy Portfolio

A Supplemental Curriculum
for Middle School Learners

A Midwest Geological Sequestration Consortium Program Funded by the U.S. Department of Energy

Activity Set 3

The Science of CCS: One Option in a Balanced Energy Portfolio

Lessons L–N

- **L** What is Under Your Feet? Basic Geology of the Illinois Basin
- **M** Exploring Porosity and Permeability
- **N** What are CCUS and EOR and How Do They Work?

Lesson L

What is Under Your Feet? Basic Geology of the Illinois Basin

Quick Overview

Lesson Scope: Identifying key components and properties of the Earth beneath our feet

Length: One ~45-minute class period

Grade Level: 5–8

Materials: Hand lenses, rock core samples, demonstration pieces of shale and sandstone, dropper, water

Materials Needed per Student

Hand Lenses

Core Samples

Piece of Shale

Dropper

Water

Each student will need paper and instructions to record trials.

Guiding Question

How can new technologies, such as carbon capture and storage (CCUS), help reduce the impact of increasing levels of carbon dioxide (CO_2) in the atmosphere?

Many students (or adults) may not have a clear understanding of what the Earth is composed of below their feet. To fully understand **carbon capture and storage**, the individual properties of rocks beneath the surface need to be identified and examined. This activity is designed to increase student knowledge of the geological storage of carbon dioxide (CO_2) as a potential technology to mitigate CO_2 release into the atmosphere, using the Illinois Basin as an example.

Key Understandings

- CCUS is the process of capturing CO_2 created from energy generation and other human activities—before it is released into the atmosphere—and storing it safely underground.
- Properties of the subsurface rocks make them a natural long-term storage place.

Objectives

- Students will identify key components of the subsurface.
- Students will be able to identify properties of the subsurface rocks of the Illinois Basin.

- Students will describe why the Illinois Basin is a natural storage place for CO_2 based on the unique arrangement of subsurface rocks.

Vocabulary

- Porosity
- Carbon capture and storage (CCUS)
- Caprock
- Permeability
- Stratigraphic layers
- Aquifer
- Surficial deposit

Background Information for the Teacher

Midwest Geological Sequestration Consortium (2011) *http://www.sequestration.org/*

The basic idea behind geological storage, or sequestration, is simple—to return the CO_2 released from the combustion fossil fuels back to the Earth where it was originally concentrated as organic carbon through the decay of ancient plant and animal life. However, the science behind geological storage raises many questions. How can carbon be captured? Which geological formations are best suited for long-term storage? How do we make sure the CO_2 stays safely stored? These are only a few of the questions that require a better understanding of the science of geologic storage.

CCUS is currently being studied in central Illinois. The Midwest Geological Sequestration Consortium (MGSC) is one of seven national research partnerships funded by the U.S. Department of Energy's National Energy Technology Laboratory that is working to find a balance between our growing energy needs and rising climate concerns by capturing CO_2 created in energy production and industrial processes and storing it safely underground in natural geological formations.

The Midwest Geological Sequestration Consortium (MGSC) is led by the Illinois State Geological Survey (ISGS) at the University of Illinois and, working in partnership with the Archer Daniels Midland Company (ADM) has stored one million tonnes of CO_2 (over a three-year period) into the Mt. Simon Sandstone of the Illinois Basin approximately 7,000 feet (2,134 meters) below the ground level. ADM provides the project site and the CO_2 for storage from their ethanol production plant. The CO_2 that is produced in this process is compressed into a dense liquid and injected into the Mt. Simon Sandstone.

The Illinois Basin is a large geological feature that lies beneath the land surface of most of Illinois, significant parts of Indiana, and parts of Kentucky. The Basin initially formed near the beginning of the Cambrian Period (542 million years ago) as a shallow, depression that opened to the seas to the south-southeast. By the time of the

Early Pennsylvanian (323 million years ago), the Basin's areal extent continued southward into eastern Arkansas, northern Mississippi, and northwestern Alabama. During the very end of the Pennsylvanian Period (299 million years ago) and into the Permian Period (299–251 million years ago), the southern end of the Basin was uplifted, forming the present-day, spoon-shaped Basin. (For a more in-depth study of Illinois geology and its history, see Build Illinois *http://www.isgs.illinois.edu/?q=outreach/geology-resources/build-illinois-last-500-million-years*.)

The geologic materials that trap and permanently store CO_2 safely in the Earth found in the Illinois Basin and sedimentary basins around the world provide a unique research opportunity. The target for geologic storage in the Illinois Basin is the Cambrian Mt. Simon Sandstone. The Mt. Simon is a regional rock unit that underlies large parts of the Midwest and almost all of the Illinois Basin.

In most of the central and southern parts of the Illinois Basin, the Mt. Simon Sandstone is very deep and is well below the depth of potable water aquifers. The Mt. Simon Sandstone reaches a thickness of about 2,600 feet (792 m) and is very thick in northern Illinois, east-central Illinois, and west central Indiana. The Mt. Simon Sandstone overall is a very fine- to coarse-grained quartz sandstone. The sandstone typically has a relatively high **porosity** (many open voids between the sand-size grains filled with salt water). The CO_2 can be stored in these small pore spaces.

The Eau Claire Shale is the primary seal, or **caprock**, that prevents upward migration of CO_2 from the Mt. Simon Sandstone attributable to its low **permeability**. The thickness of the Eau Claire Shale ranges from 300 feet (91 m) in the western and northern parts of the Basin to over 1,000 feet (305 m) in the southern part of the Basin. This formation is composed of a dolomitic sandstone or sandy dolomite in northern Illinois and siltstone or shale in the central part of the Basin. Secondary seals include the Upper Ordovician Maquoketa Shale and the Upper Devonian New Albany Shale, both of which are relatively thick, widespread, fine-grained **stratigraphic** layers.

Many natural geological resources are located within the Illinois Basin. Limestone, dolomite, and sand, gravel, and clay deposits are found near the surface of the Illinois Basin. Oil, produced from various periods of Illinois geological history, can be found in some limestone and sandstone formations. Natural gas is much less abundant in the Illinois Basin, but it is a by-product when the oil is brought to the surface. Natural gas also occurs within the coal layers in the Illinois Basin. Layers of coal are plentiful under the surface of a very large portion of the Illinois Basin. Water is another resource found in the Illinois Basin. CO_2 is stored far below these resources and is separated by multiple layers of caprock.

Common Student Misconceptions/ Student Challenges

Many students may not have a clear idea of what the subsurface of the Earth is like, even a few hundred feet, much less thousands of feet, underground. Student drawings show that they think soil will continue deeply underground, when in actuality it is very thin (115 feet (35 m) on average) compared with the distance the crust continues in the Earth's layers. They also think fossils, gold, or magma will be found. When asked to identify different layers, of rocks few may be able to identify what lies beneath their feet.

Students also have the misconception that rock layers are flat. Core samplings and seismic readings show the layers are more like a ribbon that varies in shape and depth.

Many believe that underground storage of CO_2 could affect their water supply. In reality, much of the drinking water in the Illinois Basin area comes from two sources: surface water, such as rain and snow melt, and shallow **aquifers** located in **surficial deposits** lying over bedrock. These bedrock aquifers in Northwestern Illinois range in depth from near the surface (0–150 feet or 0–50m) to depths greater than 1,000 feet (305 m). CCUS occurs at much greater depths, nearly 7,000 feet (2,134 m), and is separated from potable water by many layers of non-permeable rock. *http://sequestration.org/science/geologyofdeepsaline.html*

Another common misconception is that an earthquake in this area could create a fault or crack that would allow the CO_2 to escape into the atmosphere. Actually when choosing an underground storage site with a unit, like the Mt. Simon Sandstone, very extensive and specific site characterization is conducted out to those sites where CO_2 injection might cause a rock slippage, and therefore an earthquake. These sites are identified and avoided. Also, a big earthquake to a significant level (above magnitude 3) that humans can feel requires a very large slip, which is generally possible only through the Earth's natural forces acting on a fault. Such earthquakes would be extremely difficult to create during CO_2 injection, which would require breaking rocks through injection or withdrawal of fluids underground.

Next Generation Science Standards

Performance Standard 4-ESS2-2:

Analyze and interpret data from maps to describe patterns of Earth's features.

Disciplinary Core Idea

Measurements of a variety of properties can be used to identify materials.

Science and Engineering Practices

Analyze and interpret data to make sense of phenomena using logical reasoning.

Crosscutting Concepts

Patterns can be used as evidence to support an explanation. (4-ESS1-1) (4-ESS2-2)

Materials

- Hand lenses (2/group)
- Student worksheet (1/student) Appendix 2.7.1
- Core samples
- CCUS Diagram Appendix 2.7.1 (one per group)
- Stratigraphic column of the Illinois Basin Appendix 2.7.1 (one per group)
- For the demonstration: piece of shale and sandstone, dropper, and water

Setup

- Prepare core samples and equipment (for groups of 4)
- Worksheet (Appendix 2.7.1)
- CCUS diagram
- Stratigraphic column of the Illinois Basin (Appendix 2.7.1)

Length of Lesson

This lesson will take one 45-minute class period.

Teaching Strategy

Predict, working in groups, discussion, and demonstration.

Questions to Pose to Students

The following questions are designed to encourage students to recall information or assumptions they have on subsurface of the Earth. These results can be used as a pre-assessment or basis for further exploration in the geology of the Earth.

- Think about the Earth underground. What is under our feet?
- What do you think is below the grass? Make your predictions on the worksheet.
- Draw what you think scientists have found recently when they drilled 7,000 feet (2,134 m) into the ground in Decatur, Illinois.

Procedure

Activity

- Encourage students to use a key or legend when recording their predictions.
- Students should fold their worksheet in half lengthwise and fill in the left side. The right side will be used to record what scientists actually found.

1. Allow approximately 5 minutes for students to individually complete worksheet. (Appendix 2.7.1)
2. Use CCUS diagram to explain the basic procedure of carbon capture and storage. (Appendix 2.7.1)

3. Show students core samples that are labeled with the depth in which they were found. Students will work in groups to complete the right side of their worksheet using the core samples and the stratigraphic column of the Illinois Basin to accurately depict the underground layers found at the drill site.

4. The teacher will demonstrate the properties necessary for caprock and storage rock with pieces of sandstone and shale. Using a dropper filled with water, a small drop of water is placed on each sample. Students will observe how the drop is absorbed into the sandstone but sits on the surface of the shale.

   Note: *The following questions are to determine if students grasp the concept of the different properties of subsurface rocks.*

   - What is happening with the drops of water?
   - Can you use this information to describe which would be better as a storage rock and which would be better as caprock? Give evidence to support your answer.

5. Place students in small working groups (3 to 4 students) and provide each team with rock cutting samples taken at varying depths. Students should examine rock cuttings with a hand lens and record the depth in which they were found on the right side of their worksheet using an appropriate key to identify components. The stratigraphic column of the Illinois Basin can also be used to give a picture of the layers underground.

6. Within their group, allow students time to compare their original predictions with the data represented by the core samples. In a class discussion, share their findings.

   - Were you surprised by what is actually under your feet?
   - Do you think this layering occurs everywhere underground? Why or why not?
   - Why is the Illinois Basin a natural storage place for CO_2?

Wrap-Up/Student Feedback

(Including Assessment Opportunities)

- Participation in the activity.
- Completed: *What is under your feet?* Student Sheet (Appendix 2.7.1)
- Appropriate discussion responses to properties of layers within the Illinois Basin as a natural storage place for CO_2.

Extensions

Illinois State Geological Survey. Reinertsen, D. L. (n.d.). *Guide to the geology of the Decatur area, Macon and Christian counties.* Illinois State Geological Survey Fieldtrip Guidebook No. 1991 C. Champaign, IL: Illinois State Geological Survey.

Illinois State Geological Survey. (n.d.) *Build Illinois: The last 500 million years.* Retrieved from http://www.isgs.illinois.edu/?q=outreach/geology-resources/build-illinois-last-500-million-years.

Supplemental Materials and References

Carbon Capture and Storage, Graphics, ICO2N. (n.d.). Retrieved from http://www.ico2n.com/library/graphics.

Reinertsen, D. (n.d.). *Guide to the geology of the Decatur area, Macon and Christian counties.* Illinois State Geological Survey Fieldtrip Guidebook No. 1991 C. Champaign, IL: Illinois State Geological Survey.

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Illinois State Geological Survey (ISGS). (2010). *The geology of Illinois.* D. R Kolata and C.K. Nimz, editors. ISBN: 978-0-615-41739-4. Retrieved from https://shop.inrs.illinois.edu/isgs-books.html.

STRATIGRAPHIC COLUMN OF THE ILLINOIS BASIN

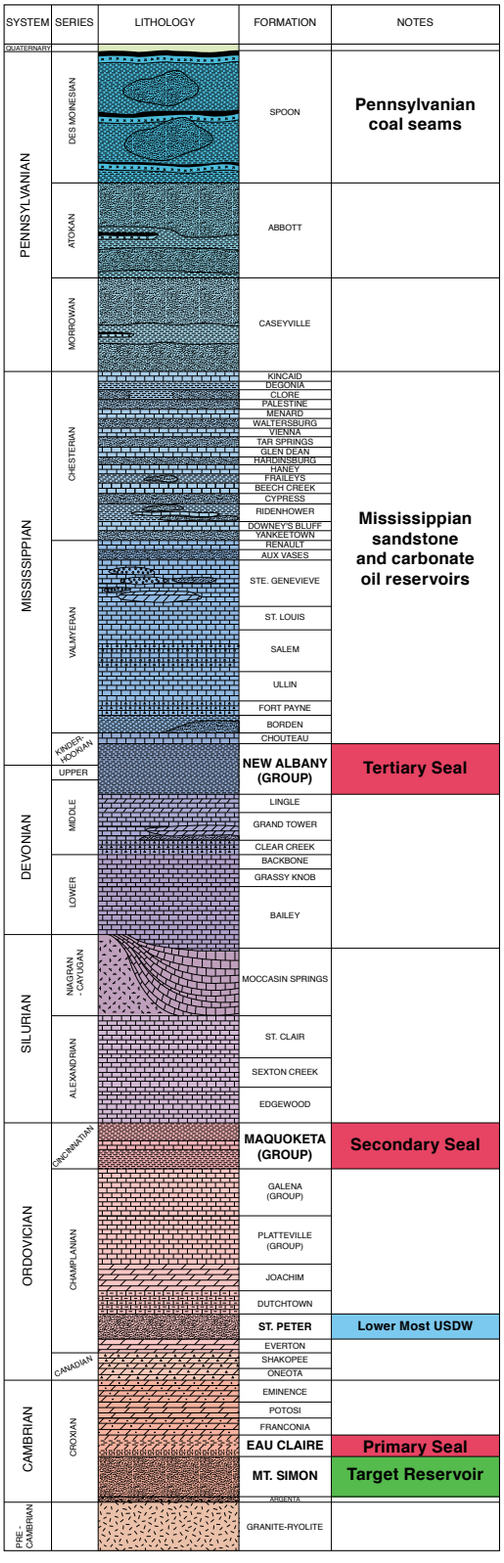


Figure 1. Stratigraphic column of the Illinois Basin at Decatur, IL USA.

CARBON CAPTURE AND STORAGE (CCUS)

CAPTURE

CO_2 is captured from large emission sources, including:

- Coal-fired power plants
- Oil and gas operations (including oil sands upgraders and refineries)
- Other industrial activities such as chemicals, fertilizers, and cement

1. Capture carbon dioxide (CO_2) emissions from large industrial facilities before the CO_2 is emitted to the atmosphere.

2. Transport the CO_2 through pipelines, and

3. Permanently store the CO_2 in deep, secure underground rock formations.

STORAGE

CO_2 is injected at a depth of 1 to 5 kilometers into deep underground rock formations where it is permanently stored. A barrier of impermeable rock and natural chemical interactions ensures long-term storage. Geological formations used for storage include:

- Depleted oil or gas reservoirs
- Saline formations
- Coal seams

CO_2 is transported using the same safe and reliable pipeline systems that have moved natural gas and oil for decades.

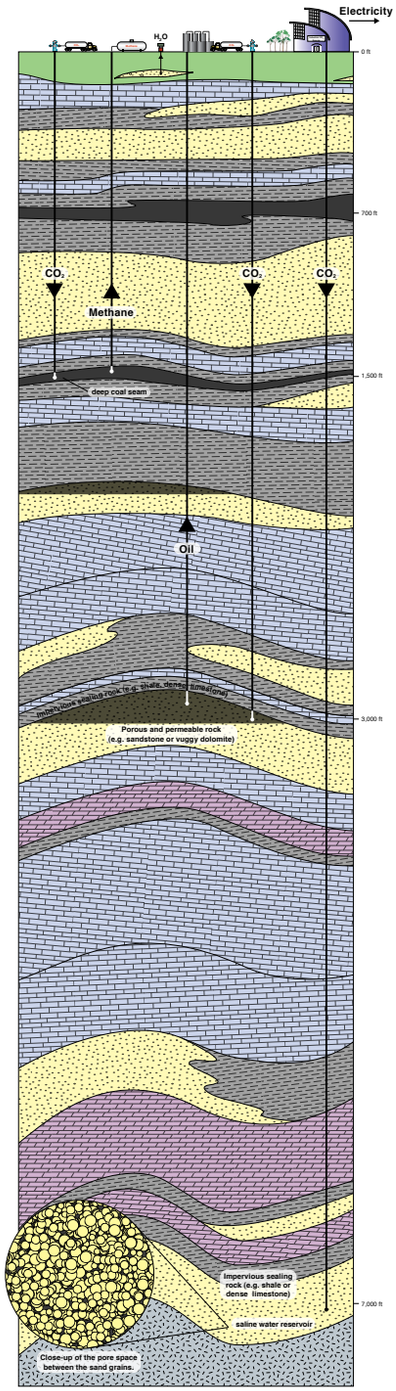
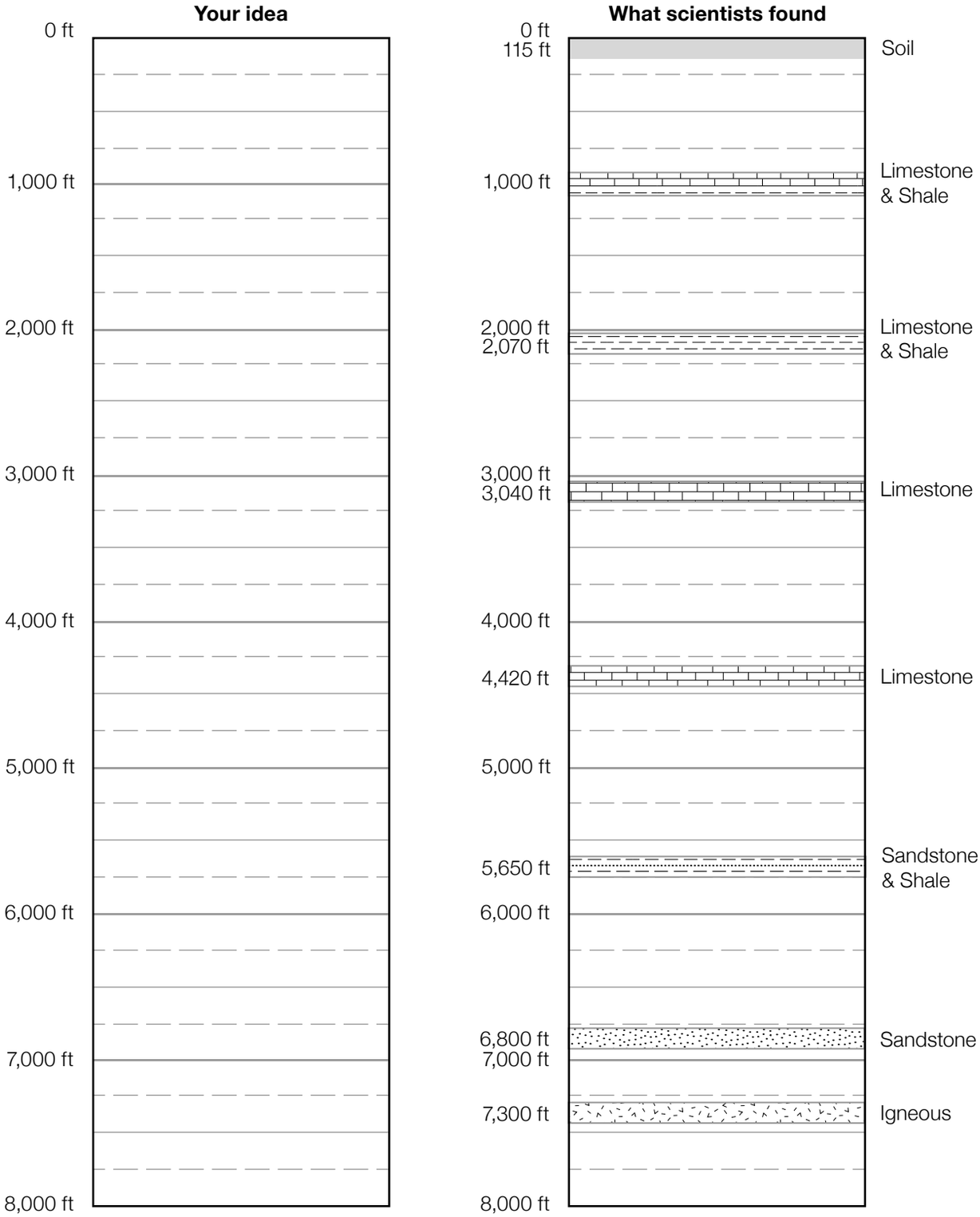


Figure 2. Stratigraphic column of the Illinois Basin at Decatur, IL USA.

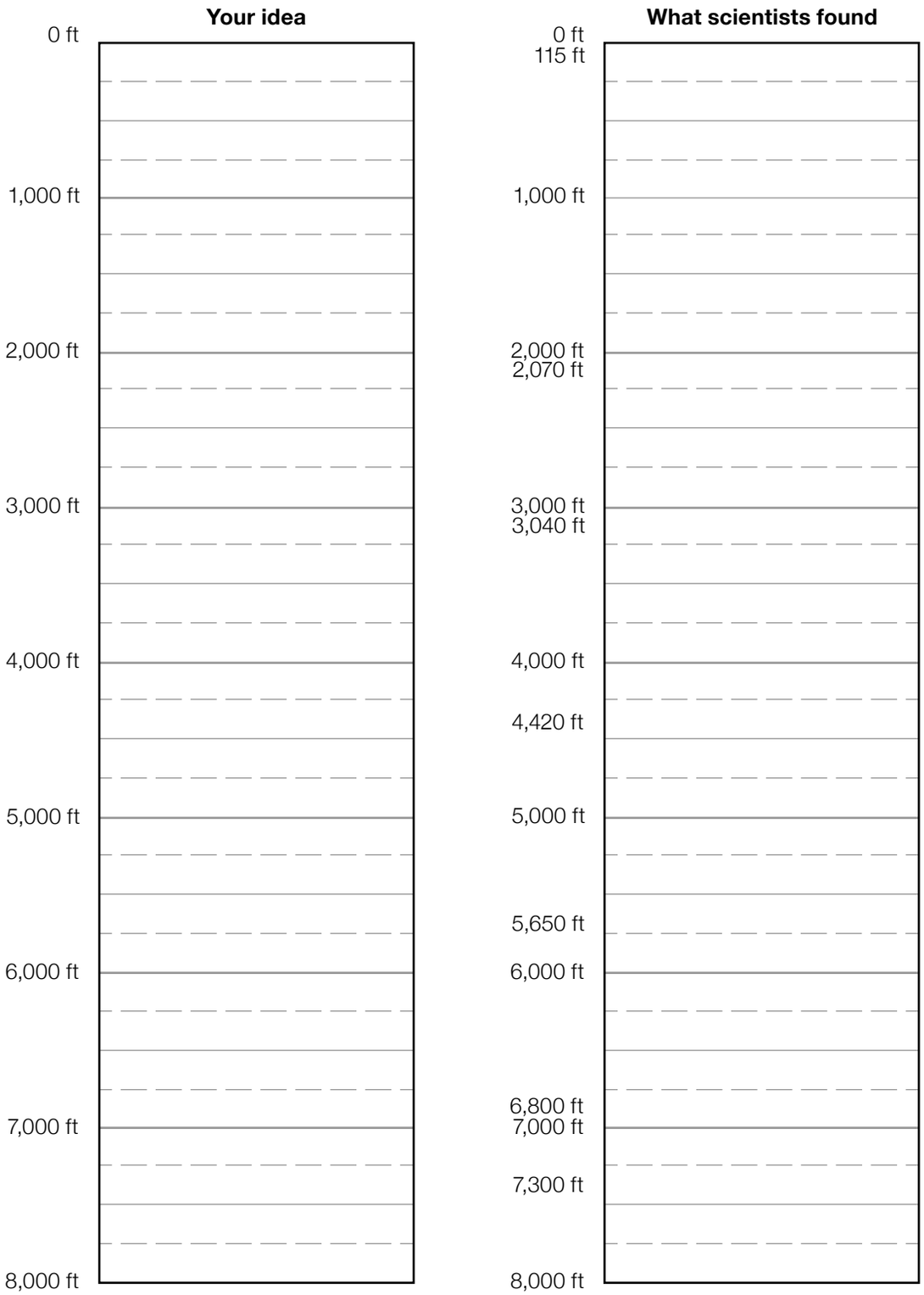
Draw what you think the Earth looks like down to 8,000 feet. Describe each of the layers and tell how deep it is.

each rectangle = 250 feet



Draw what you think the Earth looks like down to 8,000 feet. Describe each of the layers and tell how deep it is.

each rectangle = 250 feet



Lesson M

Exploring Porosity and Permeability

Quick Overview

Lesson Scope: Distinguishing between porous, permeable, and impermeable rock types

Length: One ~45-minute class period

Grade Level: 5–8

Materials: Pint jar of marbles, pint jar of sand, pint jar of mixed sand and marbles, 2-cup measuring cup, water, drinking straw

Materials Needed per Student

Pint jar of marbles

Pint jar of sand and marbles

Pint jar of sand

2 cup measuring cup

Water

Drinking straw

Each student will need paper and instructions to record trials.

Adapted from ***Vocabulary in a Jar, GeoActivity*** (WATR-2) from the Illinois State Geological Survey.

Guiding Question for the Unit

How can new technologies, such as carbon capture and storage (CCUS), help reduce the impact of increasing carbon dioxide (CO_2)?

This lesson is designed to provide students with knowledge of the properties of **porosity** and **permeability** of subsurface materials. This knowledge will provide a basis for understanding how we store CO_2 in rock formations.

Key Understandings

- **Porosity** *and* **permeability** are related properties of any rock or loose sediment. Both are related to the number, size, and connections of openings in the rock and are key factors in geological storage of CO_2.
- Some rock types with a high porosity are a natural storage place for CO_2.

Objectives

Students will be able to define and distinguish between porous, permeable, and **impermeable** rock types.

Vocabulary

- Porosity
- Permeability
- Impermeable
- Carbon capture and storage (CCUS)
- Reservoir rock
- Permeable
- Porous
- Pore space
- Caprock

Background Information for the Teacher

In earlier lessons we learned that levels of CO_2 are rising at alarming rates. We also learned that increased levels of CO_2 may be a contributing factor to climate change and global warming. **Carbon capture and storage (CCUS)** is being developed as a mitigation strategy to help with this issue. CCUS is a process in which CO_2 is captured and stored in rocks beneath the surface rather than sending it into the atmosphere. CCUS stores CO_2 permanentely deep underground.

It is important to understand the geological properties of the rocks which compose the geologic formations in order to understand why CCUS is currently being researched and developed.

This activity will use different materials, such as sand and marbles, in a jar to visually display porosity, space between grains, and permeability. Imagine your jar is a place to safely trap the CO_2. In real life, this jar would be replaced with a large underground rock or reservoir—an area deep under the ground with specific types of rock that can safely store CO_2, and is made of many grains of materials.

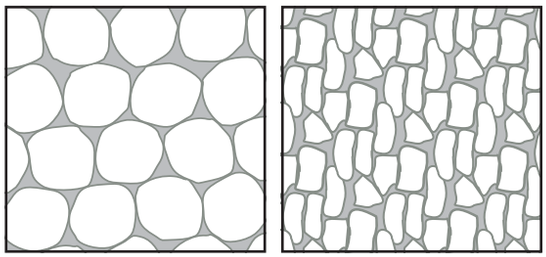


Figure 1. Different types of pore space.

In most rocks, grains touch but the rock is not completely solid. There are spaces between the grains, which are called pore spaces. Porosity is a measure of the percentage of pore spaces. Pore space can be used to store CO_2. This is often the same type of rock that has held oil or gas for millions of years. Pore spaces are very small, not big, empty cavities.

Reservoir rock is **permeable,** meaning liquids and gas can pass or flow slowly through it. Reservoir rock is also **porous**—it has tiny spaces or **pore space** where it can store liquids or gas.

Muddy sediments were layered on top of the reservoir rock and became **caprock**. Caprock is impermeable, meaning it will not let liquids or gas pass through it. This impermeable layer acts as a seal or cap,

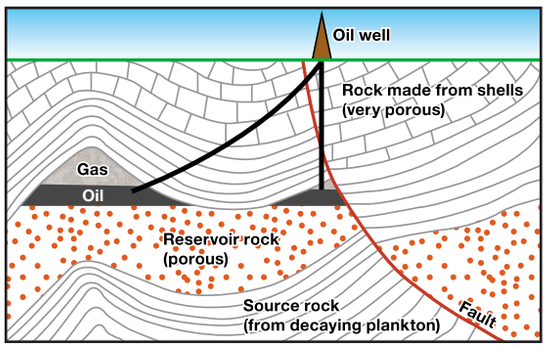


Figure 2. Geologic trapping of oil and gas.

and will trap any liquid or gas that is below, preventing the liquid or gas from making its way to the surface.

This figure shows good places under the Earth to try to find petroleum oil and natural gas.

Source: Science Buddies Staff. (2004, Oct 15). *Rocky Secrets: Where Does Oil Hide?* Science Buddies. Retrieved from http://www.sciencebuddies.org/science-fair-projects/project_ideas/Geo_p041.shtml (accessed Dec., 2015).

When looking for possible storage sites, scientists look for reservoirs that will keep the CO_2 securely within the intended storage formation. Not all reservoirs are suitable for geologic storage. Scientists look for places with special rock formations with impermeable rock above which seals the CO_2 in place. Scientists work to find places where leaks are unlikely and where rocks are thick with solid rock that "caps" the reservoir—like a lid on a jar—and prevents upward leakage or escape.

The jar shows you what you would see if you had a microscopic view of a CO_2 storage site underground. The marbles are sand grains, and the water is salt water that fills the spaces. (Teacher may want to view Porosity and Permeability http://www.youtube.com/watch?v=b0u7iG1PUwU)

Common Student Misconceptions/ Student Challenges

- Rocks, particularly bedrock, are solid and cannot hold other materials (fluids). All materials have pore spaces that can hold other materials.
- Many students believe that water and hydrocarbons lie in pools in empty cavities underground. This is not the case. The water (brine) and hydrocarbons actually are in the pore spaces (open areas) between grains that make up the rocks. These open areas are similar to the open areas in a sponge. Water and hydrocarbons can flow through the pores within the rock. They are not absorbed, rather, they move between the grains of the material through the pore spaces.

Next Generation Science Standards

Performance Standard.MS-PS1-1:

Develop models to describe the atomic composition of simple molecules and extended structures.

Disciplinary Core Idea

Substances are made from different types of atoms, which combine with one another in various ways. Atoms form molecules that range in size from two to thousands of atoms.

Science and Engineering Practices

Develop a model to predict and/or describe phenomena.

Crosscutting Concepts

Time, space, and energy phenomena can be observed at various scales using models to study systems that are too large or too small.

Materials

(For each group)

- 1 pint jar of marbles
- 1 pint jar of sand
- 2-cup measuring cup
- Water
- Drinking straw
- *Optional: a third pint jar filled with a mixture of sand and marbles*

Setup

- Prepare material and equipment for each group of students

Length of Lesson

This lesson will take one 45-minute class period.

Teaching Strategy

Group participation and discussion

Suggested Procedure

Activity

1. Ask the students what they see in the jar of marbles. They should mention marbles. When they mention air space, be sure to equate that with porosity.
2. Ask the students to estimate (hypothesize) if any water can be added to the jar of marbles, and if so, how much water can be added.
3. Slowly pour a measured amount of water into the jar. Stop when the jar is half full and point out the zone of saturation, water table, zone of aeration, and groundwater.
4. Finish filling the jar with water. You should be able to pour about 1 cup of water in each jar. Record how much water was added.
5. Explain how to measure porosity, based on your results. The pint jar holds 2 cups. You were able to add 1 cup of water to the jar with marbles. Therfore 1 cup is ½ or 50% of the capacity, which equals your porosity. Calculate the porosity of the materials using this formula: volume of water/porosity = volume of material × 100.
6. On a separate piece of paper, students should repeat the process with a jar of marbles and a jar of sand, which has a lower permeability than the marbles. Explain that porosity is

not related to grain size, but that permeability is. Larger grain size may have higher permeability.

7. Record data and calculate porosity using the equation.
8. When the investigation is completed, compare and discuss the results.
9. Which material would be better to store CO_2?
10. Which would be better as a caprock?
11. Relate this information to the rocks found in the subsurface of your area.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

- Participation in the activity
- Completed Porosity/Permeability student sheet

Supplemental Materials and References

Science Buddies. (2013, October 8). *Rocky secrets: Where does oil hide?* Retrieved from http://www.sciencebuddies.org/science-fair-projects/project_ideas/Geo_p041.shtml#background.

Videos/Animations

Baldwin, P. (2013, March 13). *Porosity and permeability lab.* Retrieved May 16, 2014, from http://www.youtube.com/watch?v=b0u7iG1PUwU.

Google Images. (n.d.). *Porosity and permeability - google search*. Retrieved May 16, 2014, from https://www.google.com/search?q=porosity+and+permeability&client=firefox-a&hs=p5j&rls=org.mozilla:en-US:official&channel=sb&tbm=isch&tbo=u&source=univ&sa=X&ei=UWV2U-TNAoSYqAbEjoG4Cg&ved=0CEgQ7Ak&biw=1352&bih=579.

Lesson N

What are CCUS and EOR and How Do They Work?

Quick Overview

Lesson Scope: Practicing the concept of using a scientific model

Length: One ~45-minute class period

Grade Level: 5–8

Materials: 100 ml beaker, two-hole rubber stopper with plastic tubes, 250 ml flasks, clear glass marbles, rubber tubing, safety glasses, 100 ml vinegar, 12 ml syringe, water, food coloring, baking soda

Materials Needed per Student

100 ml beaker

Two-hole rubber stopper with plastic tubes

250 ml flasks

Clear glass marbles

Flexible plastic tubing

Safety glasses

Each student will need paper and instructions to record trials.

Adapted from *CSI: Climate Status Investigations*, The Keystone Center.

Guiding Question for the Unit

How can technologies, such as carbon capture, utilization and storage (CCUS), help reduce the impact of increasing levels of carbon dioxide (CO_2) in the atmosphere?

This lesson is designed to use engineering principles to model **enhanced oil recovery (EOR)** to improve student understanding of CCUS as a way to mitigate our current levels of CO_2 emissions.

Key Understandings

- Students will have an opportunity to practice the concept of using a **scientific model** to simulate situations they cannot practically experience in a classroom.
- Engineers improve existing technologies or develop new ones to increase their benefits, decrease known risks, and meet societal demands.

Objectives

- Students will simulate EOR, a potential form of oil production and carbon storage, as an idea being

considered to use CO_2 instead of releasing it into the atmosphere.

Vocabulary

- Enhanced oil recovery (EOR)
- Scientific model

Background Information for the Teacher

The basic idea behind carbon capture and storage is simple—to return the CO_2 released from burning fossil fuels back into the earth where it originated from as organic carbon created through the decay of ancient plant and animal life. However, the science behind geological storage raises many complex questions. How can the CO_2 be captured? Which geological formations are best suited for long-term storage? And how do we make sure the CO_2 stays safely stored? These are only a few of the questions that require a better understanding of the science of geological storage.

Most people do not realize that, as part of the carbon cycle, CO_2 remains in the atmosphere for an average of 200–300 years. This means that climate changes are underway and will continue for a thousand years or more, even if we could end CO_2 emissions immediately. The goal behind CCUS is to provide a way to hold CO_2 emissions to present-day levels or keep global carbon emissions flat until new technologies can be developed that will meet current energy needs with less environmental impact. This goal includes finding uses for CO_2 that reduce the amount of CO_2 we release into the atmosphere while also realizing a benefit from CO_2 usage.

Note: *See Wedge Game in Lesson Q for a practical application of this in relation to global carbon emissions.*

Enhanced Oil Recovery (EOR) is the process of obtaining oil from rocks, which is difficult to produce. EOR can increase the production of oil from existing oil fields, especially when oil fields have been producing for many years. The oil that remains in the field is more difficult to recover. Using EOR methods increases the flow and production.

By using CO_2 captured from industrial and power plant sources, we have the potential to produce billions of additional barrels of trapped oil and gas. Additionally, by using the CO_2 produced from an industrial or power plant source, we have the added environmental benefit of not releasing the CO_2 into the atmosphere. After completion of EOR activities, the CO_2 used in oil recovery can be permanently stored in the former oil formation.

People may incorrectly think there is a big, open space or underground cave where CO_2 can be stored. This activity uses a scientific model (a mental or physical representation of a process or structure) to show how CO_2 can be stored underground in the pore spaces in rock.

This activity will allow students to build a model that will show how CO_2 can help produce oil from a rock where it has been trapped for millions of years.

Common Student Misconceptions/ Student Challenges

- Students may want to extend the model beyond its known range of applicability. Sometimes this works and sometimes it does not. Thorough testing of the model is critical to ensure the validity of the results. Likewise, identifying variables and changing one at a time is important.
- Students may think they can "fix anything" through development of a model. Models are close approximations that work because they follow known patterns of nature. Models are still valuable because they embody hypothesizing and theory building-trial and error.

Next Generation Science Standards

Performance Standard. MS-ETS1-1:

Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

Disciplinary Core Idea - ETS1.A: Defining and Delimiting Engineering Problems

The more precisely a design task's criteria and constraints can be defined, the more likely it is that the designed solution will be successful. Specification of constraints includes consideration of scientific principles and other relevant knowledge that are likely to limit possible solutions.

Science and Engineering Practices - Constructing Explanations and Designing Solutions

Define a design problem that can be solved through the development of an object, tool, process, or system and includes multiple criteria and constraints, including scientific knowledge that may limit possible solutions.

Developing and Using Models

Develop a model to generate data to test ideas about designed systems, including those representing inputs and outputs.

Crosscutting Concepts - Influence of Science, Engineering, and Technology on Society and the Natural World

All human activity draws on natural resources and has both short and long-term consequences, positive as well as negative, for the health of people and the natural environment. The uses of technologies and limitations on their use are driven by individual or societal needs, desires, and values; by differences in such factors as climate, natural resources, and economic conditions.

Materials

Each student group will need:

- 2 - #6, two-hole rubber stopper with rigid plastic tubes inserted
- 2 - 250 ml Erlenmeyer flasks
- 1 - 400 ml beaker
- 2 - 11 cm (~4.5 in.) rigid plastic tubing
- 2 - 14 cm (~5.5 in.) rigid plastic tubing
- 2 lengths of flexible plastic tubing, 30 cm (~12 in.) long each
- 1 - 12 ml syringe (no needle)
- Clear glass marbles from a hobby or garden supply store
- 100 ml (~1 cup) of vinegar
- Supply of water
- Food coloring to represent oil
- Box of baking soda
- Safety glasses for each student
- Copies of How Does EOR Work? Student Sheet - 1 per student

Setup

- Prepare appropriate number of lab stations for your class (usually 4 in a group)
- Prepare student worksheets (Appendix 3.3.1)
- Review procedure

Length of Lesson

This lesson will take one 45-minute class period.

Teaching Strategy

Group participation and discussion

Suggested Procedure

Activity

1. Explain that students will perform an activity to learn about the method of enhanced oil recovery and geologic carbon storage using the concepts of porosity and permeability.
2. Explain that this model requires CO_2 and that the students will be making CO_2 by mixing vinegar and baking soda. It is important to note safety procedures and wear safety glasses. Divide students into groups. They should then move to a lab station with the appropriate materials needed to complete the lab.
3. Hand out *How Does EOR Work?* Lab procedure. Review.
4. Have students write their hypothesis on the *How Does EOR Work?* student sheet
5. Allow students to conduct the lab while you roam the room and help.
6. When students have completed the lab, ask them to clean their lab

materials and station so that the next class can use the materials.

7. Discuss experiment results using the student sheet as a guide.

Wrap-Up/Student Feedback

(Including assessment opportunities)

- Participation in the lab activity
- Completed How Does EOR Work? student sheet

Following the class discussion related to CCUS, ask students to answer these questions independently:

- *Based on your studies of carbon capture and storage so far, do you think this is an effective way to utilize CO_2?*
- *What modifications would you make?*
- *What are some other possible solutions to reduce global carbon emissions?*
- *How else is CO_2 used?*

Appendix 3.3.1 | Lab Procedure – How Does EOR Work?

Name _____ Date _____

1. Be sure you have the materials listed below at your lab station:
   - 2 - #6, two-hole rubber stopper with rigid plastic tubes inserted
   - 2 - 250 ml Erlenmeyer flasks
   - 1 - 400 ml beaker
   - 2 - 11 cm rigid plastic tubing
   - 2 - 14 cm rigid plastic tubing
   - 2 lengths of flexible plastic tubing, 30 cm long each
   - 1 - 12 ml syringe (no needle)
   - Clear glass marbles from a hobby or garden supply store
   - 100 ml of vinegar
   - Supply of water
   - Food coloring to represent oil
   - Box of baking soda
   - Safety glasses for each student

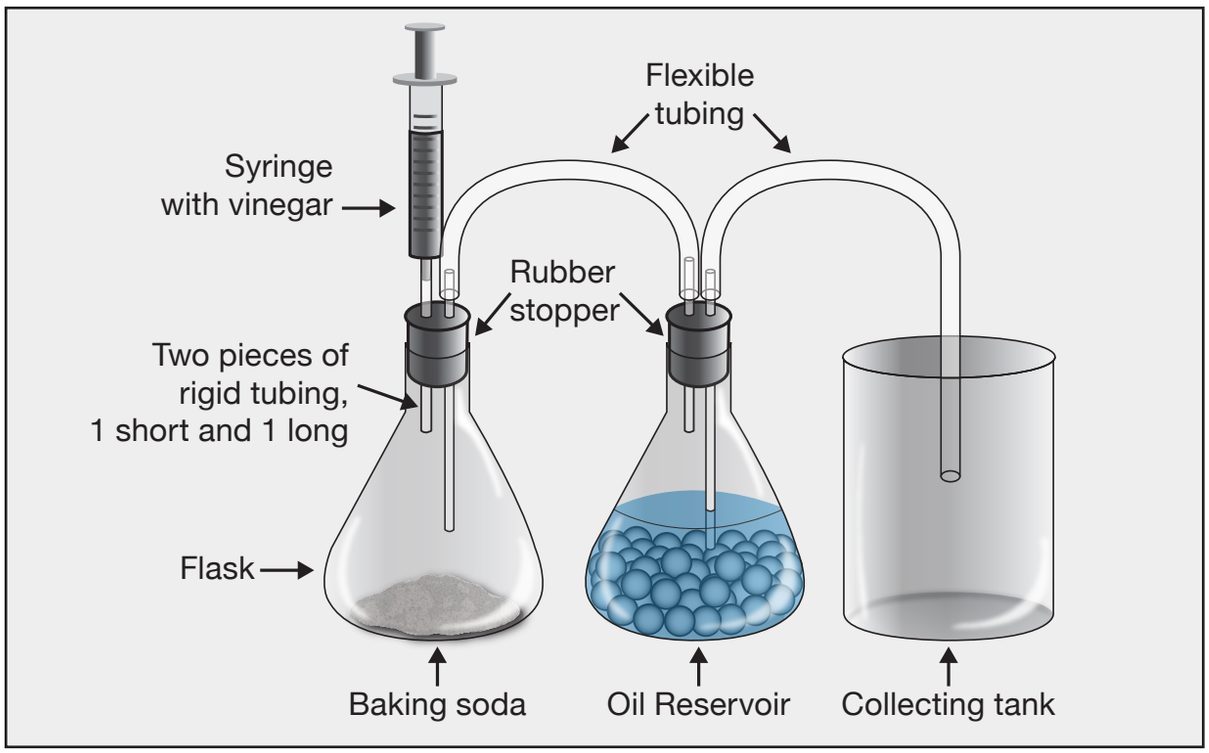


Figure 1. Enhanced oil recovery lab setup.

2. Assemble the CO_2 generator, oil reservoir, and collecting tank using Figure 1 (modified from The Keystone Center) above. Make sure that all connections are airtight. Place enough baking soda in the flask to cover the bottom (see step 6).

3. Put on your safety glasses.

4. Write your hypothesis on the Geologic Sequestration student worksheet.

5. Pour about 40 ml of vinegar into a cup. Put the tip of the 12 ml syringe into the vinegar, making sure that the plunger is all the way down. Keep the tip of the syringe below the surface as you pull back on the plunger to fill it to the 10 ml mark. If you get air bubbles in the syringe, empty it and repeat the procedure again.

6. Place two to three tablespoons of baking soda in the first flask.

7. Put the marbles in the second flask and add about 200 ml of water and some food coloring, which represents oil. This will represent an underground oil deposit.

8. Connect tubing as shown in the picture.

9. Place the syringe into the straw on the rubber stopper on the first flask and slowly add 10 ml of vinegar to the baking soda. The gas that is being produced (carbon dioxide) will push oil out of the underground deposit and into the collecting tank.

10. Slowly add more vinegar to the baking soda until the oil stops flowing into the collecting tank.

11. When you have finished this activity, your instructor will tell you how to clean up your materials.

12. Answer the questions on the Geologic Sequestration student worksheet.

Name _____ Date _____

Hypothesis: If I add CO_2 to the "oil reservoir" I think…

1. Why would EOR be used in an oilfield?

2. What caused the oil to leave the reservoir? Be specific.

3. What percentage of the oil was recovered from the reservoir?

4. Why can you not recover all the oil from your reservoir in this exercise?

5. Where does the CO_2 for this type of sequestration come from?

6. What happens to the CO_2 once it is pumped into the ground?

7. Do you think that this is a practical method to reduce CO_2 emissions into the atmosphere? Explain your answer.

Activity Set 4

TAKING IT TO THE NEXT STEP

A Supplemental Curriculum
for Middle School Learners

A Midwest Geological Sequestration Consortium Program Funded by the U.S. Department of Energy

Activity Set 4

Taking It to the Next STEP

Lessons O–R

Starting Local:
The Energy Saving
Challenge

Local Energy Debate

Career Opportunities

Science Fair
Opportunities

Lesson 0

Starting Local: The Energy Saving Challenge

Quick Overview

Lesson Scope: Reducing carbon CO_2 emissions through energy awareness

Length: Three ~45-minute class periods

Grade Level: 5–8

Materials: Computers, post-it notes, kill-a-watt meter, energy bills, measuring tape, digital thermometer

Materials Needed per Student

Computers

Post-it Notes

Kill A Watt Meter

Each student will need paper and instructions to record trials.

Guiding Question

How can we become agents of change to address current and future energy and environmental issues?

This lesson requires students to take action to reduce carbon dioxide (CO_2) emissions through an energy awareness campaign. Students complete an energy audit with recommendations for action steps to support **energy conservation** and **energy efficiency** measures.

Key Understandings

- Students' personal energy use connects to global issues.
- Energy use contributes to CO_2 emissions. Students can limit CO_2 emissions by changing behaviors.
- Energy saving behaviors and utilization of energy efficient technologies also saves money.

Objectives

- Students will estimate the energy consumption and the cost of energy at their school and in their classroom
- Students will collect data through an energy audit at their school.
- Students will design a method for reducing human impact on the environment.

Vocabulary

- Energy conservation
- Energy efficiency

Background Information for the Teacher

Energy efficiency and energy conservation are two ways to reduce the amount of energy we use. Energy conservation means that you use electricity (or another energy source) only when needed. Examples of energy conservation include turning off the light when you leave a room or walking to school instead of driving. Another way to conserve energy is by choosing to invest in energy efficient technologies. Energy efficiency refers to the amount of energy that is used to power a machine or appliance, versus the amount of energy that is lost as heat. For example, a compact fluorescent light bulb (CFL) produces the same amount of light (or lumens) as the incandescent light bulb but uses less energy.

Again, energy conservation is the reduction of or elimination of unnecessary energy use and waste. Energy efficient technology can get the same job done but uses less energy to do it. The Environmental Protection Agency (EPA) and US Department of Energy set guidelines for energy efficient products. Products that are deemed energy efficient receive the Energy Star label. More information can be found at *http://www.energystar.gov/*

In this lesson, students will complete a walk-through energy audit. An energy audit is an analysis of a building's energy use to pinpoint where energy can be saved through energy efficiency and energy conservation measures.

Common Student Misconceptions/ Student Challenges

Students may think their actions have little impact. This lesson allows students to become the specialist, giving them the opportunity to make a significant impact—both in cost saving and increasing energy awareness.

Next Generation Science Standards

Performance Standard.MS-ESS3-3:

Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment.

Disciplinary Core Idea

Human activities have significantly altered the biosphere, sometimes damaging or destroying natural habitats and causing the extinction of other species. But changes to Earth's environments can have different impact (negative and positive) for different living things.

Typically as human populations and per capita consumption of natural resources increase, so do the negative impacts on Earth, unless the activities and technologies involved are engineered otherwise.

Science and Engineering Practices

Apply scientific principles to design an object, tool, process, or system.

Crosscutting Concepts

The uses of technologies and any limitations on their use are driven by individual or societal needs, desires, and values; by the findings of scientific research; and by differences in such factors as climate, natural resources, and economic conditions. Thus technology use varies from regions to region over time.

Performance Standard.HS-ESS3-4:

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.

Disciplinary Core Idea

Scientists and engineers can make major contributions by developing technologies that produce less pollution and waste, and that preclude ecosystem degradation.

When evaluating solutions it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, as well as to consider social cultural, and environmental impacts.

Science and Engineering Practices

Design or refine a solution to a complex real-world problem, based on scientific knowledge, students-generated sources of evidence, prioritize criteria, and tradeoff considerations.

Crosscutting Concepts

Feedback (negative or positive) can stabilize or destabilize a system. Engineers continuously modify these technological systems by applying scientific knowledge and engineering design practices to increase benefits while decreasing cost and risks.

Materials

- Computer with internet access
- ~100 Post-it notes
- Kill-a-Watt meter
- 12 months energy bills (Electric and Natural Gas)
- Measuring tape
- Digital thermometer

Student worksheet – Estimated Energy Usage and Cost (Appendix 4.1.1)

Student worksheet – Energy Conservation and Energy Efficiency (Appendix 4.1.2)

Student handout – The Charge (Appendix 4.1.3)

Setup

Note: *It is best to encourage students to request the information from district officials.*

- At least, two weeks prior to teaching this lesson, request electric and natural gas bills from your school district office.
- Make copies of all student handouts and worksheets
- Speak to your school administrator about possibly funding student programs with money saved due to potential outcomes of the energy audit. You could solicit input from the students.

Length of Lesson

This lesson will take a minimum of three class periods and could take as long as a week depending on how much time you spend on the energy audits and presentations.

Suggeted Procedure

This lesson is closely related to Plugged into CO_2 (Lesson E), which allowed students to calculate the amount of electricity needed to run various home appliances. That lesson discussed wattage (a unit of power, measured as the amount of electricity converted to work) and allowed student to calculate total daily use. You may need to remind student of the concepts used in that lesson or conduct lessons close together.

Day 1

Review leading to the Energy Audit. (Although this lesson may seem repetitive, it is important to ask students to revisit energy consumption prior to initiating the Energy Audit.)

1. Initiate discussion by asking students the following question: What are all the ways our school uses energy?

   This question should serve as a review for students. They have been exploring energy uses throughout the unit. At this point, it is important to remind students that these uses (lighting, heating, etc.) may also be termed energy services. Energy services are conveniences that are afforded by our ability to harness and use energy. Not all societies have access to the same energy resources and technologies.

2. Identify classroom devices powered by electricity (5 minutes). Give each student 2 sticky notes. Ask them to locate and label two devices that are powered by electricity. Each post-it note should include the student's name, identification of device, and estimated wattage. (Students can use the Kill-a-Watt Meter to get wattage). Remind students that watts are a unit of power (energy used [joule] divided by the time).

3. Estimated Cost (20 minutes). Next, provide each student with a copy of Handout 1 (Appendix 4.1.1). Before students begin working on the handout, ask each to share the information on his/her device (name and wattage). Then, ask students to work with a partner to complete the remainder of the handout. This should serve as a review for the students.

4. After students have figured the cost to power the classroom for one week, find the average and multiply by 4. This will provide an approximate cost for 1 month.

5. Discussion (5 minutes). Ask the students to consider the cost to power their classroom for one month.

   Did they originally think the cost would be higher or lower? And why?

   Most likely, students will think that the cost is minimal.

   Now, pass out the school or district electric bills for the last twelve months (1 month's bill to each pair of students). Designate a class leader to record the kWh and cost (have the table projected onto a Smartboard® or drawn on a whiteboard).

6. Now, ask students to reflect on and describe the electricity consumption and cost.

   Students are often surprised by the significant cost to the school and district. Take a minute to listen to and record student responses.

Month	Electricity (kWh)	Cost $
January		
February		
March		
April		
May		
June		
July		
August		
September		
October		
November		
December		
Total		

Day 2

1. Introduction (5 minutes): Begin class by reviewing student responses to the district utility bills for electricity. Then, ask the students: Is money (or cost) the only concern in relation to electricity consumption?

   From previous lessons, students should have an understanding of climate change and the importance of CO_2 emissions.

2. Review the basic energy flow provided by the National Academies' What You Need to Know About Energy at *http://needtoknow.nas.edu/energy/interactive/understanding-efficiency/*

   Click on the "At Home" link. This diagram demonstrates energy source and loss throughout process of getting electricity to our homes and schools.

3. Handout 2 (Appendix 4.1.2) (15 minutes). Understanding Energy Conservation and Energy Efficiency. This handout requires students to have access to a demonstration provided by the National Academies at the above link.

4. Once students have an opportunity to explore energy efficiency and energy conservation, it is important to ask them how they can make a difference. Allow students time to brainstorm avenues for increasing awareness and reducing energy usage within their school. At this point, encourage them to focus on electricity. The challenge is for the teacher is to step back and let students take charge.

5. Energy Audit: Collecting Data. Explain to students that, when an auditor enters a building, he/she is looking for signs of energy use, energy loss, and opportunities for change (either through behavior or energy efficient technologies). They have already explored the different devices that are powered through electricity. Their job now is to explore ways the school is losing and wasting energy. Focused only on electricity at this point, break students up into two groups.

   - Watt busters
   - Lights-on brigade

6. See Handout 3 (Appendix 4.1.3) for instructions.

   Broader Impacts – Students will analyze their findings to determine the potential reduction in CO_2 emissions.

   Students will use the equation to convert energy usage into mass of CO_2:

$$(X)\ \text{KwH} \times \frac{1.52\ \text{lbs}\ CO_2}{1\ \text{KwH}} = (X)\ \text{lbs}\ CO_2$$

Days 3 to 4

7. Have students present their findings to an audience of administrators, school board members, other classes, and/or community members.

It would be a good idea to practice these presentations in class first

It will take a class period to present to each other and give feedback, and another similar amount of time to present to others.

8. Have students reflect on the process and the product by journaling. Questions students could answer include:

   a. What are three things you learned about the school in this process?

   b. What surprised you? What was not surprising?

   c. Give an example of a part of this process that you could take home or into the community?

   d. What is the next step you would like to take?

Wrap-Up/Student Feedback

(Including Assessment Opportunities)

Student feedback should be provided through the use of an authentic audience. It is important for students to report energy audit findings and recommendations to school administrators, faculty, and students. Students should present clear data and informed recommendations.

Supplemental Materials and References

Energy Works Michigan. (2014). *Helping to build a sustainable, equitable, prosperous energy future in Michigan*. Retrieved from http://energyworksmichigan.powerdash.com/home/

National Academy of Science. (2012). *What You know about energy*. Retrieved from http://needtoknow.nas.edu/. This site provides information and valuable visuals for understanding energy conversions (and loss) and energy efficiency.

Science Buddies. (2014). *Engineering Design Process*. Retrieved from http://www.sciencebuddies.org/engineering-design-process/engineering-design-process-steps.shtml#theengineeringdesignprocess. This site provides an excellent flow diagram of the Engineering Design Process.

University of Wyoming and Montana State University. (2014). *E3A: Exploring energy efficiency and alternatives*. Retrieved from http://e3a4u.info/

U.S. Department of Energy. (n.d.). *Do-It-Yourself Energy Audits*. Retrieved from http://energy.gov/energysaver/articles/do-it-yourself-home-energy-audits. This website helps teach quick tips for conducting an energy audit and prioritizing energy efficiency upgrades.

U.S. Environmental Protection Agency and the Department of Energy. (n.d.). *Energy Star*. Retrieved from http://www.energystar.gov. This site provides information on the Energy Star program and qualifying products.

Handout 1 – Estimated Energy Usage and Cost

Name: _____ **Date:** _____

Instructions: Your task is to complete an energy audit. You and your classmates are charged with estimating the total electrical energy in your classroom. Each of you identified devices that are powered by electricity and estimated the wattage. Share your device and wattage with the class. Now, determine the cost of powering these devices for one school week (or 5 days).

	Device	Est. Watts	Est. kW	Est. Time (hours)	Est. Energy kWh	Cost kWh *12 cents/kwh
1.	Light bulb	60 W	.06 kW	10 h	.60 kWh	7.2 cents
2.	Projector	250 W	.250 kW	4h	1 kWh	12 cents
3.	Pencil Sharpener	80 W	.08 kW	.25h	.02 kWh	.24 cents
4.						
5.						
6.						
7.						
8.						
9.						
10.						
11.						
12.						
13.						
14.						
15.						
16.						
17.						
18.						
19.						
20						
21.						
22.						
23.						
					Total Cost Per Day	$
					Total Cost Per Week	$

Handout 2 - Energy conservation and energy efficiency

: _____ : _____

Take 5 minutes to review the differences in energy used, CO_2 vented, and pounds of coal burned in the demo provided by the National Academies at the following link: *http://needtoknow.nas.edu/energy/interactive/understanding-efficiency/*

Based on the information provided, answer the following questions:

How much energy do you save in kWh per month if you select an energy efficient CFL bulb instead of an incandescent bulb? 24 kWh – 7.2 kWh = 16.8 kWh

How much coal is burned to power an incandescent bulb for 1 week? One month?
5.6 lbs. (week) and 24 lbs. (month)

What is the difference in pounds of CO_2 vented per month when using a CFL versus an incandescent light bulb? 50.4 lbs. – 12.6 lbs. = 37.8 lbs.

What is the difference between energy efficiency and energy conservation? Energy efficiency and energy conservation are two ways to reduce the amount of energy we use. Energy conservation means that you use electricity (or another energy source) only when needed. Energy efficiency refers to the amount of energy that is used to power a machine or appliance, versus the amount of energy that is lost as heat.

Challenge questions: Homework

What is the name of the utility company that provides electricity to your school?

What is the primary source of the electricity?

Handout 3 - The Charge

You have been asked to make a difference!! Your school administrators are looking for ways to reduce utility bills but they are not sure where to start. You have been contacted for your Energy Audit skills and expertise. Can your Energy Audit Team reduce the amount of electricity consumed? If so, think about how these potential savings may support student driven programs. THINK BIG and PROMOTE CHANGE!

First, you need to collect the data that will help you to raise awareness in your school regarding the energy/electricity usage. Then, you need to figure out recommendations and how to make others value these recommendations.

What recommendations will your team make?

Energy Audit Teams

Watt Busters: All sorts of devices use electricity. You will use the kilowatt meter to measure how many watts each device uses when on and possible phantom loads. Phantom loads are the result of a device pulling electricity even when it isn't in use. Collecting information on the number of kilowatts a device uses will allow you to estimate the cost of usage to inform your energy conservation and efficiency recommendations.

Lights-on Brigade: Much of the electricity used in a school is for the lights. The Lights-on Brigade will determine what type of bulbs the classroom has and how many watts each type of light bulb uses. This information will allow the Lights-on Brigade to estimate costs for lighting a classroom and to make energy conservation and efficiency recommendations.

Recommendations:

Each team is responsible for collecting the information, synthesizing the data, and forming recommendations into a report for the school administrator. Don't forget to include all the cost savings and reduced environmental footprint!

Each team will develop a report that will include:

Current data

Recommended action items

Reduced energy usage

Reduced CO_2 emissions

To get amount of CO_2:

$$(X)\ KwH \times \frac{1.52\ lbs\ CO_2}{1\ KwH} = (X)\ lbs\ CO_2$$

Watt Busters:

You are tasked with determining the electricity used by devises and figuring out ways to reduce electricity usage. Good luck.

Materials:

Kilowatt Meter

Questions:

What devices are powered by electricity in your classroom?

What is the wattage for each of these devices?

How often are each of these devices used?

Does the devise have a phantom load?

Using the Kilowatt Meter:

1. Turn off the device
2. Unplug the device
3. Plug in the kilowatt meter
4. Plug the device into the kilowatt meter
5. Push the 'watt' button
6. Record the number of watts in your data chart
7. Repeat for each device

Watt Buster Data			
Device	**Watts (Powered On)**	**Hours in Use**	**Watts (Phantom)**

Credit: Edited Energy Audit from Michigan Energy Works

Lights-on Brigade:

Your task is to determine the wattage and lumens of current light bulbs and make recommendations for more energy efficient options. You are also charged with the task of figuring out ways to get lights turned-off when not in use. Good luck!

EXAMPLE: Compact Fluorescent Lamp (CFL) and Incandescent Cost and Energy Use			**Conduct Your Comparison in These Columns**	
Bulb Type	**CFL**	**Incandescent**	**Current**	**Recommended**
Equivalent Light Output Wattage (**same Lumens**)	23 watts	100 watts		
Bulb Cost	$7.00	$0.25		
Bulb Lifespan	10,000 hours	1,000 hours		
Bulb Cost for 10,000 Hours	**$7.00**	$0.25 x 10 = **$2.50** (takes 10 bulbs to get 10,000 hours)		
Energy Used in kWh	10,000 hours of light at 23 watts = 230,000 watt-hours 230,000 ÷ 1,000 watts = **230 kilowatt-hours (kWh)**	10,000 hours of light at 100 watts = 1,000,000 watt-hours 1,000,000 ÷ 1,000 watts = **1,000 kilowatt-hours (kWh)**		
Utility Electricity Cost for 10,000 hours of light at $0.12 / kWh	230 kWh x $0.12 = **$27.60**	1,000 kWh x $0.12 = **$120.00**		
Store Purchase Cost + Electricity/ Operation Cost = **Total Cost**	$7.00 + $27.60 = **$34.60**	$2.50 + $120.00 = **$122.50**		
	That is a savings of over $87.90			

Credit: E3ACurriculum Materials

Lesson P

Local Energy Debate

Quick Overview

Lesson Scope: Reducing carbon dioxide (CO_2) emissions through energy awareness

Length: Three ~45-minute class periods

Grade Level: 5–8

Materials: Computers, debate sheets

Materials Needed per Student

Computer

Debate sheets

Each student will need paper and instructions to record trials.

Adapted from the National Energy Education Development Project (2013). *Great Energy Debate.*

Guiding Question

How can we become agents of change to address current and future energy issues?

This lesson is intended for students to explore the positive and negative aspects of some of the United States' main energy sources. Students will explore these aspects for their assigned energy source. During the presentation and debate, student teams will be required to defend advantages and implications of their energy source's disadvantages. This may illuminate other factors, such as the economy and environment, as students weigh the tradeoffs of using a particular energy source.

Key Understandings

- In the United States, most of our electricity is generated from six major sources of energy (coal, nuclear power, petroleum, natural gas, hydroelectricity, and wind).
- These sources represent both **renewable** and **non renewable energy**.
- Each energy source has advantages and disadvantages that can be identified at different stages of production.
- Economic, environmental, and social factors need to be considered when evaluating the tradeoffs of each energy source.

Objective

- Students will investigate, research, discuss, and present the economic and environmental benefits and challenges of using the major energy sources.

Vocabulary

- Renewable energy
- Non renewable energy

Background Information for the Teacher

- This activity has been adapted from the NEED's Great Energy Debate. This document is available as a resource and provides additional options for classroom use at *https://the-need-project.myshopify.com/products/great-energy-debate*.

The lesson will allow students to research, discuss, and present information about a specific energy source. Each student group will be making a presentation to a board of community leaders (could be other students, parents, or teachers) at a mock community meeting. While students are giving presentations, the other students will be serving as community members who are attending the meeting to hear about potential energy sources for their community. As community members, students can ask questions of the other groups at the end of the energy presentations by other groups.

Note: *The intention is for students to challenge or call attention to some of the potential drawbacks of each energy source.*

Although the *Energy Source Debate Sheets* will provide students with the information needed to discuss their energy sourc,e it is not intended to be an exclusive list. Student groups should be encouraged to conduct their own research to supplement the material. Please note that the information provided on the debate sheets can be open to a reader's interpretation of what constitutes a pro, con, or fact. Some students will be tempted regard all pieces of information as facts. In this case, it will be necessary to remind students that although they are advocating for the energy source that they were assigned, they will also need to present a balanced view of their source, including disadvantages. They should also explore some pros and cons of other energy sources so that they can contribute to the discussion.

Another option for the debate sheet is to remove the fact column and use only the disadvantage and advantage columns to avoid confusion.

The goal at the end of the debate/presentations is for students to have some basic understanding of the complex nature of energy. There is no one right or wrong answer, but the solution will likely be a portfolio of options. Some strategies will focus on efficiency and conservation; some will focus on reducing greenhouse gas emissions; and some will rely on renewable sources.

Common Student Misconceptions/ Student Challenges

Note: *This activity allows students the opportunity to discuss various energy sources from multiple perspectives.*

Students may believe that because renewable energy is better for the environment, it should automatically be considered as our primary source of energy. They may not understand that our energy needs far exceed the amount of energy that renewables can provide at this time. Additionally, they may not understand how economics plays a role in our energy portfolio.

Next Generation Science Standards

Performance Standard.HS-ESS3-4:

Evaluate or refine a technological solution that reduces impacts of human activities on natural systems.

Disciplinary Core Idea

Scientists and engineers can make major contributions by developing technologies that produce less pollution and waste and that preclude ecosystem degradation.

When evaluating solutions, it is important to take into account a range of constraints, including cost, safety, reliability, and aesthetics, as well as to consider social, cultural, and environmental impacts.

Science and Engineering Practices

Design or refine a solution to a complex real-world problem, based on scientific knowledge, student-generated sources of evidence, prioritized criteria, and tradeoff considerations.

Crosscutting Concepts

- Feedback (negative or positive) can stabilize or destabilize a system.
- Engineers continuously modify these technological systems by applying scientific knowledge and engineering design practices to increase benefits while decreasing cost and risks.

Materials

- Set of energy source debate sheets for each team (two or three students per team) (Appendix 4.2.1)
- SWITCH videos accessed at *http://www.switchenergyproject.com/topics/alltopics*
- Access to computers for research

Setup

- Copy the energy source debate sheets (enough copies for each team) (Appendix 4.2.1)
- Access to computers for background research.
- Recruit some adults to serve as community board members and give them the commissioner handout to explain their role. (Appendix 4.2.2)

- Set a table in front of the room for the community board members (usually three)

Length of Lesson

This lesson will take approximately three 45-minute class periods.

Suggested Procedure

Day 1

1. Start by watching the first section of the video SWITCH at *http://www.switchenergyproject.com/topics/alltopics* to introduce the lesson. If time allows, you can also watch the segment on coal, natural gas, nuclear, and wind sources. Tell students they will be investigating the challenges and solutions to the major energy sources in a scenario, the *Great Energy Debate*. It will be up to them to defend, debate, and decide which energy source should power their community.

2. Divide the class into four groups and assign (or allow them to randomly select) one of the main energy sources (coal, nuclear, natural gas, and wind). Note that we will not be using petroleum as a source for this activity since it is rarely used for electricity production in the U.S. Explain that students will be researching their own energy source to make a presentation to the community board, but they also will need to research other energy sources so they can ask good questions. Answer any questions students may have.

Day 2

1. Hand out the *Energy Source Debate Sheets (Appendix 4.2.1)*. Each student or group should have a set including all energy sources being used in the game. Explain that each group should start by completing the sheet for the energy source sheet that they were assigned. For each sentence, students need to determine if the statement is a *pro* or a *con* (ignore *fact* to avoid confusion) for their energy source.

2. Instruct students that they will need to be ready to present the advantages of their energy source during a brief (three minute) presentation to the community board. Their goal is to convince the commissioners to select their energy source. Partly, this will be decided based on their ability to argue for their source. Remind students that they want to advocate the positive attributes of their energy source and may want to call attention to the negatives of other sources.

1. Explain that they will need to do research their energy source and the other teams' energy sources. If you did not have time in class to watch the Switch video clips on individual energy sources, encourage students to visit the website *http://www.switchenergyproject.com/topics/alltopics to watch short video clips about the various energy resources*.

This video will prepare them for the debate and will give them information for when they want to challenge another energy source. Suggest that students take notes about the various resources to be used as talking points in the debate.

Day 3

(The community meeting)

1. Ask three colleagues, school administrators, community leaders, or parents to serve as city council members for the debate. If no volunteers are available, you may assign three students to serve as the council members.

2. On the day of the mock community meeting, place a table at the head of the room for the commissioners. Student groups will be making a presentation to the class and the commissioners. Commissioners and community members need to hear facts about each energy source so that they can make decisions about future energy needs. The groups should devise "solutions" or the advantages of using their energy source to solve the energy crisis.

3. Each group is then given a set amount of time (about three minutes) to present their energy source as a solution to the problem and why the community needs to use this source to provide electricity. After the presentation, the remaining groups have an opportunity to ask questions and voice concerns about the group's energy source. This discussion should be moderated, and students need to be reminded to (a) back up claims with evidence, and (b) be respectful by only criticizing ideas and not people. After all groups have made presentations, the commissioners will leave the room to discuss their options.

4. While the commissioners are out of the room, the teachers should engage students in a discussion/vote to see which source they feel will win. You can take a vote to see how the students feel (note that because of the competitive nature of the activity, students may automatically vote for their own source). Engage the students with probing questions about the complex nature of our energy situation based on information presented and students' comment.

Did you agree with the results of the game? Was one energy source perceived as the best? Are there concerns for having only one energy source? Why? Was this a difficult process? Why? Finally, do you foresee an energy source that will better meet the demands for a sustainable future? How will technology play a role? Will human behaviors play a role in this sustainable future? When the commissioners return to the classroom they should tell the students that they have decided to employ elements of each of the energy sources presented:

Coal – Because it is inexpensive and abundantly available.

Natural gas – Because it is less expensive than coal and has some advantages over coal (less industrial processing and handling) and is slightly better for the environment.

Wind – Because it is renewable and applicable for some smaller-scale applications in the community in places where wind is significant.

Nuclear – Because it is economical, safe, and fossil fuels do not emit greenhouse gasses.

Solar – Because it is readily available, easy to implement, and is renewable in areas with significant sunshine.

You want students gain some basic understanding of the complex nature of energy. There is no one right or wrong answer, but the solution will be a portfolio of options. Some strategies will focus on efficiency and conservation; some will focus on reducing greenhouse gas emissions; and some will rely on renewables sources.

Wrap-Up/Student Feedback

(Including Assessment Opportunities)

Group work and discussion are possible assessment opportunities. After the debate, the opportunity for a written reflection can help students and the teacher determine what students learned throughout this lesson. Other questions that could be posed to the students include the following:

1. Is there one energy source that is the "best" source? Why does our country use a variety of energy sources?
2. What do you think will be the greatest energy challenge the world will face in the next 50 years?

Supplemental Materials and References

Arcos Films LLC. (2012). *SWITCH energy project*. Retrieved from http://www.switchenergyproject.com/topics/alltopics This website contains videos necessary for this lesson regarding energy sources and their benefits and challenges.

National Energy Education Development Project. (2013). *Great energy debate*. Retrieved from https://the-need-project.myshopify.com/products/great-energy-debate. This resource contains handouts and instructions for conducting the debate for this lesson.

Coal

	Relevance	
	Advantage	Disadvantage
1. Coal is one of the most abundant fuels in the U.S. We have a 168–239-year supply based on the current rate of consumption.		
2. Although coal is still being formed today, we use it thousands of times faster than it is formed.		
3. Coal generates 37 percent of the electricity in the U.S.		
4. The U.S. exports about 10 percent of the coal it produces to other countries.		
5. Coal has been burned to cook food and heat living spaces and water for thousands of years.		
6. Today, over 90 percent of the coal consumed in the U.S. is used by the electric power sector to generate electricity.		
7. When coal is burned, carbon dioxide, sulfur dioxide, nitrous oxide, and other pollutants are produced.		
8. To remove coal buried deep in the earth, mine shafts are constructed to bring coal to the surface.		
9. An easier way to mine coal near the Earth's surface is to remove the layers of earth to uncover the coal. This is called surface mining.		
10. Large amounts of land are disturbed in the process of surface mining.		
11. Surface mines can be restored to grasslands or parks after the coal is removed.		
12. More than two-thirds of the nation's coal is produced from surface mines.		
13. The water that filters through abandoned mines picks up chemicals that pollute the water if the mines are not closed correctly.		
14. Coal is used to smelt iron into steel and by the paper and building supply industries.		
15. Coal can be turned into a gas to make it burn cleaner. This process is expensive.		
16. Coal mining can be dangerous for miners due to gases and explosion hazards.		
17. Ash from coal plants can be reycycled and used for cement additives, roadway materials, and even habitat restoration for oysters.		
18. Some cleaner coal technologies require less coal to produce the same amount of electricity.		
19. The methane gas that is found around much of the coal in the U.S. is a valuable resource.		
20. The electricity industry uses items like scrubbers to reduce harmful emissions from coal plants.		

Credit: Adapted from the National Energy Education Development Project (2013). *Great Energy Debate*.

Natural Gas

	Relevance	
	Advantage	Disadvantage
1. Natural gas was formed from the decomposition of tiny sea plants and animals that lived hundreds of millions of years ago.		
2. Natural gas is mostly of a chemical called methane.		
3. Natural gas is odorless; an odorant called mercaptan is added for safety.		
4. Natural gas can be processed and other products, such as propane and the materials in plastics, can be recovered from it.		
5. Natural gas is considered to be the cleanest-burning fossil fuel. It produces almost no sulfur or nitrogen oxides.		
6. Natural gas and petroleum are often found together in underground deposits.		
7. In the past, oil drillers were not interested in the natural gas that was found at the site of an oil well. Today, it is as valuable as the oil.		
8. The invention of high-pressure pipelines has made it possible to transport natural gas all over the U.S.		
9. Leaks can occur in natural gas pipelines. Fires and explosions can result from these leaks if proper safety precautions are not taken.		
10. About 8 percent of the natural gas we produce comes from federal offshore wells in the Gulf of Mexico.		
11. Natural gas is a nonrenewable resource.		
12. Today, the U.S. has a large supply of natural gas. There are also large reserves of natural gas offshore on the outer continental shelf and in the Gulf of Mexico.		
13. Natural gas is used equally by homes and businesses, industry, and for electric power generation.		
14. Natural gas can be used as a cleaner-burning transportation fuel.		
15. Natural gas supplies will likely last about 100 years at today's prices and consumption rate.		
16. Natural gas accounts for 27.4 percent of total U.S. energy consumption.		
17. If a higher price is charged for natural gas, supplies could last as long as 200 years.		
18. Fifty-seven percent of homes in the U.S. use natural gas as their main heating fuel.		
19. Natural gas is used to produce peak load electricity because natural gas furnaces can be bought on line and shut down quickly and efficiently to generate steam for electricity.		
20. Burning methane produces carbon dioxide. Both methan and carbon dioxide are greenhouse gases that trap heat energy. Increasing the levels of greenhouse gases in the atmosphere can affect the global climate.		

Credit: Adapted from the NEED Project. Source of content: www.NEED.org

Nuclear

	Relevance	
	Advantage	Disadvantage
1. In 1939, scientists discovered that certain atoms could be split. The splitting of atoms releases a great amount of energy.		
2. Over 100 nuclear power plants operate in the U.S.		
3. Nuclear plants provide about 19 percent of the electricity generated in the U.S.		
4. A nuclear reactor can supply a large amount of energy using a very small amount of fuel.		
5. The construction of nuclear power plants is very expensive compared with fossil fuel plants.		
6. Nuclear reactors do not burn uranium or fuel to generate electrical power. They split the uranium atoms, so their emissions are minimal.		
7. Uranium is easy to transport.		
8. Uranium is inexpensive.		
9. The U.S. has abundant supplies of uranium. However, we import most of the uranium used in power plants because it is cheaper than mining it.		
10. Nuclear power plants produce electricity by heating water into steam in the same way as fossil fuel plants.		
11. Workers at nuclear power plants receive less radiation from the plant than they do from other sources like medical x-rays.		
12. Some parts of reactors become radioactive after they have been used.		
13. Radioactive waste from nuclear power plants is stored underground in spent fuel pools or dry cask containers at the plant sites.		
14. Uranium has a very high energy density, producing a large amount of energy from a small amount of mass or space.		
15. Uranium is a nonrenewable energy source.		
16. A nuclear power plant produces a lot of wasted heat. If this heat is used in a moving water system, the water temperature can increase.		
17. The main health risk from a nuclear power plant is potential radiation exposure.		
18. Nuclear power plants in the U.S. are highly regulated.		
19. An accident at a nuclear power plant could cause widespread damage if people or the environment were exposed to high levels of radioactivity.		
20. There has been renewed interest in nuclear power in the U.S. in the past few years as concern over global climate change has increased.		

Credit: Adapted from The NEED Project. Source of content: www.NEED.org

Wind

	Relevance	
	Advantage	Disadvantage
1. Wind is air in motion caused by the uneven heating of the Earth's surface by the sun.		
2. Wind turbines do not cause air or water pollution because no fuel is burned to generate electricity.		
3. Wind is a renewable source of energy.		
4. Wind turbines operate on average about three-fourths of the time, though not always at capacity.		
5. For hundreds of years, windmills were used to grind wheat and corn, to pump water, and to cut wood at sawmills.		
6. Wind turbines have turning blades to harness the wind's kinetic energy. The blades are connected to drive shafts that turn generators to make electricity.		
7. Wind plants can typically convert 30 to 40 percent of the wind's kinetic energy into electricity.		
8. When the wind is not blowing, other sources of energy must be used to generate needed electricity.		
9. The locations of wind farms are carefully planned. Good sites include the tops of smooth, rounded hills, open plains or shorelines, and mountain gaps.		
10. Offshore turbines produce more electricity than turbines on land.		
11. Wind power plants, or wind farms, are clusters of several wind turbines spread over a large area. The land around the wind turbines can also be used for grazing or growing crops.		
12. Wind farms are often owned and operated by business people who sell the electricity to utility companies because the turbines can be expensive to build.		
13. Wind turbines can be used in remote areas that do not otherwise have access to electricity.		
14. Almost every state has the capacity to produce electricity from wind.		
15. The U.S. generates about 20 percent of the world's wind energy.		
16. Older wind turbines are very noisy; new technologies have eliminated most noise.		
17. Wind turbines can injure birds and bats that fly into the spinning blades.		
18. Producing electricity from wind power plants costs about $0.05/kWh. The average cost of electricity (generated mostly from fossil fuels) in the U.S. today is about $0.12 cents/kWh.		
19. Wind turbines provide the U.S. with enough electricity to power over 13 million homes.		
20. Offshore turbines cost more money to build and operate than turbines on land.		

Credit: Adapted from the National Energy Education Development Project (2013). Great Energy Debate.

Energy Debate Game

Game premise

Our "hypothetical" community receives its energy from an energy co-op. Every couple of years the energy co-op evaluates its current energy portfolio and considers changes. This year, five groups are asking to make a presentation to the energy commission. There will be a presentation on coal, natural gas, wind, solar, and nuclear energy.

The role of the energy commission

As a commissioner, you are asked to listen to the presentations and then ask questions (just as would happen in a real community/city council meeting). After the commissioners are finished with their questions, we will open the floor to citizens in the gallery. These citizens are actually students who will be making competing presentations so their questions may be slanted towards their energy source, or they could be the type of comments you would hear at a regular community meeting.

Process

At the end of the presentations and question/answer period, the commissioners will adjourn to discuss and make their recommendations. While you are "discussing" the issues, the classroom teacher will be conducting an assessment by asking the students probing questions about the complex nature of our energy situation. The teacher will also ask them to vote on how they think the commissioners will rule. This is a game, so some students may vote for their energy source or they may feel strongly about another source. When you return to the meeting, you will tell the students that you have decided to employ elements of each of the energy sources presented:

- Coal – Because it is inexpensive and abundantly available
- Natural Gas – Because it is less expensive than coal and better for the environment
- Wind – Because it is renewable and usable for some smaller-scale applications in the community
- Solar – Because it is an abundant resource and renewable
- Nuclear – Because it is economical, safe, and cleaner for the environment than fossil fuels

Appendix 4.2.1 | Energy Debate Game

Goal

We want students to gain some basic understanding of the complex nature of energy. There is no one right "or" wrong answer, but the solution will likely be a portfolio of options. Some strategies will focus on efficiency and conservation; some will focus on reducing greenhouse gas emissions; and some will rely on renewables.

Lesson Q

Career Opportunities

Quick Overview

Lesson Scope: Opportunities and challenges to be found within renewable energy careers

Length: Two to three ~45-minute class periods

Grade Level: 5–8

Materials: Internet access, student rubric

Materials Needed per Student

Internet access

Student rubric

Each student will need paper and instructions to record trials.

Guiding Question

How can we become agents of change to address current and future energy issues?

The demand for skilled personnel in the fields of energy, engineering and more are creating new opportunities for people to use their unique talents, skills, and interests workin in jobs that have a positive impact on our environment and quality of life. This activity will provide an overview of those opportunities and challenges in energy careers.

Key Understandings

- Energy is now and will continue to be a major consideration in all parts of our lives. As the evolving energy sector becomes more complex, new opportunities will develop for careers in the energy sector.
- Students will be able to explore areas they may not have previously considered as a potential career choice.

Objectives

- Students will investigate careers in the energy sector, specifically in the areas of geology, climate change mitigation, and carbon capture and storage (CCUS) as a means to understand our complex dependence on the energy sector and to promote emerging technologies for the future.

Background Information for the Teacher

As the number of humans increases world-wide, our activities and energy needs increase and affect the land,

oceans, and atmosphere. Our energy needs have even altered global climate patterns. Burning fossil fuels, releasing chemicals into the atmosphere, reducing the amount of forest cover, and the rapidly expanding farming, development, and industrial activities are releasing carbon dioxide (CO_2) into the atmosphere, occuring at a rate that cannot be balanced by natural methods. New methods of energy production and CO_2 emission management are necessary to change the current trend.

Teachers of our future leaders can help students be better prepared and equipped to move into jobs in both established and emerging fields of science, technology, engineering, and mathematics, (STEM) providing solutions to current energy and environmental challenges. Many of these careers may be new to students but interesting once they become aware of the many possibilities. There are several web sources cited in the Supplemental Materials and Reference section.

Next Generation Science Standards

Performance Standard MS-ESS3-3:

Apply scientific principles to design a method for monitoring and minimizing a human impact on the environment.

Disciplinary Core Idea - Human Impacts on Earth Systems

Human activities have significantly altered the biosphere, sometimes damaging or destroying natural habitats and causing the extinction of various species. But changes to Earth's environments can have different impacts (negative and positive) for different living things.

Typically, as human populations and per-capita consumption of natural resources increase, so do the negative impacts on Earth unless the activities and technologies involved are engineered otherwise.

Science and Engineering Practices - Constructing Explanations and Designing Solutions

Apply scientific principles to design an object, tool, process, or system.

Crosscutting Concepts

- **Cause and Effect** - Relationships can be classified as causal or correlational, and correlation does not necessarily imply causation.
- **Stability and Change** - Stability might be disturbed either by sudden events or gradual changes that accumulate over time.

Materials

- Access to web sites and resource materials
- Student rubric (Attachment 4.4.1)

Setup

- Prepare material and equipment

Length of Lesson

This lesson will take two to three 45-minute class periods.

Suggested Procedure

Teaching Strategy

Activity

1. Students are to choose one career in the energy sector that interests them and research the following five different aspects of the job/career:
    - Day-to-day activities
    - Skills required
    - Education/training
    - Salary
    - Job outlook
2. Encourage students to think beyond geoscience and engineering.
3. Students choose a product to present to the class at an assigned date. Choices include a brochure, a TV announcement, or a written report advertising this career.
4. Invite school administrators, parents, grandparents, or other classes to hear the presentations.

Wrap-Up/Student Feedback

(Including Assessment Opportunities)

- Participation in the activity
- Completed product and presentation

Extensions

- Invite a guest(s) who works in one of the geology-related careers to speak on the job description requirements for their position. Before the classroom visit, students should prepare a list of five questions to ask the visitor(s) about their jobs.
- Divide the students into pairs. Using the job descriptions, have one person of each pair role play the applicant and one person role play the employer. The students will conduct a mock interview by taking turns asking questions. After 15 minutes, the students should switch roles.
    - Ask the students who played the role of employer whether they would have hired the applicant and name specific reasons why or why not.

Supplemental Materials

"Professional Career Pathways in the Geosciences" includes the American Geological Institute's Careers site on the internet, an interactive CD-ROM, and a video. *http://www.americangeosciences.org/workforce/geoscience-careers-brochure*

American Geological Institute (AGI) developed these materials in cooperation with its member societies and other geoscience organizations through a grant from the Alfred P. Sloan Foundation.

The Directory of Geoscience Departments lists more than 800 degree-granting geoscience departments in North America; the Guide to Geoscience Departments gives in-depth

profiles of selected geoscience departments. Both books are available from AGI.

Contact the organizations listed here, as well as other geoscientific societies and government agencies that employ geoscientists.

American Geological Institute
Address: 4220 King Street
Alexandria, Virginia 22302-1502
Phone: 703/379-2480 FAX: 703/379-7563
Web: *https://www.americangeosciences.org*

- **American Association of Petroleum Geologists**
  Address: P.O. Box 979
  Tulsa, Oklahoma 74101-0979
  Phone: 918/584-2555 FAX: 918/560-2636
  Web: *http://www.aapg.org*
- **Geological Society of America** Address:
  3300 Penrose Place
  P.O. Box 9140
  Boulder, Colorado 80301
  Phone: 303/447-2020 FAX: 303/447-1133
  Web: *http://www.geosociety.org*

References

Campus Explorer. (2014). *Career information: Mining and geological engineers*. Retrieved April 16, 2014, from *http://www.campusexplorer.com/careers/6811960E/mining-and-geological-engineers/*

Eden Project. (2008). Retrieved from *http://www.edenproject.com*

Green Energy Jobs (2014). *Green Energy Jobs - Career Guide*. Retrieved May 16, 2014, from *http://www.greenenergyjobs.com/career-guide/*

Career Product Rubric

Name:_____ **Date:**_____

Standards: WHST.9-12.2 RST.11-12.1 RST.11-12.7		Absent	Tolerable	OK	Decent	I'm Convinced!	
		0	1	2	3	4	5
CRITERIA	**Organization** Follows proper narrative composition format						
	Content Reflects accurate knowledge of job/career						
	Answers 5 different aspects of the job/career: day to day activities, skills required, education /training, salary, & job outlook						
	Written/Oral Text Must be in your own words Follows conventions						
	Bibliography At least 2 different types of sources used						
	Conventions (Mechanics) Correct spelling Correct grammar and punctuation Correct paragraph structures						
	Appearance (Neatness) Legible letter forms Effective word and letter spacing Clean corrections						
Comments: (grading scale should be adapted to your needs)		0	1 8	9 10–15 16	17 18–23 24	25 26–31 32+	
		F	D	C	B	A	Grade

Lesson R

Science Fair Opportunities

Quick Overview

Lesson Scope: Opportunities developing science fair projects

Length: One to two ~45-minute class periods

Grade Level: 5–8

Materials: Internet access, student rubric

Materials Needed per Student

Internet access

Each student will need paper and instructions to record trials.

Guiding Question

How can we become agents of change to address current and future energy issues?

Application is part of the learning process. A science project with a culminating event lets students use what they know and conduct a real investigation, which may lead to continued interest in the subject, a career in science, or even the starting point for a solution to a worldwide problem.

Key Understandings

- A high quality science fair project done well puts good science into practice.
- A science fair project must contain research and testing to arrive at specific conclusions.
- Experimentation is driven by inquiry and is a method for testing ideas.

Objectives

- Students choose a hypothesis, design and conduct an investigative experiment (investigation), or design and engineer a practical solution to a problem (invention).

Background Information for the Teacher

An exciting component of a middle school science curriculum can be a science fair. This activity gives students a chance to practice science as they investigate testable questions. It also gives students a chance to explore and expand their interests in a particular area of science.

Science fairs work well near the end of the year after instruction has covered scientific procedures and opportunities for students to work with investigative activities in the classroom. A science fair

is a long-term project that should be carefully guided to ensure quality science is being practiced. Timelines with explicit due dates for milestones help students unfamiliar with working on long- term projects manage their time. Rubrics also help define expectations of the activity overall. There are several good resources to help teachers, which are cited in the Resources and References section.

A science fair can be part of an open house for the school. It is also a good way to involve administrators, parents, and community members. It can be an opportunity for students to become excited and interested in science as a career.

Next Generation Science Standards

Performance Standard MS-ETS 1-1:

Define the criteria and constraints of a design problem with sufficient precision to ensure a successful solution, taking into account relevant scientific principles and potential impacts on people and the natural environment that may limit possible solutions.

Disciplinary Core Idea ETS1.A: Defining and Delimiting Engineering Problems

The more precisely a design task's criteria and constraints can be defined, the more likely it is that the designed solution will be successful. Specification of constraints includes consideration of scientific principles and other relevant knowledge that are likely to limit possible solutions.

Science and Engineering Practices - Asking Questions and Defining Problems

Define a design problem that can be solved through the development of an object, tool, process, or system and includes multiple criteria and constraints, including scientific knowledge that may limit possible solutions.

Developing and Using Models

Develop a model that generates data to test ideas about designed systems, including those representing inputs and outputs.

Crosscutting Concepts - Influence of Science, Engineering, and Technology on Society and the Natural World

All human activity draws on natural resources and has both short and long-term consequences, positive as well as negative, for the health of people and the natural environment.

The uses of technologies and limitations on their use are driven by individual or societal needs, desires, and values by differences in such factors as climate, natural resources, and economic conditions.

Materials

- Access to web sites and resource materials

Setup

- Prepare material and equipment

Length of Lesson

This lesson will take one or two 45-minute class periods to explain the principles of a science fair and the components of a quality project. Consider reviewing some of the websites listed in the resource section of this lesson with the students. Include a discussion of the rubric for the projects. Most of the science fair work is done outside of the classroom with periodic due dates for a progress check.

Suggested Procedure

Activity

Students are to choose a hypothesis that interests them and investigate that question using the scientific method:

- Defining a problem
- Formulating a hypothesis
- Designing on experiment
- Collecting data
- Formulating a conclusion
- Presenting results

Give a defined due date. This may also include target dates for project components.

Invite school administrators, parents, grandparents, or other classes to hear the presentations.

Wrap-Up/Student Feedback
(Including Assessment Opportunities)

- Participation in the activity
- Completed product and presentation

Extensions

Participate in a judged science fair.

Supplemental Materials

Cool-Science-Projects. (n.d.). *Cool science projects: We deliver with guides, tips, and examples*. Retrieved May 16, 2014, from http://www.cool-science-projects.com/index.html.

Discovery Education. (2013). Science fair central *Science fairs offer students an opportunity to practice science investigation and invention*. Retrieved May 16, 2014, from http://school.discoveryeducation.com/sciencefaircentral/Getting-Started.html.

Education.com. (2014). *Science fair project ideas: Over 2,000 free science projects*. Retrieved May 16, 2014, from http://www.education.com/science-fair/help-child-with-science-fair-projects.

Ipl2: Information You Can Trust, Internet Public Library. (n.d.). *Science fair project resource guide*. Retrieved May 16, 2014, from http://www.ipl.org/div/projectguide/gettingstarted.html.

MiniScience.com. (n.d.). *Experimental science projects: An introductory level guide*. Retrieved May 16, 2014, from http://www.miniscience.com/SciProjIntro.asp.

Rowan, T. (n.d.). *Science fair primer.* Retrieved May 16, 2014, from http://users.rcn.com/tedrowan/primer.html.

Smoak, E., Williamson, R. (n.d.) A student's guide to keeping science in your science project. Retrieved May 16, 2014, from http://www.ag.ncat.edu/extension/programs/dte/science.pdf.

Stryjewsk, E. (n.d.) Fathermag.com, *Science fair project guidelines*. Retrieved May 16, 2014, from http://www.fathermag.com/904/science/guidelines.shtml.